Data Science

Lecture 02: Probability Theory

Prof. Dr. Michael Granitzer

Speaker notes

Converted from LNKB-tex part-en-ds-chp2-probability-and-statistics.tex. The content units of the original chapter are included below in the original order, as shared units from content/units/math/stochastic/ (the same units feed the probabilistic machine learning course), plus the Bernoulli/binomial split-off, sampling and estimation before the test, and course-local parts (module takeaways, the churn shop).

Two sessions. Session 1 covers Modules 0–4 (orientation, events, counting, the axioms, conditioning and Bayes); session 2 covers Modules 5–7 (random variables, describing a sample, testing). The Module 5 slide marks the boundary.

Where We Left Off

Lecture 01 ended on a claim: data does not speak for itself — an assumption always speaks for it, and a pattern is knowledge only once it has been checked. Today builds the language in which both are written.

Assume

Modules 1–5 — a model of chance: outcomes, probabilities, random variables, the joint

→

Observe

Module 6 — a sample, described, with its error

→

Check

Module 7 — would chance alone produce this?

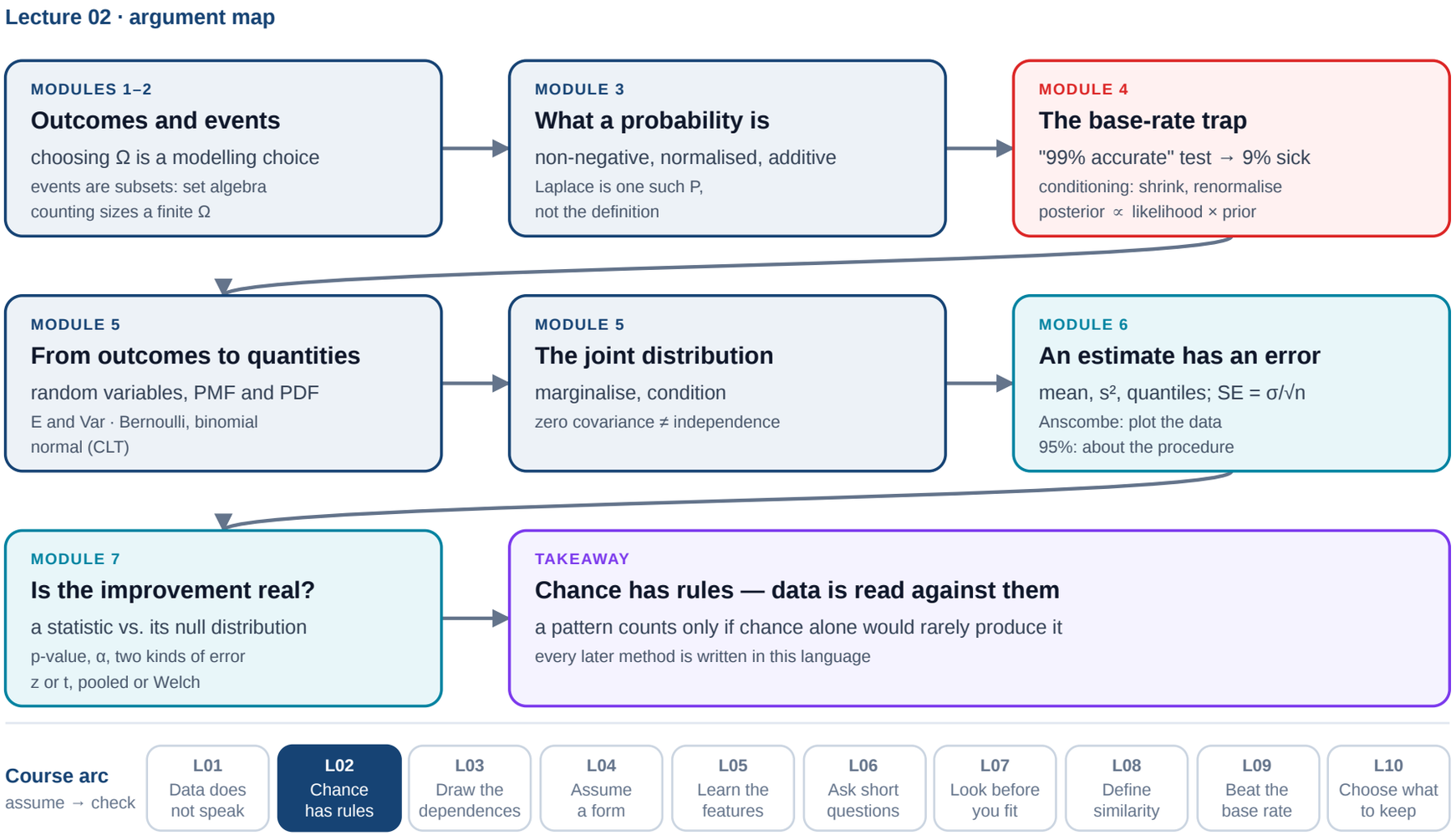
> The app users of Lecture 01 churn less. By the end of today you can say whether that difference is real — not yet *why* (that is Lecture 03).

Speaker notes

Pick up the thread from the roadmap table of Lecture 01. Say which module equips which step. Say why the formal treatment comes first: every later check — a test, a validation error, lift — is a probability statement, so the language has to exist before the first check can be written down.

The lecture spans two sessions; the Module 5 slide marks the boundary. Recall this three-card flow there.

Argument Map



Speaker notes

The lecture on one page. Top row: the objects (outcomes, events), what makes a number a probability, and the one operation that matters, conditioning — with the trap it resolves: the 99% test that is right only 9% of the time. Middle row: from outcomes to quantities to the joint, then to a sample and the error of what we compute from it. Bottom row: the check, a test statistic against its distribution under "chance alone". Point at the strip: every later lecture writes its assumption and its check in this language.

Module 0

Orientation

> ***Guiding Question:*** *What does a data scientist actually need from probability theory — and why does the formal treatment come first?*

The Landscape

Probability theory
Given a model of chance, what follows?
Kolmogorov axioms, distributions, limit theorems

Mathematical statistics
Given data, which model of chance is plausible?
Estimation, hypothesis testing

Data science
Given data and a decision, what should we do?
Prediction, learning, evaluation

The arrow runs left to right: this lecture builds the vocabulary the rest of the course spends its time using.

Speaker notes

Orientation slide. The three columns are not three fields to memorise but three questions — deduction from a model, inference of a model, decision with a model. Say explicitly that we start on the left because everything on the right is stated in its language.

What We Need From Probability

In this lecture

- sample spaces and events
- counting outcomes
- probability as a measure
- conditioning and independence
- Bayes' theorem
- random variables and distributions
- describing data numerically
- testing a hypothesis

Where it comes back

- naive Bayes and Bayesian decisions
- impurity measures in decision trees
- likelihoods in probabilistic models
- sampling and error estimates in evaluation
- independence assumptions in graphical models



Speaker notes

Set expectations: this is the one heavily formal lecture, and every item on the left has a named consumer on the right. If students ask "why do we need this", point at the right-hand column rather than defending rigour in the abstract.

Example: Students, Courses and Exams

Probabilistic ML — 40 students
a quarter of them fail the exam

Databases — 60 students
one in ten fails the exam

Each of 100 students sat exactly one of the two exams.

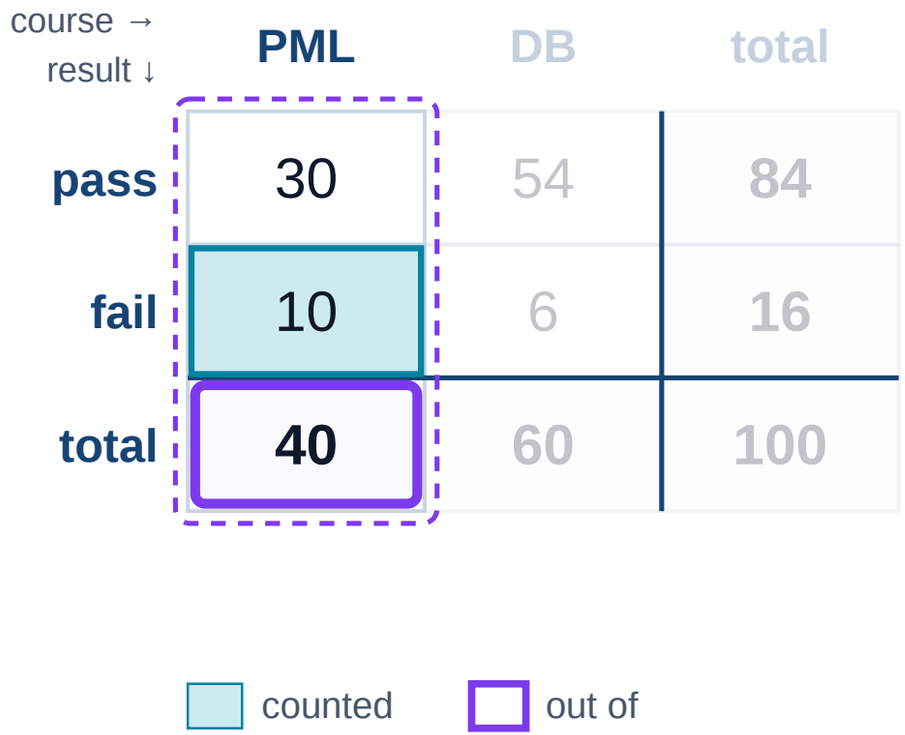
- **Forward:** how many students fail overall?
- **Backward:** a student failed — how probable is it that they took PML?

Speaker notes

The example is Bishop's two-box experiment (*Pattern Recognition and Machine Learning*, §1.2) with a course in place of a box and an exam result in place of a fruit. The experiment has two stages: a course $C \in \{\mathrm{PML}, \mathrm{DB}\}$ with $P(C = \mathrm{PML}) = 0.4$, then a result $R \in \{\mathrm{pass}, \mathrm{fail}\}$ with $P(R = \mathrm{fail} \mid \mathrm{PML}) = 0.25$ and $P(R = \mathrm{fail} \mid \mathrm{DB}) = 0.1$. The forward question asks for a marginal, $P(\mathrm{fail}) = 0.4 \cdot 0.25 + 0.6 \cdot 0.1 = 0.16$; it runs in the direction of the experiment. The backward question asks for $P(\mathrm{PML} \mid \mathrm{fail})$; it runs from the observed result back to its cause and needs Bayes' theorem. Both are answerable by counting alone.

Joint, Marginal, Conditional — One Table

? Of the PML students, what share failed?



P(fail | PML)

= **10** / **40** = 0.25

conditional: a cell, out of its own column

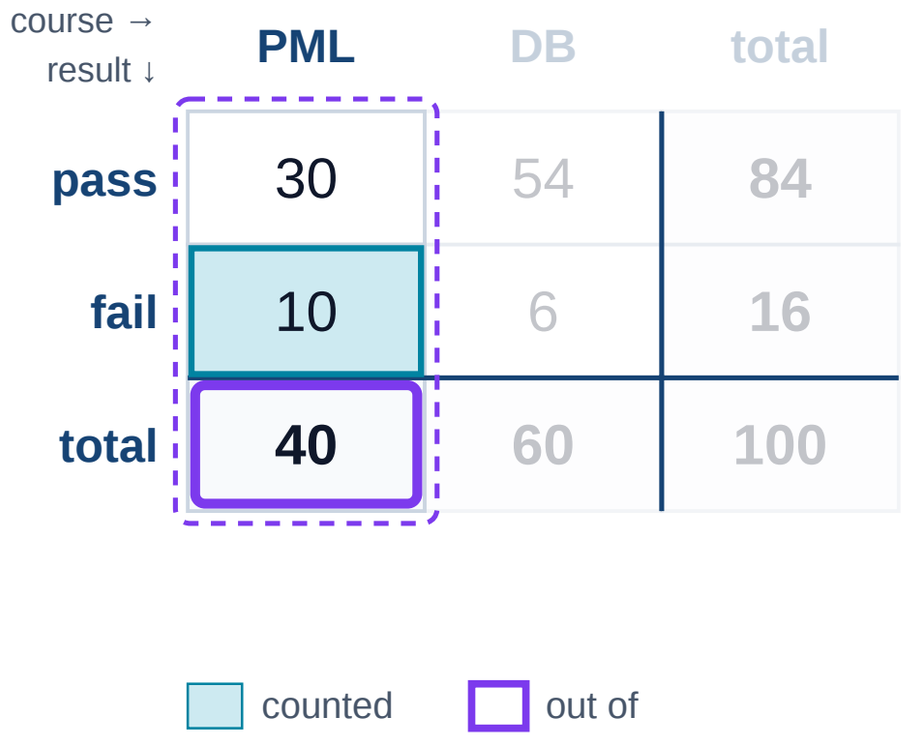
Speaker notes

Each question is shown first; the room answers before the next click marks the counted cells (teal) and the count they are taken out of (violet). Joint: $P(\mathrm{PML}, \mathrm{fail}) = 10/100$, one cell over the grand total. Marginal: $P(\mathrm{fail}) = 16/100$, a row total over the grand total. Conditional: $P(\mathrm{fail} \mid \mathrm{PML}) = 10/40$, a cell over its own column total; the DB column is not taken into account at all.

The joint probabilities sum to 1 over the four cells; the conditionals sum to 1 only within their own column, $P(\mathrm{pass} \mid \mathrm{PML}) + P(\mathrm{fail} \mid \mathrm{PML}) = 30/40 + 10/40 = 1$. The formal definitions follow later; on a finite table they reduce to exactly these divisions.

Sum Rule and Product Rule — On the Table

? 40% took PML, and a quarter of them failed. What share failed PML?



… of those, failed: P(fail | PML) = **10** / **40**

Product rule

P(PML, fail) = P(PML) · P(fail | PML)

= 40/100 · 10/40 = 10/100

a share of a share — the 40s cancel

Speaker notes

Sum rule: the row total is the sum of the row's cells, so $P(\text{fail}) = P(\text{PML}, \text{fail}) + P(\text{DB}, \text{fail}) = 0.10 + 0.06 = 0.16$; summing over the course removes it. Product rule: a cell is its column's share of the table times the cell's share of that column, $P(\text{PML}, \text{fail}) = P(\text{PML})\,P(\text{fail} \mid \text{PML}) = \frac{40}{100} \cdot \frac{10}{40}$. The column total appears once as a numerator and once as a denominator and cancels.

The Two Rules

Sum rule

$$P(X) = \sum_{Y} P(X, Y)$$

Add along a row or column to remove a variable.

Product rule

$$P(X, Y) = P(Y \mid X) P(X)$$

A cell is its column total times its share of that column.

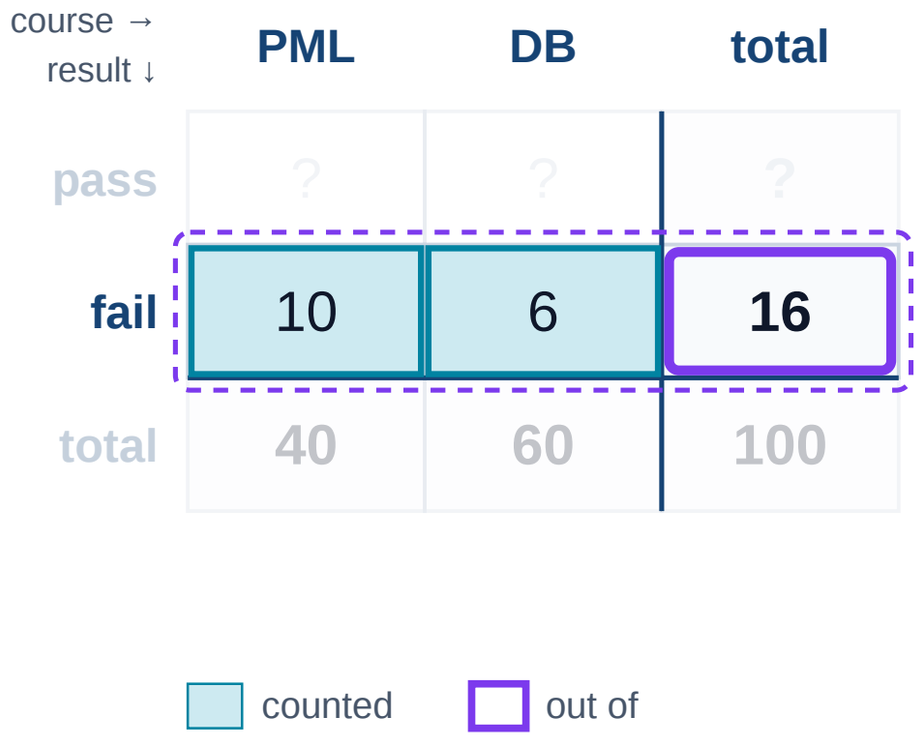
Almost all of probability theory is these two rules applied repeatedly.

Speaker notes

X and Y stand for the course and the result. $P(X)$ is shorthand for the whole distribution of X, so each rule is one equation per value. For continuous variables the sum becomes an integral, $p(x) = \int p(x, y)\, dy$. The product rule holds in either order, $P(X, Y) = P(Y \mid X)P(X) = P(X \mid Y)P(Y)$; equating the two forms gives Bayes' theorem. The formal treatment justifies both rules from the axioms and the definition of conditional probability and extends them beyond finite tables.

Bayes' Theorem — On the Table

? And which course did they most probably take?



P(PML | fail) = **10** / **16** ≈ 0.63

P(DB | fail) = **6** / **16** ≈ 0.37

PML — odds 10 : 6

same normaliser: comparing needs only the cells

Speaker notes

Steps 1–3 answer the question by counting: the evidence "failed" removes the pass row, 16 students remain, 10 of them from PML, so $P(\mathrm{PML} \mid \mathrm{fail}) = 10/16 = 0.625$, up from the prior $P(\mathrm{PML}) = 0.4$ because PML students fail more often (a quarter against a tenth).

Steps 4–8 rebuild the same answer from what is usually known instead of a table: the course sizes (the prior) and the fail rate per course (the likelihoods). The cell counts are hidden and appear only once they can be inferred. Prior times likelihood gives the cell, $0.4 \cdot 0.25 = 0.10$ (product rule); summing that over both courses gives the normaliser $P(\mathrm{fail}) = 0.10 + 0.06 = 0.16$ (sum rule); their ratio is the posterior. The pass row is never needed. Step 9 asks the comparative question, which course is more probable: both posteriors share the normaliser, so the comparison needs only the cells, $10 : 6$ in favour of PML.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

$$P(\text{PML} \mid \text{fail}) = \frac{0.25 \cdot 0.4}{0.25 \cdot 0.4 + 0.1 \cdot 0.6} = \frac{0.10}{0.16} \approx 0.63$$

Speaker notes

With equal fail rates, $P(\text{fail} \mid \text{PML}) = P(\text{fail} \mid \text{DB}) = 0.25$, the likelihood cancels and the posterior equals the prior: $0.10/(0.10 + 0.15) = 0.4$. Course and result are then independent. In general, evidence changes a belief only to the extent that the hypotheses assign it different probabilities; the ratio $P(\text{fail} \mid \text{PML})/P(\text{fail} \mid \text{DB})$ (2.5 originally, 1 in the variant) measures how much.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

$$P(\text{PML} \mid \text{fail}) = \frac{0.25 \cdot 0.4}{0.25 \cdot 0.4 + 0.1 \cdot 0.6} = \frac{0.10}{0.16} \approx 0.63$$

And if one in four DB students failed too?

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_Y P(X \mid Y)P(Y)$$

$$P(\text{PML} \mid \text{fail}) = \frac{0.25 \cdot 0.4}{0.25 \cdot 0.4 + 0.1 \cdot 0.6} = \frac{0.10}{0.16} \approx 0.63$$

And if one in four DB students failed too?

Then $P(\text{PML} \mid \text{fail}) = 0.4$: the result says nothing about the course.

Takeaway

Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. What follows makes them precise and extends them beyond counting.

Speaker notes

The terms used informally here — joint, marginal, conditional probability, Bayes' theorem — are all derived later from three axioms plus the definition $P(A \mid B) = P(A \cap B)/P(B)$. The sum rule is additivity (the third axiom) applied to the disjoint events $\{Y = y\}$; the product rule is that definition rearranged; combined, they give the theorem of total probability. The formal treatment does not contradict the counting picture; it removes the restriction to finite tables of counts.

Module 1

Events and the Algebra of Sets

> ***Guiding Question:*** *If probability is going to be a measure, what exactly are the objects it measures — and how do they combine?*

What Is a Random Experiment?

A coin flip, tomorrow's rainfall, whether a user clicks a link. What makes each of them a *random experiment* — and what would not be one?

Speaker notes

The definition does not require the world to be indeterministic; it requires only that the outcome cannot be predicted from what is specified. Repeatability "in theory" means the configuration and procedure can be re-created, not that the experiment is actually repeated. Many random experiments in data analysis are not designed by anyone — daily rainfall, whether a user clicks a link, the length of a queue at noon. There the configuration is whatever system produced the data, and treating it as repeatable is a modelling assumption, typically that the observations come from the same unchanged process.

What Is a Random Experiment?

A coin flip, tomorrow's rainfall, whether a user clicks a link. What makes each of them a *random experiment* — and what would not be one?

> **Definition (Random experiment).** A procedure that can — at least in theory — be repeated infinitely often, characterised by:
>
> 1. **Configuration** — a precisely specified system that can be reconstructed.
> 2. **Procedure** — an instruction for executing the experiment.
> 3. **Unpredictability** of the outcome.

Coin flip: a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

Randomness as Missing Information

> A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.
>
> Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

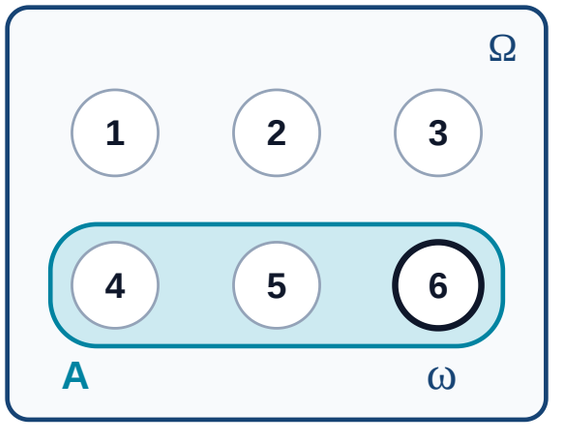
Speaker notes

Calling something random is a statement about the observer's knowledge, not only about the system. A coin flip is governed by classical mechanics; given initial velocity, spin and height, the outcome is determined, and Diaconis, Holmes and Montgomery (2007) built a coin-tossing machine that produces the same face every time. The flip is random only because these quantities are not measured.

For a model the implication is that its "noise" term collects everything it does not represent: genuine indeterminacy, measurement error, and unmodelled signal. These cannot be told apart from the residuals alone; adding a relevant input variable can turn part of what looked like noise into explained variation.

Sample Space and Events

Definition (Sample space, event space). A set $\Omega = \{\omega_1, \ldots, \omega_n\}$ is the **sample space** of a random experiment if every outcome is associated with exactly one element $\omega \in \Omega$, an **outcome**.
For finite Ω, every subset $A \subseteq \Omega$ is an **event**; A **occurs** iff the observed $\omega \in A$. All events together, $\mathcal{P}(\Omega)$, form the **event space**.



Ω	sample space: all outcomes
ω = 6	an outcome: an element
A = {4, 5, 6}	an event: a subset

ω ∈ A: the event "at least 4" occurs

An outcome is an element; an event is a subset — a *question you can ask* about the result.

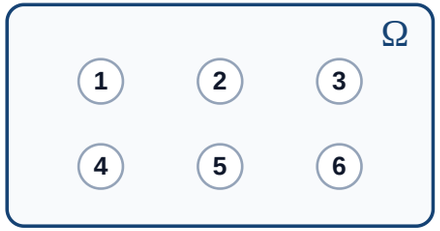
Speaker notes

"Exactly one" means the outcomes are mutually exclusive (no result maps to two elements) and exhaustive (every possible result maps to some element). Which results are distinguished is a modelling decision: for a die, $\{1, \ldots, 6\}$ and $\{\mathrm{even}, \mathrm{odd}\}$ are both valid sample spaces, but the second cannot express the event "a six".

An outcome ω is an element of Ω; an event is a subset. The elementary event $\{\omega\}$ is a set containing one outcome and is not the same object as ω. Probabilities are assigned to events, so one writes $P(\{\omega\})$, often abbreviated $P(\omega)$.

The figure is the die panel used for all event types that follow: the roll showed 6, and since $6 \in \{4, 5, 6\}$ the event "at least 4" occurred. For infinite Ω — the drug dosage below — the event space is a suitable family of subsets rather than the full power set; this returns with continuous distributions.

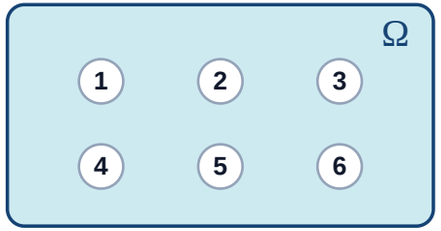
Important Event Types



$\varnothing$

impossible event

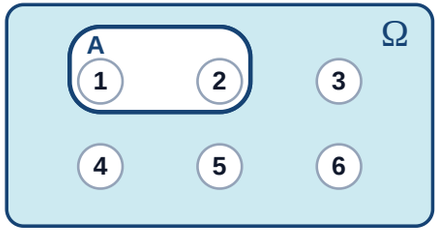
no outcome belongs to it



Ω

certain event

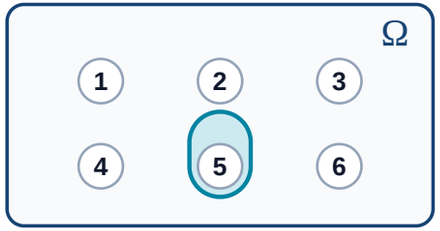
every outcome belongs to it



$\bar{A} = \Omega \setminus A$

complementary event

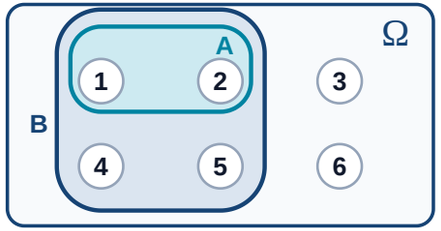
A = {1, 2}, $\bar{A}$ = {3, 4, 5, 6}



$|A| = 1$

elementary event

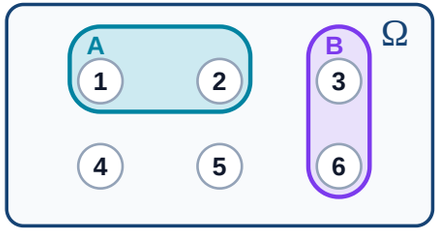
A = {5}: one outcome



$A \subseteq B$

A entails B

A = {1, 2}, B = {1, 2, 4, 5}



$A \cap B = \varnothing$

incompatible events

A = {1, 2}, B = {3, 6}

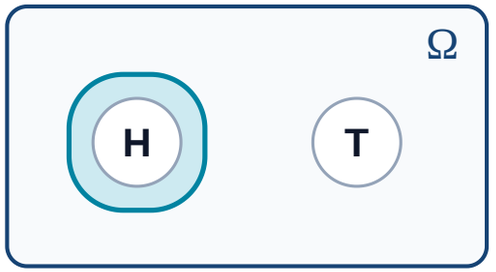
Speaker notes

All six panels use the die, $\Omega = \{1, \ldots, 6\}$. The impossible event $\emptyset$ contains no outcome and never occurs; the certain event Ω always occurs. The complement $\bar{A} = \Omega \setminus A$ occurs exactly when A does not. An elementary event contains one outcome. $A \subseteq B$ means A entails B: whenever A occurs, B occurs too. Incompatible (disjoint) events, $A \cap B = \emptyset$, cannot occur together.

Set operations correspond to logical connectives between the statements the events express: inclusion is implication, intersection is "and", union is "or", complement is "not". This correspondence is why identities between events can be read off Venn diagrams.

Examples

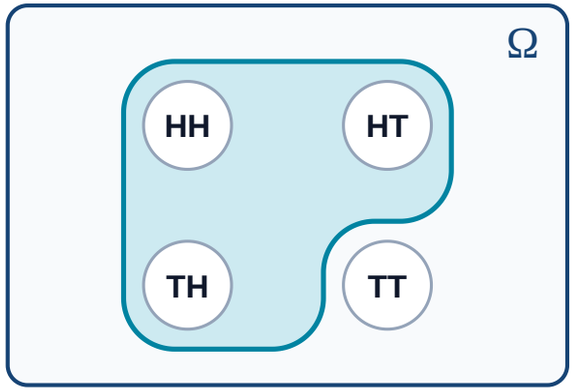
One coin flip



E = {H}: "heads"

events: ∅, {H}, {T}, {H, T}

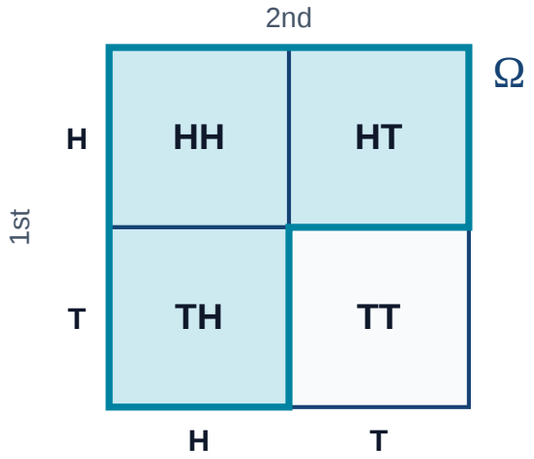
Two coin flips: the set



E = {HH, HT, TH}: "at least one head"

4 outcomes, $2^4 = 16$ events

… as a grid: 1st × 2nd



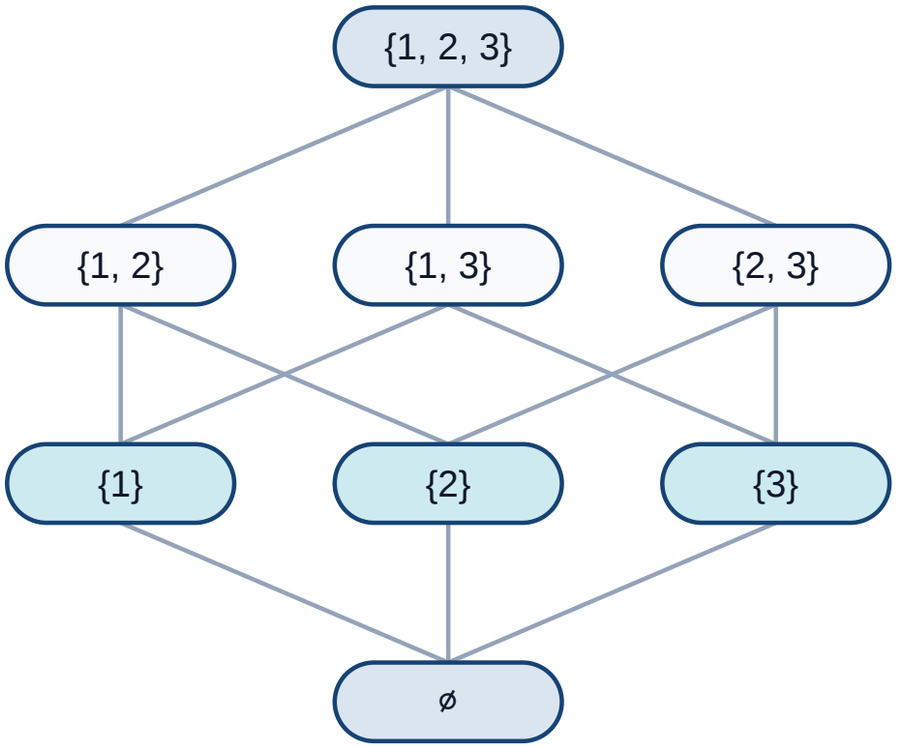
same outcomes, arranged by flip

The same event, twice: as a subset of Ω, and as cells of the 1st × 2nd grid. The sample space fixes which questions can be asked.

Speaker notes

For two flips, $\Omega = \{HH, HT, TH, TT\}$. "At least one head" is $\{HH, HT, TH\}$, "exactly one head" is $\{HT, TH\}$, and "the first flip is heads" is $\{HH, HT\}$: same sample space, different questions. If the sample space recorded only the number of heads, $\{0, 1, 2\}$, the event "the first flip is heads" could not be expressed at all, and the three outcomes would not be equally likely ($\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$).

The Event Space Grows Fast



$|\Omega| = n \rightarrow 2^n$ events

1	2
2	4
3	8
6	64
10	1,024
20	1,048,576

lines: one element added

For $|\Omega| = n$ the event space has 2^n elements — far too many to list, which is why probability is done by counting, not by enumeration.

Speaker notes

Each event is determined by an independent in/out choice for each of the n outcomes, so there are 2^n events. For three outcomes these are $\emptyset$, three singletons, three pairs and Ω, ordered by inclusion in the figure. Already a modest $n = 300$ gives more events than the estimated 10^{80} atoms in the observable universe.

A probability measure on a finite Ω is nevertheless fixed by only n numbers, the probabilities of the elementary events; additivity gives every other event. For infinite Ω, in particular $\Omega = \mathbb{R}$, the full power set contains sets to which no consistent "length" can be assigned (Vitali sets), which is why a smaller family of events, a σ-algebra, is used there.

Not Every Sample Space Is Finite

Dosage of a drug: Ω = (0, ∞)



uncountably many outcomes — probability can only live on intervals, not on points

Race with 7 teams: all orderings



7! = 5,040 outcomes; shaded: the event "team 3 wins"

Choosing Ω is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.

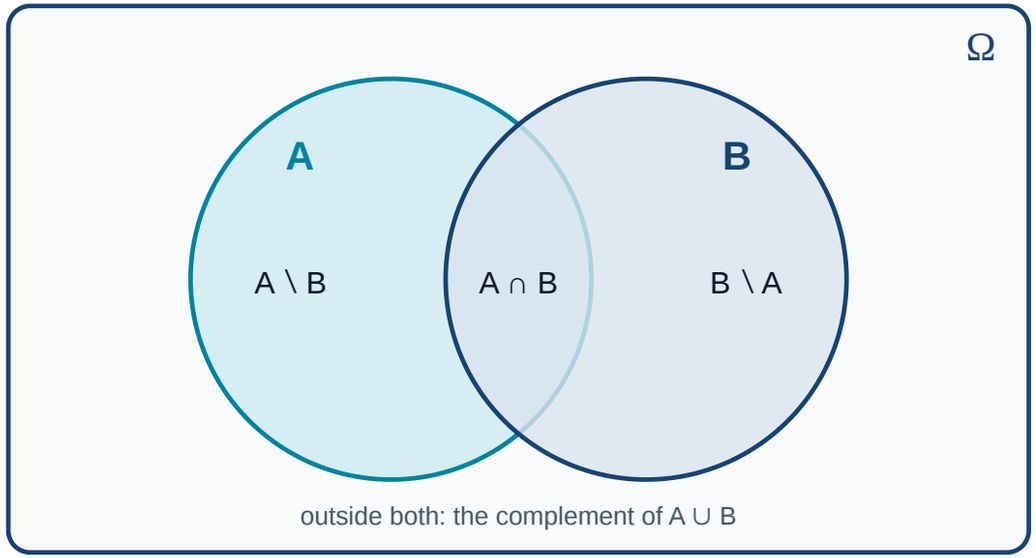
Speaker notes

A drug dosage modelled as a positive real number gives an uncountable sample space $\Omega = [0, \infty)$; events are intervals such as $[5, 10]$ mg, and any single exact value has probability zero under a continuous model. The model ignores that real dosages are measured with finite precision and administered in fixed steps; the idealisation is chosen because intervals and densities are easier to work with than a fine discrete grid.

The race example is finite but large: seven teams can finish in $7! = 5{,}040$ orders, so the sample space of full rankings has 5,040 outcomes and 2^{5040} events. The same modelling choice — what counts as an outcome — arises for every attribute of a dataset.

Venn Diagrams

Venn diagrams are the graphical representation of events:



- the **rectangle** is the sample space Ω
- a **circle** is an event $A \subseteq \Omega$
- every **region** is a set — and every set operation is a region you can point at

Speaker notes

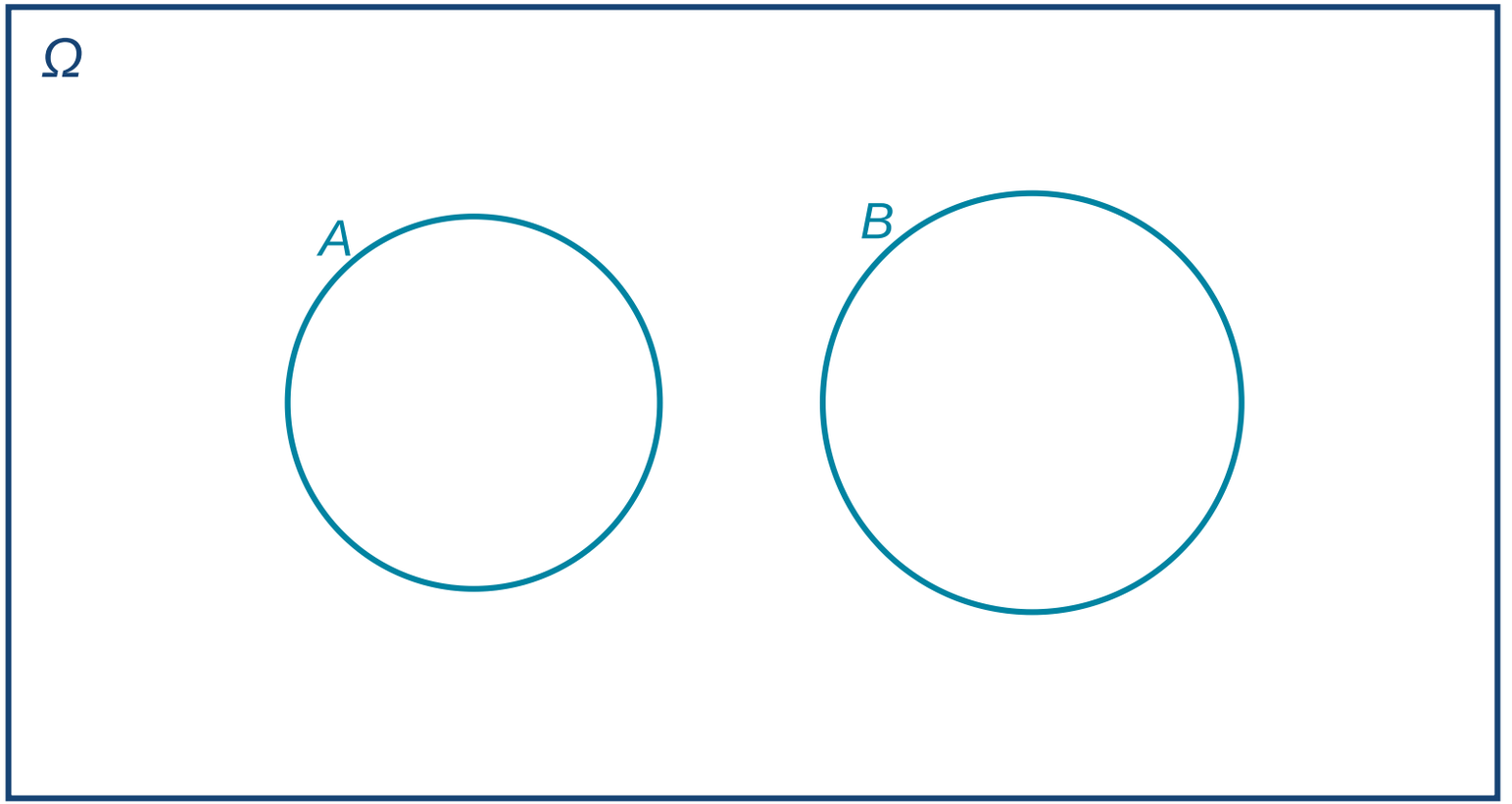
Two circles divide the rectangle into four regions: $A \setminus B$, $A \cap B$, $B \setminus A$ and $\overline{A \cup B}$. These four are pairwise disjoint and together make up Ω, so every event built from A and B by set operations is a union of some of them; there are $2^4 = 16$ such events. Three circles give eight regions.

The sizes of the circles carry no meaning in a plain Venn diagram. Once probabilities are attached and areas are drawn proportional to them, laws such as $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ become statements about areas.

Important Event Types

Before each step: which region will light up?

$$A \cap B = \emptyset$$



A and B are disjoint

Speaker notes

The four steps highlight: the event A; its complement $\bar{A} = \Omega \setminus A$, everything outside the circle; a sub-event $B \subseteq A$, drawn as a circle inside A; and two disjoint events, $A \cap B = \emptyset$, drawn as non-overlapping circles.

Disjointness is a property of the sets themselves, fixed before any probability is assigned. It is often confused with independence, which is a property of the probabilities; two disjoint events with positive probability are never independent, since $P(A \cap B) = 0 < P(A)P(B)$.

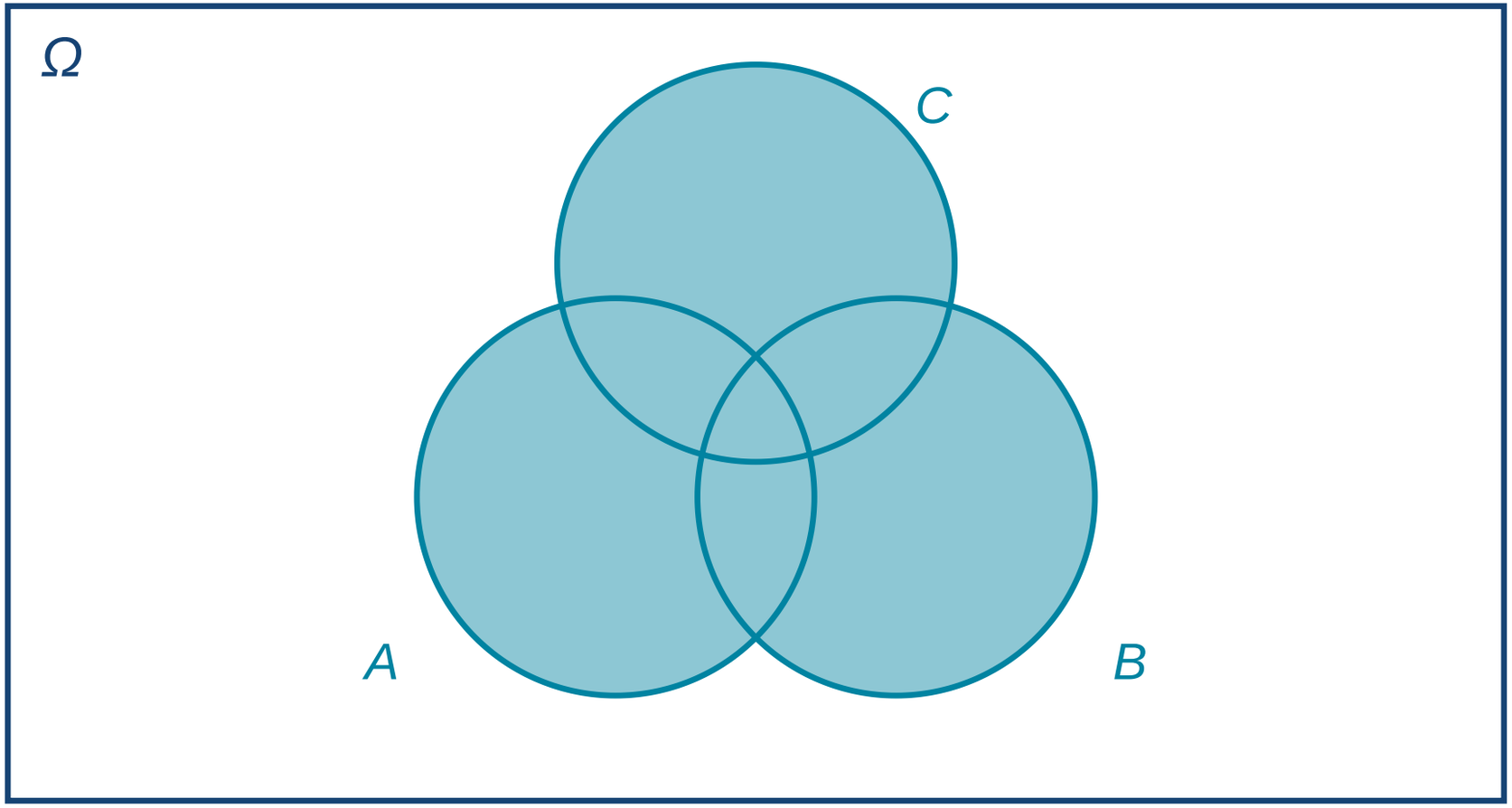
Algebra of Events

Law	Union form	Intersection form
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$	$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

Events form a set algebra — the same laws you know, but now every identity is a picture.

Set Operations as Regions

A ∪ B ∪ C



Union of three events

Speaker notes

The four steps highlight, with three sets: $A \cup B$ (everything in either circle), $A \cap B$ (the lens shared by A and B, including the part also in C), $A \cap B \cap C$ (the single central region), and $A \cup B \cup C$ (all seven regions inside the circles). Union enlarges and intersection shrinks: $A \cap B \cap C \subseteq A \cap B \subseteq A \cup B \subseteq A \cup B \cup C$.

De Morgan's laws complete the algebra: $\overline{A \cup B} = \bar{A} \cap \bar{B}$ and $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

Mini-Exercise

Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

Speaker notes

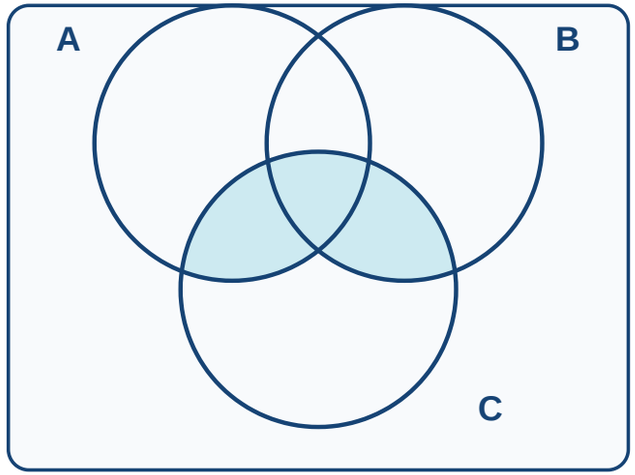
The outer operation decides. $(A \cup B) \cap C$ is an intersection with C, so it lies inside C. $(A \cap B) \cup C$ is a union with C, so it contains C. Expanded with the distributive laws: $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ and $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$.

Since one is a subset of C and the other a superset, they are equal exactly when both equal C, i.e. when $A \cap B \subseteq C \subseteq A \cup B$. A frequent error is to move the brackets freely; the associative law allows this only when both operations are the same.

Mini-Exercise

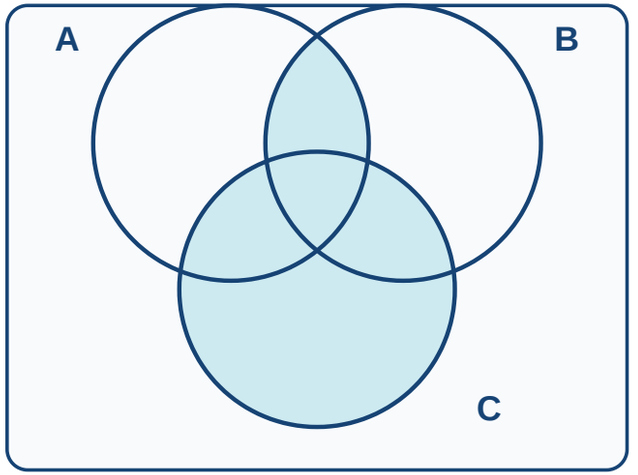
Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

(A ∪ B) ∩ C



inside C only

(A ∩ B) ∪ C



all of C, plus the lens A ∩ B

No. The first lies entirely inside C; the second is all of C plus the lens $A \cap B$. The distributive laws expand them into different expressions.

Takeaway

An event is a **subset** of the sample space, and combining statements about outcomes is set algebra: *or* is a union, *and* an intersection, *not* a complement. Choosing Ω decides which events exist at all.

Speaker notes

Module 1 closes here (course-local slide: the Venn unit ends on its exercise). The point to carry into Module 3: a probability will be a number attached to each of these subsets, so the algebra of events is what the rules of probability are written in.

Module 2

Counting

> ***Guiding Question:*** *How do we determine the size of a sample space we could never write down?*

Why Counting Comes First

If every outcome in a finite sample space is equally likely, then the probability of an event is a **ratio of counts**:

$$P(A) = \frac{|A|}{|\Omega|}$$

So before we can measure probability, we must be able to count $|A|$ and $|\Omega|$ — often without enumerating them.

Speaker notes

Motivate combinatorics as the arithmetic of sample spaces, not as a separate topic. Flag the assumption doing the work here — "equally likely" — which the axiomatic definition of probability replaces with an axiom system that does not need it.

Basic Principle of Counting

Generalised basic principle of counting. If an experiment consists of r stages, where stage i has n_i possible outcomes regardless of the earlier stages, then the whole experiment has

$$n_1 \cdot n_2 \cdots n_r$$

possible outcomes.

Example. How many 7-place licence plates, if the first 3 places are letters (26) and the last 4 are digits (10)?

$$26^3 \cdot 10^4 = 175{,}760{,}000$$

Speaker notes

The phrase "regardless of the earlier stages" is the hidden assumption — it fails as soon as choices interact (no repeated letters, say), which is exactly what the four cases below systematise.

Three Questions Decide the Formula

Order?

Does the sequence matter?
matters → **permutation** (rankings)
does not → **combination** (committees)

Repetition?

May an element be picked twice?
without — teams of people
with — letters in a word

How many?

Select k of n elements, or arrange all n?

Speaker notes

Give students a problem and have them answer these three questions *before* reaching for a formula. Nearly every counting error is a misread of one of these three, not an algebra slip.

The Four Basic Cases

Ordered, without repetition

k-permutation of n:

$$\frac{n!}{(n-k)!}$$

From $\{a, b\}$, $k = 2$: ab, ba

Ordered, with repetition

$$n^k$$

From $\{a, b\}$, $k = 2$: aa, ab, ba, bb

Unordered, without repetition

k-combination of n:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

From $\{a, b\}$, $k = 2$: $\{a, b\}$

Unordered, with repetition

$$\binom{n+k-1}{k}$$

From $\{a, b\}$, $k = 2$: $\{a, a\}, \{a, b\}, \{b, b\}$

Speaker notes

2x2 grid — the whole unit in one slide. Keep it visible while working examples; students should learn to locate a problem in this square rather than recall four formulas in isolation.

Permutations

Definition (k-permutation). An ordered selection of k elements from a set of n distinct elements without repetition. Counting position by position: n choices, then $n-1$, …, down to $n-k+1$, giving $\frac{n!}{(n-k)!}$.

Example. A class of 6 men and 4 women is ranked after an exam, no ties. How many possible rankings? $10! = 3{,}628{,}800$.

Speaker notes

The stage-by-stage argument *is* the basic counting principle applied with a shrinking pool — worth making explicit so the formula is derived, not given.

Combinations

Each k-combination corresponds to $k!$ different k-permutations — the same k elements in every possible order. Therefore

$$\binom{n}{k} = \frac{\text{number of } k\text{-permutations}}{k!} = \frac{n!}{k!(n-k)!}$$

Example. A committee of 3 from 20 people: $\binom{20}{3} = 1140$.

Dividing by $k!$ is the standard move for "forgetting" an ordering we counted.
Remember it as *over-count, then correct*.

Speaker notes

The over-count-then-divide pattern recurs constantly. Ask why we divide by k! and not by something else — because every unordered selection was counted exactly k! times, uniformly.

Properties of the Binomial Coefficient

$$\binom{n}{k} = \binom{n}{n-k}$$

Choosing which k to include is the same as choosing which $n-k$ to leave out.

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Split by a fixed element i: groups **containing** i, plus groups **not containing** i.

Binomial theorem.

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

Speaker notes

Both identities are proved by counting the same set two ways — the standard combinatorial proof technique. The binomial theorem's proof is the same idea: expanding the product gives 2^n terms, and x^k y^(n-k) appears exactly as often as there are ways to pick which k factors contribute x.

Application: How Many Subsets?

A set with n elements has

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

subsets — sum over all subset sizes.

> **Alternative intuition.** Each element is either in the subset or not: a binary choice, n times, so $2 \cdot 2 \cdots 2 = 2^n$.

This is the size of the event space $\mathcal{P}(\Omega)$ from the sample-space unit.

Speaker notes

Close the loop with the sample-space unit: the 2^n we asserted there is now derived twice. Ask which of the two arguments they would trust more under pressure — the binary-choice one generalises better.

Mini-Exercise

From 5 people, how many ways are there to pick a **chair and a deputy**? And how many ways to pick a **two-person committee**?

Speaker notes

One point: the same selection, counted with and without order, differs by the factor $k!$ — over-count, then correct. Ask which cell of the four-case square each answer sits in.

Mini-Exercise

From 5 people, how many ways are there to pick a **chair and a deputy**? And how many ways to pick a **two-person committee**?

Chair and deputy: order matters, $5 \cdot 4 = 20$. Committee: each pair was counted twice (as ab and ba), so $20/2! = 10 = \binom{5}{2}$.

Takeaway

> Counting reduces to three questions — **order? repetition? how many?** — and one habit: when an ordering was counted that should not have been, divide it out.

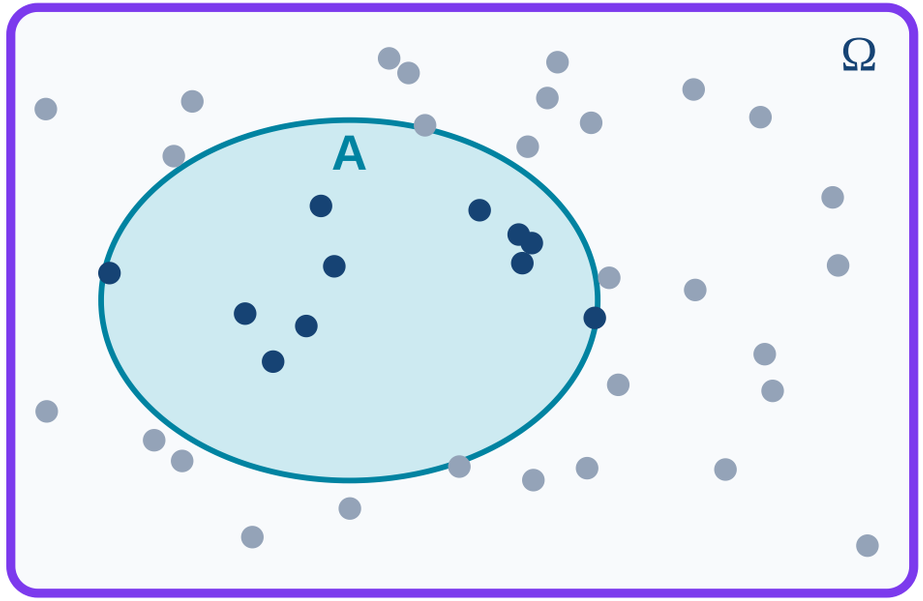
Module 3

What Is a Probability?

> ***Guiding Question:*** *Every attempt to define probability turns out to be circular. What do we do instead?*

An Intuition: Probability as Area

Throw 40 stones at random into Ω



P(A) = area of A / area of Ω ≈ 11 of 40 stones

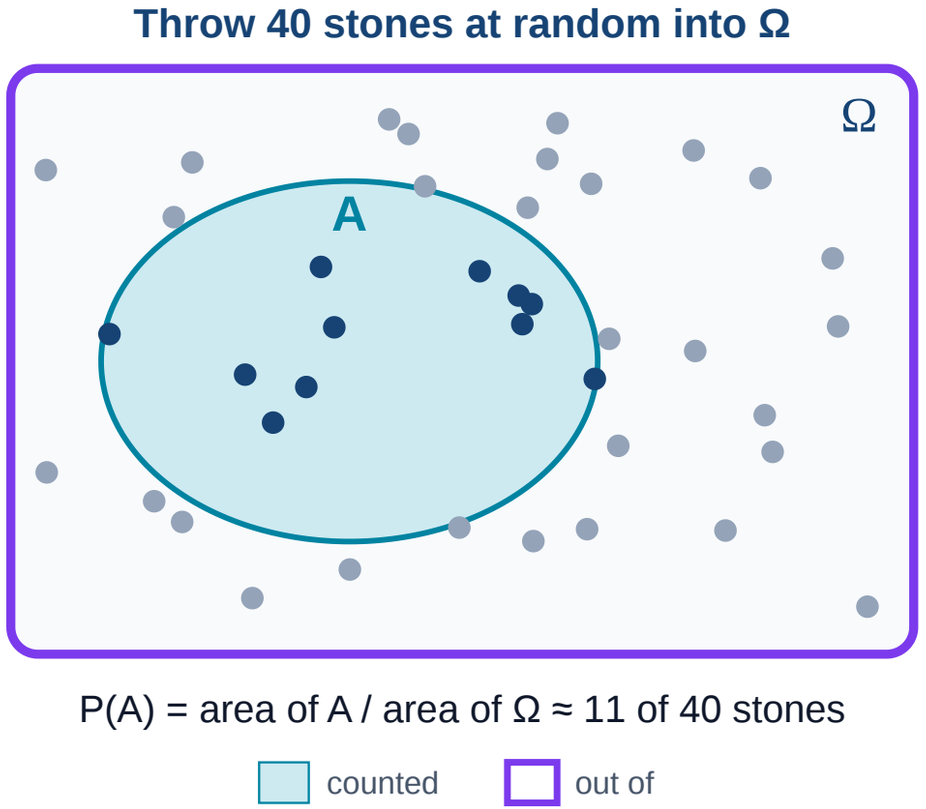
counted out of

Draw the sample space Ω on the ground, with each event as a region. Throw a stone at random.
How likely does it land in A?

Speaker notes

In the figure A covers about 27% of Ω, and 11 of 40 stones (27.5%) land in it; with more stones the share converges to the area ratio, which is the frequentist reading on the next slide. Under the two assumptions, $P(A)$ is the area of A divided by the area of Ω. Assumption 1 corresponds to normalisation, $P(\Omega) = 1$. Assumption 2 makes all elementary events equally likely, which is the Laplace setting. Dropping assumption 2 means that some parts of Ω carry more weight than their area: the stone is thrown with a non-uniform aim. Area is then replaced by weighted area, and the axioms keep exactly the properties that any such weighting shares with plain area — non-negativity, total one, and additivity over disjoint regions.

An Intuition: Probability as Area



Draw the sample space Ω on the ground, with each event as a region. Throw a stone at random.

How likely does it land in A?

$$P(A) = \frac{\text{area of } A}{\text{area of } \Omega}$$

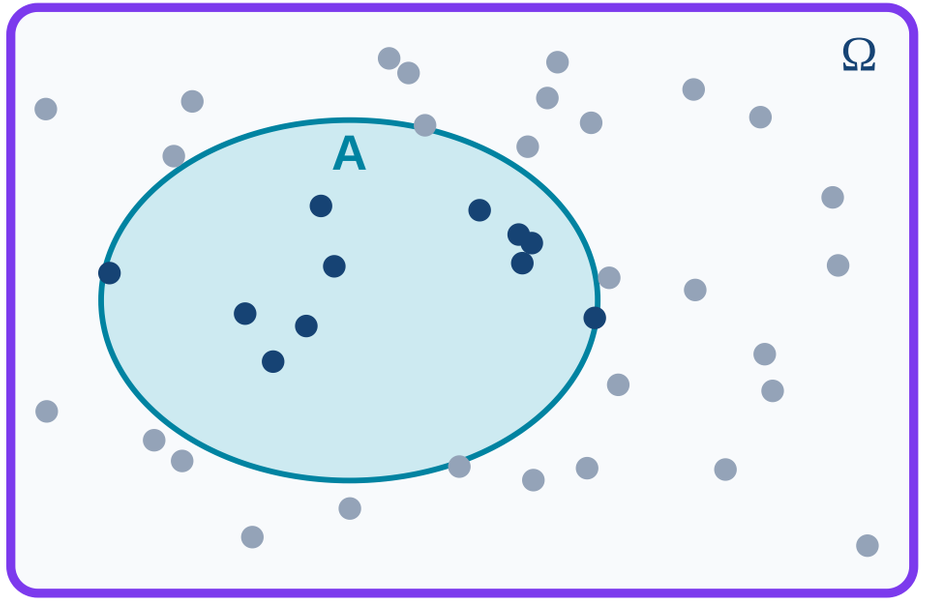
provided that

1. the stone always lands in Ω, and
2. every part of Ω is equally likely.

Probability measures **how much of** Ω an event occupies.

An Intuition: Probability as Area

Throw 40 stones at random into Ω



P(A) = area of A / area of Ω ≈ 11 of 40 stones

counted out of

Draw the sample space Ω on the ground, with each event as a region. Throw a stone at random.

How likely does it land in A?

$$P(A) = \frac{\text{area of } A}{\text{area of } \Omega}$$

provided that

1. the stone always lands in Ω, and
2. every part of Ω is equally likely.

Probability measures **how much of** Ω an event occupies.

> Assumption 2 is the one we will have to give up.

Two Ways to Read a Probability

Frequentist
The probability of an elementary event a is the fraction of repetitions of the experiment that yield a.
Requires the experiment to be repeatable.

Bayesian
The probability of a is the degree of certainty that a will happen — even without conducting the experiment.
Requires a stated state of knowledge.

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

Speaker notes

"The probability that this machine part fails within a year" has a natural frequentist reading: the fraction of parts of the same type that fail within a year. "The probability that this patient has the disease" refers to one person in one state; it has no repetition, so it is read as a degree of belief, or frequentistically only by placing the patient in a reference class (all patients with these symptoms), whose choice changes the number.

The frequentist reading assigns probabilities to repeatable events, not to fixed but unknown quantities such as a model parameter. The Bayesian reading assigns probabilities to both, which is what makes a prior distribution over a parameter meaningful. Both readings obey the same axioms, so every theorem holds under either.

The Classical (Laplace) Definition

Definition (Classical / Laplace probability). If every elementary event in Ω is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment**; its probabilities need not be estimated, they are implied by the setup.

Speaker notes

Counting (combinatorics) exists to evaluate $|A|$ and $|\Omega|$ when they are too large to list. The formula is valid only if the elementary events really are equally likely, and that depends on the choice of Ω. For the sum of two dice, $\Omega = \{2, \ldots, 12\}$ has 11 outcomes but they are not equiprobable; $P(\text{sum} = 7) = 6/36$, not $1/11$. The Laplace formula applies to the 36 ordered pairs, not to the 11 sums.

Where no symmetry of the setup justifies equal probabilities — a loaded die, a click-through rate — the probabilities have to be estimated from data or specified by a model.

Why That Definition Is Not Enough

> **The circularity.** "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

So what do we do? Stop trying to *define* what probability is, and instead specify what it must *do*.

Speaker notes

"Equally likely" already presupposes a notion of probability, so the Laplace formula cannot serve as its definition. The frequentist definition $P(A) = \lim_{n\to\infty} n_A/n$ has two problems: no experiment is repeated infinitely often, and the limit is not guaranteed to exist for an arbitrary sequence of results. The law of large numbers states that relative frequencies converge, but only with probability one — it presupposes probability rather than defining it.

The axiomatic approach, due to Kolmogorov (1933), avoids both problems: it specifies the rules every probability assignment must obey and leaves the origin of the numbers — symmetry, frequencies, or belief — to the application.

The Kolmogorov Axioms

Definition (Probability measure, Kolmogorov 1933). Let Ω be a sample space and $\mathcal{P}(\Omega)$ its event space. A function $P : \mathcal{P}(\Omega) \to \mathbb{R}$ is a **probability measure** if

I	$P(A) \geq 0$	non-negative
II	$P(\Omega) = 1$	normalised
III	$A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$	additive

INTUITION

Probability is **one kilogram of sand** spread over Ω; $P(A)$ is the sand lying in A.

- **I** — no region holds negative sand.
- **II** — all the sand lies somewhere in Ω.
- **III** — two regions that do not overlap hold their sand together.

The axioms say how *any* assignment must behave — not which probabilities the events get. How the sand is spread is the modeller's choice.

For infinite Ω the event space is a σ-algebra and Axiom III holds for countably many disjoint events — measure theory; nothing in this course depends on it.

Speaker notes

The sand picture is the area picture of the first slide with the uniformity assumption dropped: area is replaced by mass, and the mass may pile up anywhere. All three axioms hold for any such distribution of mass — it is never negative, the total is fixed at one, and the mass in two non-overlapping regions is the sum of their masses. What the axioms deliberately leave open is the spreading itself; symmetry (Laplace), frequencies and degrees of belief are three ways to choose it.

Upper bounds such as $P(A) \leq 1$ are not axioms; they follow from them (since $P(A) + P(\bar{A}) = 1$ and $P(\bar{A}) \geq 0$). On a finite Ω, a probability measure is completely specified by the values on elementary events, $p_\omega \geq 0$ with $\sum_\omega p_\omega = 1$; additivity then gives $P(A) = \sum_{\omega \in A} p_\omega$. Countable additivity — Axiom III for countably many disjoint events — is needed, for example, for the geometric distribution, whose infinitely many probabilities must sum to 1.

Probability Space

Definition. Let Ω be a sample space, $\mathcal{P}(\Omega)$ an event space, and P a probability measure. The triple $(\Omega, \mathcal{P}(\Omega), P)$ — also written (Ω, P) — is a **probability space**.

INTUITION

Three separate modelling choices: Ω — **what can happen** (the ground); $\mathcal{P}(\Omega)$ — **what may be asked** (the regions one may draw); P — **how likely each answer is** (how the sand is spread).

Same Ω, different P

A fair and a loaded die share $\Omega = \{1, \ldots, 6\}$ but are two probability spaces.

Finite Ω

P is fixed by the elementary events: $p_\omega \geq 0$, $\sum_\omega p_\omega = 1$, and $P(A) = \sum_{\omega \in A} p_\omega$.

Speaker notes

The three components state three separate modelling choices: Ω fixes what counts as an outcome, the event space fixes which questions may be asked, and P fixes how likely each answer is. A theorem that begins with "let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space" holds for every such choice.

In practice the underlying probability space is often left implicit and only the distributions of random variables are written down; it is still assumed to exist.

What the Axioms Already Give Us

Property	Statement	From	Intuitively
Complement	$P(\overline{A}) = 1 - P(A)$	II, III	*the sand not in A lies in $\overline{A}$*
Empty event	$P(\emptyset) = 0$	complement, $A = \Omega$	*no region, no sand*
Bounds	$0 \leq P(A) \leq 1$	I, complement	*no region holds more than all the sand*
Monotonicity	$A \subseteq B \Rightarrow P(A) \leq P(B)$	I, III	*a larger region holds at least as much*
Inclusion–exclusion	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	III	*the overlap was weighed twice*
Union bound	$P(A \cup B) \leq P(A) + P(B)$	I, inclusion–exclusion	*weighing twice never weighs less*
Finite additivity	$P(A_1 \cup \ldots \cup A_p) = \sum_i P(A_i)$ for disjoint A_i	III, repeated	*disjoint regions add up*

Speaker notes

Derivations. Complement: A and $\bar{A}$ are disjoint with union Ω, so $P(A) + P(\bar{A}) = P(\Omega) = 1$. Bounds: $P(A) \geq 0$ by Axiom I and $P(A) = 1 - P(\bar{A}) \leq 1$. Empty set: $\bar{\Omega} = \emptyset$, so $P(\emptyset) = 1 - P(\Omega) = 0$. Monotonicity: for $A \subseteq B$, $B = A \cup (B \setminus A)$ disjointly, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$. Inclusion–exclusion: write $A \cup B = A \cup (B \setminus A)$ and $B = (B \setminus A) \cup (A \cap B)$, both disjoint; then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. On the Venn picture, $P(A) + P(B)$ counts the overlap twice. The union bound (Boole's inequality) drops the non-negative term $P(A \cap B)$; it extends to any finite or countable union and is often the only usable estimate when the overlaps are unknown. Finite additivity follows from Axiom III by induction over p.

Monotonicity transfers logic to numbers: if A entails B, A cannot be more probable than B. The conjunction fallacy (Tversky and Kahneman's "Linda problem") is the empirical finding that people often judge $P(A \cap B) > P(A)$, which violates it.

Takeaway

Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

Speaker notes

An axiomatic definition fixes how probabilities must behave and leaves their source to the modeller: symmetry (Laplace), long-run frequency, or a stated degree of belief. Any assignment that is non-negative, normalised and additive is a probability measure, whichever source it comes from, and every theorem derived from the axioms — total probability, Bayes' theorem, the law of large numbers — applies to it. This is why frequentist and Bayesian statistics share one mathematics and differ only in which quantities they treat as random.

Laplace Probabilities in Practice

When all outcomes in Ω are equally likely, computing $P(A)$ means **counting twice**: the favourable cases, and the possible ones.

$$P(A) = \frac{|A|}{|\Omega|}$$

The whole difficulty is choosing a sample space in which "equally likely" is actually true.

Speaker notes

The formula is trivial; the modelling is not. Every error in the examples below comes from an Ω where outcomes are not equiprobable.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

Why ordered pairs? Because $(1, 6)$ and $(6, 1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Speaker notes

The classic trap. If a student proposes the 21 unordered pairs, ask them to bet on "double six" versus "one and six" — the intuition corrects itself fast.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

- $|\Omega| = 6 \cdot 6 = 36$ ordered pairs
- $E = \{(1,6), (6,1), (2,5), (5,2), (3,4), (4,3)\}$, so $|E| = 6$

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

Why ordered pairs? Because $(1,6)$ and $(6,1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Example: Ordered or Unordered — Both Work

An urn holds **6 white** and **5 black** balls. Three are drawn. **What is the probability of exactly 1 white and 2 black?**

Variant 1 — order matters

$|\Omega| = 11 \cdot 10 \cdot 9 = 990$

Favourable: white-black-black, black-white-black, black-black-white $= 6 \cdot 5 \cdot 4 + 5 \cdot 6 \cdot 4 + 5 \cdot 4 \cdot 6 = 360$

$$\frac{360}{990} = \frac{4}{11}$$

Variant 2 — order ignored

$|\Omega| = \binom{11}{3} = 165$

Favourable: $\underbrace{\binom{6}{1}}_{\text{the white}} \cdot \underbrace{\binom{5}{2}}_{\text{the blacks}} = 6 \cdot 10 = 60$

$$\frac{60}{165} = \frac{4}{11}$$

Speaker notes

Same answer, two sample spaces — the point of the slide. Both are legitimate *because* both are equiprobable: ordered draws are equally likely, and so are unordered groups. Mixing the two (favourable counted unordered, total counted ordered) is the error to warn against.

The General Rule

> When selecting k items from n, you may count **with** or **without** order — as long as you are consistent, and as long as the chosen sample space really is equiprobable.

Pick whichever counting is easier for the event at hand.

Speaker notes

Connects straight back to the four cases in the counting unit: the choice of case is a modelling choice about Ω, not a property of the problem.

Mini-Exercise

The urn again, smaller: **2 white and 2 black** balls, two drawn. What is the probability of **one of each**? Count without order, then check with order.

Speaker notes

One point: either counting works if it is used consistently. The common error to listen for is 4 favourable unordered pairs over 12 ordered draws (1/3). The earlier ten-couples problem ("five chosen from 10 couples, none a couple", $\binom{10}{5}2^5 / \binom{20}{5} \approx 0.52$) belongs on the exercise sheet.

Mini-Exercise

The urn again, smaller: **2 white and 2 black** balls, two drawn. What is the probability of **one of each**? Count without order, then check with order.

Unordered: $\binom{4}{2} = 6$ pairs, $2 \cdot 2 = 4$ mixed, so $4/6 = 2/3$. Ordered: $4 \cdot 3 = 12$ draws, $2 \cdot 2 + 2 \cdot 2 = 8$ mixed, so $8/12 = 2/3$ — the same, because both sample spaces are equiprobable.

Takeaway

Laplace's formula — favourable over possible cases — is **one** probability measure, valid only in a sample space whose outcomes are equally likely. The formula is trivial; choosing that sample space is the work.

Speaker notes

Module 3 closes here (course-local slide: the examples unit ends on its exercise). Connect back to the axioms: Laplace satisfies them, which is why it is a probability — not the other way round.

Module 4

Conditioning, Independence, Bayes

> ***Guiding Question:*** *A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and what is the correct bookkeeping?*

Motivating Example

You roll a 6-sided die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

Speaker notes

Without the information, "sum is 8" is $\{(2,6),(3,5),(4,4),(5,3),(6,2)\}$, so $P = 5/36 \approx 0.14$. Knowing the first roll is 3 leaves six equally likely outcomes, of which only $(3,5)$ sums to 8: $P = 1/6 \approx 0.17$. The information raised the probability slightly; other observations lower it — after a first roll of 1, a sum of 8 is impossible.

Conditioning is not a new kind of probability. It is the old probability restricted to a smaller sample space, with the outcomes that remain keeping their relative weights and being rescaled to sum to 1.

Motivating Example

You roll a 6-sided die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

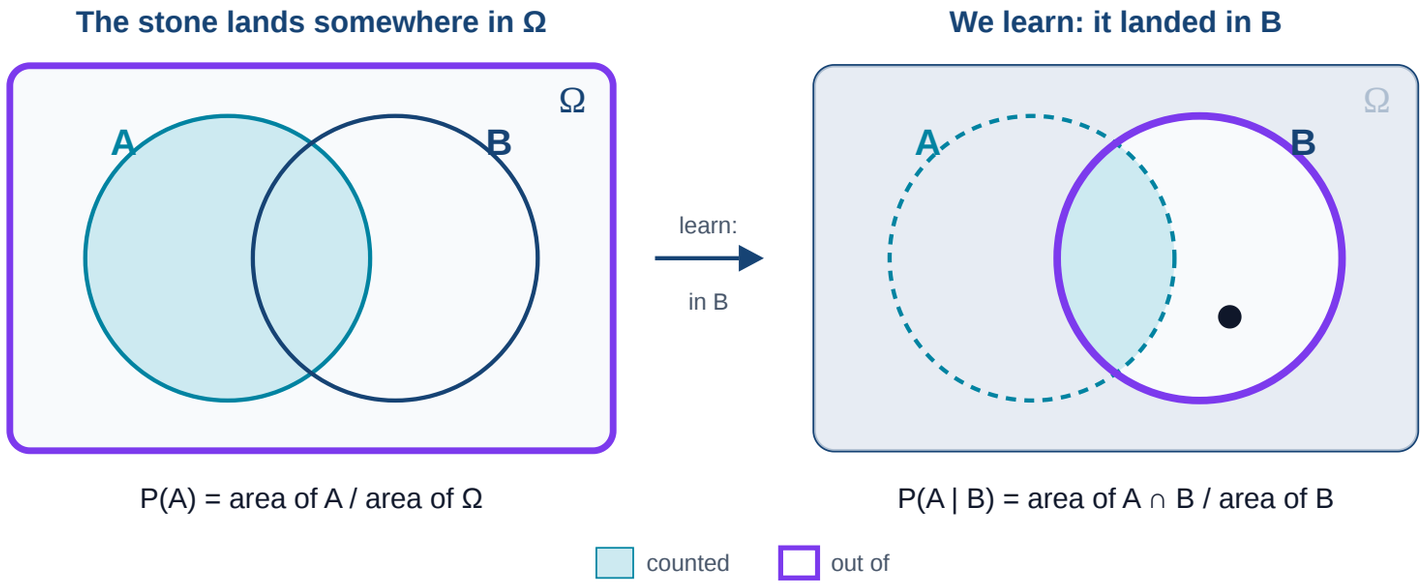
- Before the first roll: $\Omega = \{(1,1),(1,2),\ldots,(6,6)\}$, $|\Omega| = 36$
- After seeing the 3, only $\Omega_{red} = \{(3,1),\ldots,(3,6)\}$ remains
- Outcomes like $(2,4)$ have become **impossible**
- The reduced space now carries all the probability: $P(\Omega_{red}) = 1$
- "Sum is 8" inside it: $E = \{(3,5)\}$

$$P = \frac{1}{6}$$

Definition

Definition (Conditional probability of A under the condition B).

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad \text{if } P(B) > 0$$



Back to the stone: once it is known to lie in B, B **is the whole world**.

Speaker notes

The figure carries the colour code used from here on: teal is what is counted, the violet frame is what it is counted out of. Before the information, A is counted out of Ω; after it, only the part of A inside B counts, and it is measured against B.

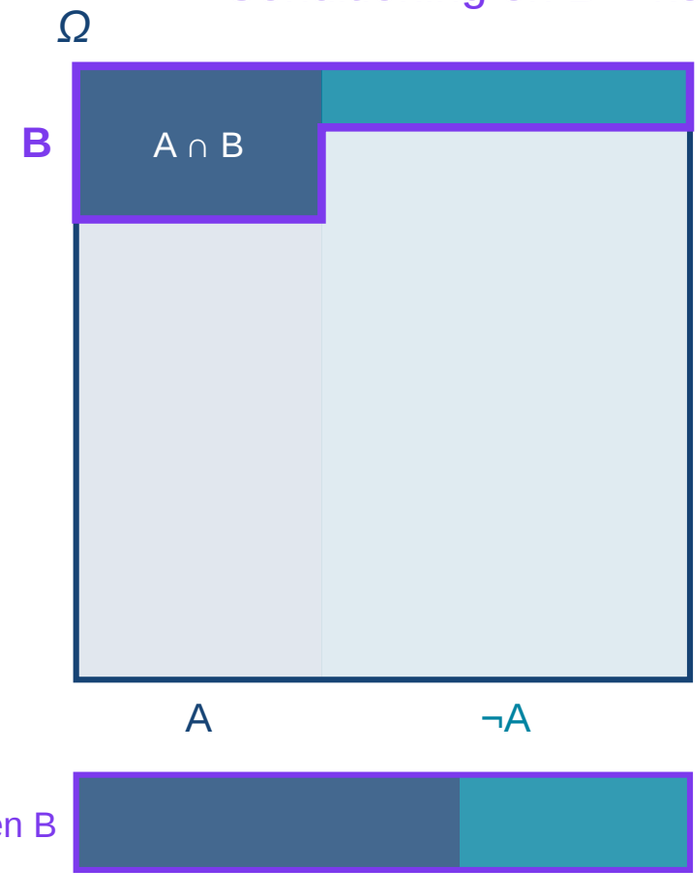
In the dice example $P(A \cap B) = 1/36$ and $P(B) = 6/36$, so $P(A \mid B) = 1/6$; dividing by $P(B)$ is the rescaling of the six remaining outcomes to total probability 1.

The condition $P(B) > 0$ is necessary because the ratio is otherwise undefined. Conditioning on events of probability zero, such as a continuous variable taking an exact value, requires densities instead and is treated with continuous distributions.

$P(A \mid B)$ and $P(A \cap B)$ are different quantities: $P(A \cap B) \leq P(A \mid B)$ always, with equality only when $P(B) = 1$ or $P(A \cap B) = 0$.

Conditioning, Drawn

Conditioning on B = keep B, stretch it back to width 1



P(A) = 0.40

P(B) = 0.10 + 0.06 = 0.16

P(A ∩ B) = 0.10

P(A | B) = 0.10 / 0.16 = 0.63

P(A | B) = 0.63

Before the last click: what share of B is A?

Then: set $P(A \cap B) = P(B)$. Now $P(A \mid B)$?

Speaker notes

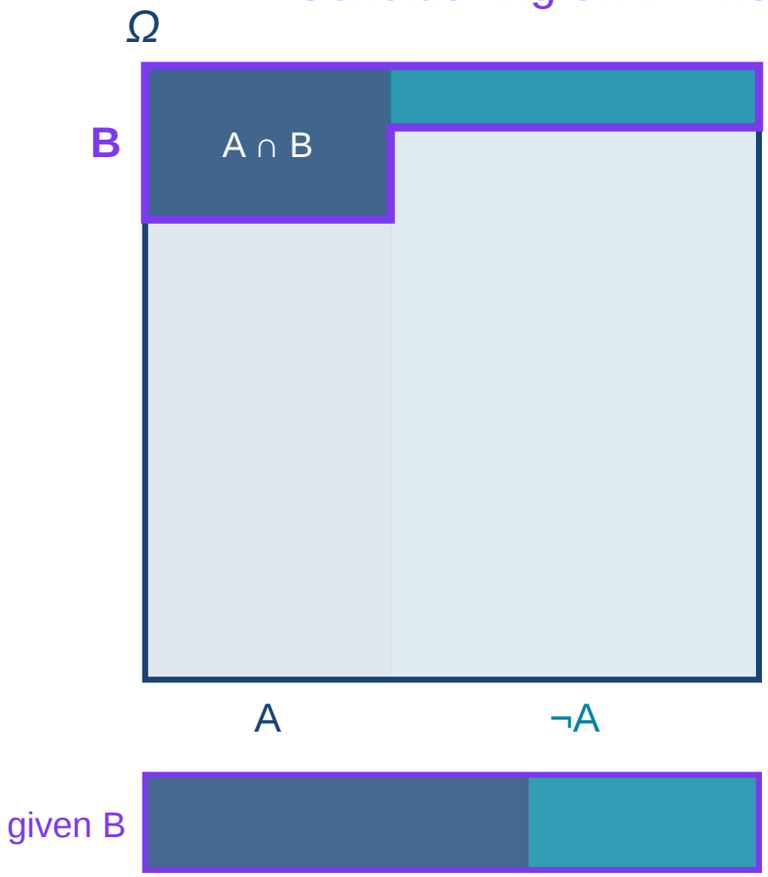
The four steps: the unit square split into a column A of width $P(A) = 0.4$; the event B, outlined in violet, made of two cells, $A \cap B$ of area 0.10 inside the column and $\bar{A} \cap B$ of area 0.06 outside; everything outside B removed; and the remainder stretched to total 1, giving $P(A \mid B) = 0.10/0.16 = 0.625$. The stretch factor is $1/P(B)$. These are the numbers of the students-and-exams example with A = took PML and B = failed.

The three sliders are areas, the quantities the definition is written in: $P(A)$, $P(B)$ and $P(A \cap B)$. They are free except that the overlap can be no larger than either event and no smaller than $P(A) + P(B) - 1$ (otherwise $A \cup B$ would exceed Ω); outside that range the overlap slider snaps back. Everything else is derived: $P(\bar{A} \cap B) = P(B) - P(A \cap B)$ and $P(A \mid B) = P(A \cap B)/P(B)$. The heights of the two cells are the conditionals $P(B \mid A)$ and $P(B \mid \bar{A})$; the picture contains them, but they are not needed to define $P(A \mid B)$.

Two settings worth showing. $P(A \cap B) = P(B)$ puts all of B inside A: $P(A \mid B) = 1$. $P(A \cap B) = P(A) \cdot P(B)$ (e.g. $0.4 \cdot 0.16 \approx 0.06$) gives both cells the same height: A takes the same share of B as of Ω, $P(A \mid B) = P(A)$, and the caption reports that learning B changes nothing — the case called independence.

Conditioning, Drawn

Conditioning on B = keep B, stretch it back to width 1



P(A) = 0.40

P(B) = 0.10 + 0.06 = 0.16

P(A ∩ B) = 0.10

P(A | B) = 0.10 / 0.16 = 0.63

P(A | B) = 0.63

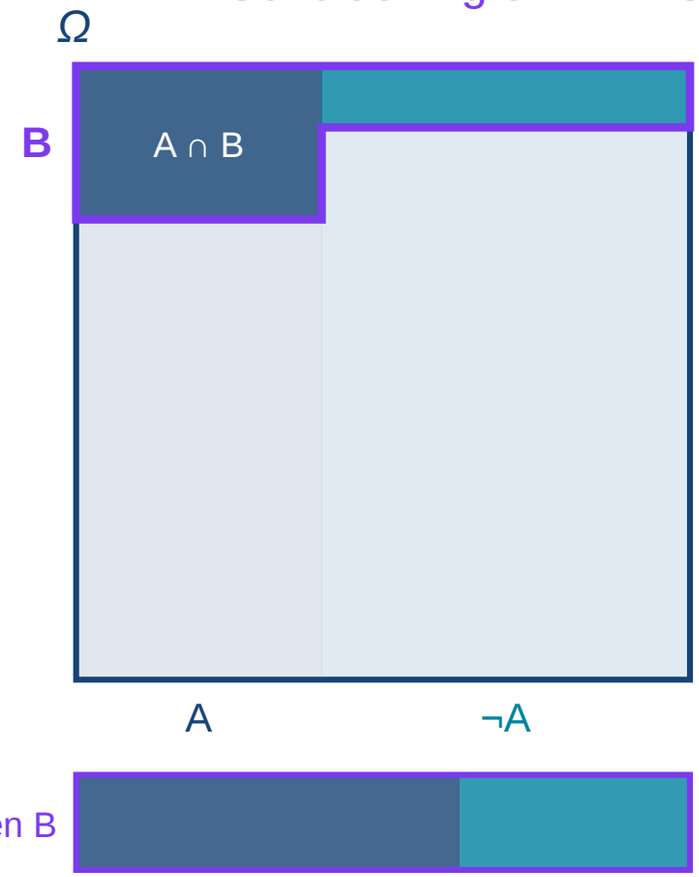
Before the last click: what share of B is A?

$P(A \cap B)/P(B) \approx 0.63$

Then: set $P(A \cap B) = P(B)$. Now $P(A \mid B)$?

Conditioning, Drawn

Conditioning on B = keep B, stretch it back to width 1



P(A) = 0.40

P(B) = 0.10 + 0.06 = 0.16

P(A ∩ B) = 0.10

P(A | B) = 0.10 / 0.16 = 0.63

P(A | B) = 0.63

Before the last click: what share of B is A?

$P(A \cap B)/P(B) \approx 0.63$

Then: set $P(A \cap B) = P(B)$. Now $P(A \mid B)$?

1: all of B lies inside A.

Example: Two Coins

Two fair coins are flipped. How likely is B = **both heads**, if we learn …

(a) "The first coin is heads"

$F = \{hh, ht\}$

(b) "At least one coin is heads"

$A = \{hh, ht, th\}$

Speaker notes

Let the room estimate both before the answers appear; (b) is most often answered with $\frac{1}{2}$.

Example: Two Coins

Two fair coins are flipped. How likely is B = **both heads**, if we learn …

(a) "The first coin is heads"

$F = \{hh, ht\}$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(b) "At least one coin is heads"

$A = \{hh, ht, th\}$

Example: Two Coins

Two fair coins are flipped. How likely is B = **both heads**, if we learn …

(a) "The first coin is heads"

$F = \{hh, ht\}$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(b) "At least one coin is heads"

$A = \{hh, ht, th\}$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Example: Two Coins

Two fair coins are flipped. How likely is B = **both heads**, if we learn …

(a) "The first coin is heads"

$F = \{hh, ht\}$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

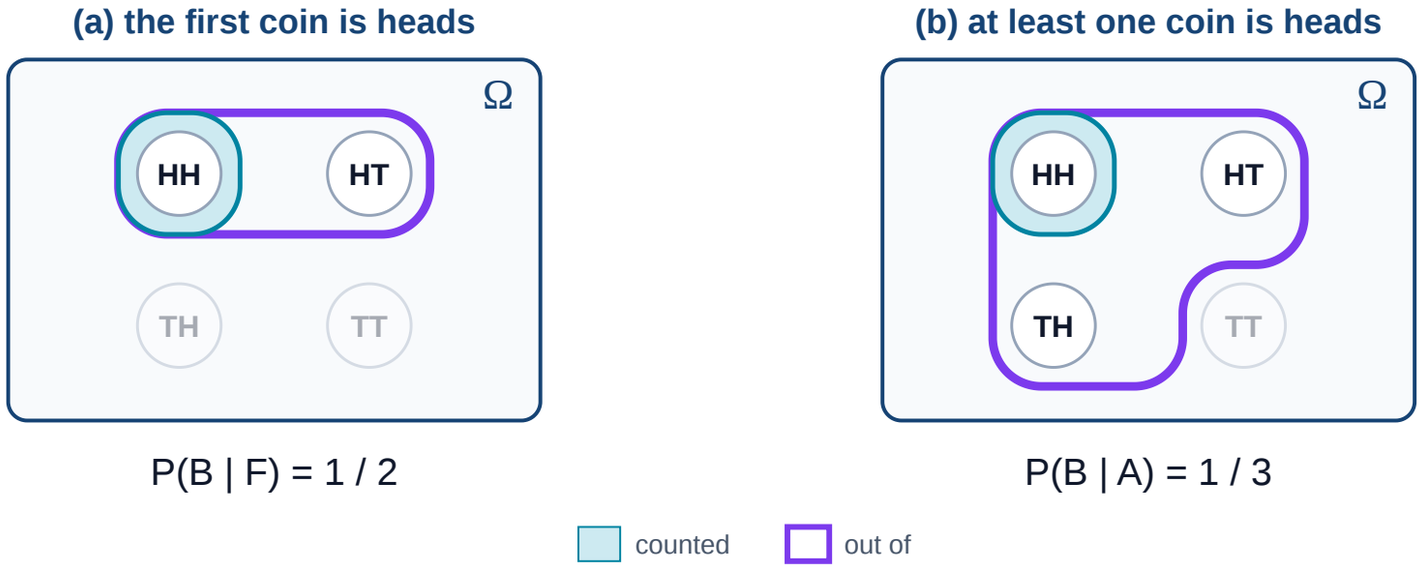
(b) "At least one coin is heads"

$A = \{hh, ht, th\}$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Same event, similar news. **Why is (b) smaller?**

Two Coins, Drawn



Both count the same outcome hh. News (a) leaves **two** possible outcomes, news (b) leaves **three**: weaker news rules out less, so hh gets a smaller share.

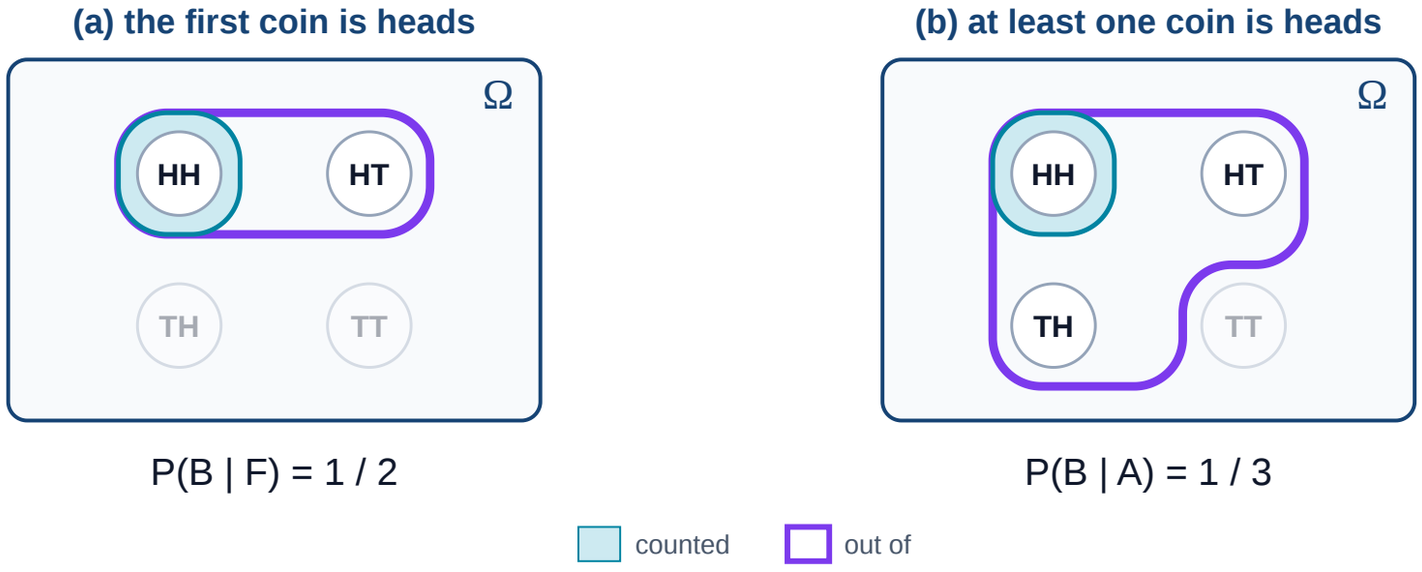
Speaker notes

(b) is smaller because the information in (a) is stronger: naming *which* coin is heads rules out two of the four outcomes, leaving $\{hh, ht\}$, whereas "at least one is heads" rules out only tt, leaving three equally likely outcomes of which one is hh.

A frequent error is to answer $\frac{1}{2}$ in (b) by reasoning "one coin is heads, the other is heads with probability $\frac{1}{2}$". That argument silently fixes which coin is known to be heads, which is the information of (a), not of (b). The figure makes the difference countable: the violet frame holds two outcomes in (a) and three in (b), and the counted outcome hh is the same.

Venn diagrams and outcome sets are drawable for two, at most three events; a model with ten binary variables has $2^{10} = 1{,}024$ joint outcomes. The definition of conditional probability applies unchanged to any number of events, which is why it, not the picture, is the tool. The same structure underlies the "boy or girl paradox" (Gardner, 1959): "a family with two children has at least one boy" gives probability $\frac{1}{3}$ for two boys, "the older child is a boy" gives $\frac{1}{2}$. The answer also depends on how the information was obtained; the $\frac{1}{3}$ assumes that "at least one" was the full content of what was learned.

Two Coins, Drawn



Both count the same outcome hh. News (a) leaves **two** possible outcomes, news (b) leaves **three**: weaker news rules out less, so hh gets a smaller share.

> Pictures stop at two or three events. **Learn the rule** — it does not run out.

Total Probability

Theorem (Total probability). Let $E_1, \ldots, E_k$ be mutually exclusive with $\Omega = E_1 \cup \ldots \cup E_k$ and $P(E_i) > 0$. Then for any event B:

$$P(B) = \sum_{i=1}^{k} P(E_i) \cdot P(B \mid E_i)$$

Proof. Decompose B along the partition and use additivity:

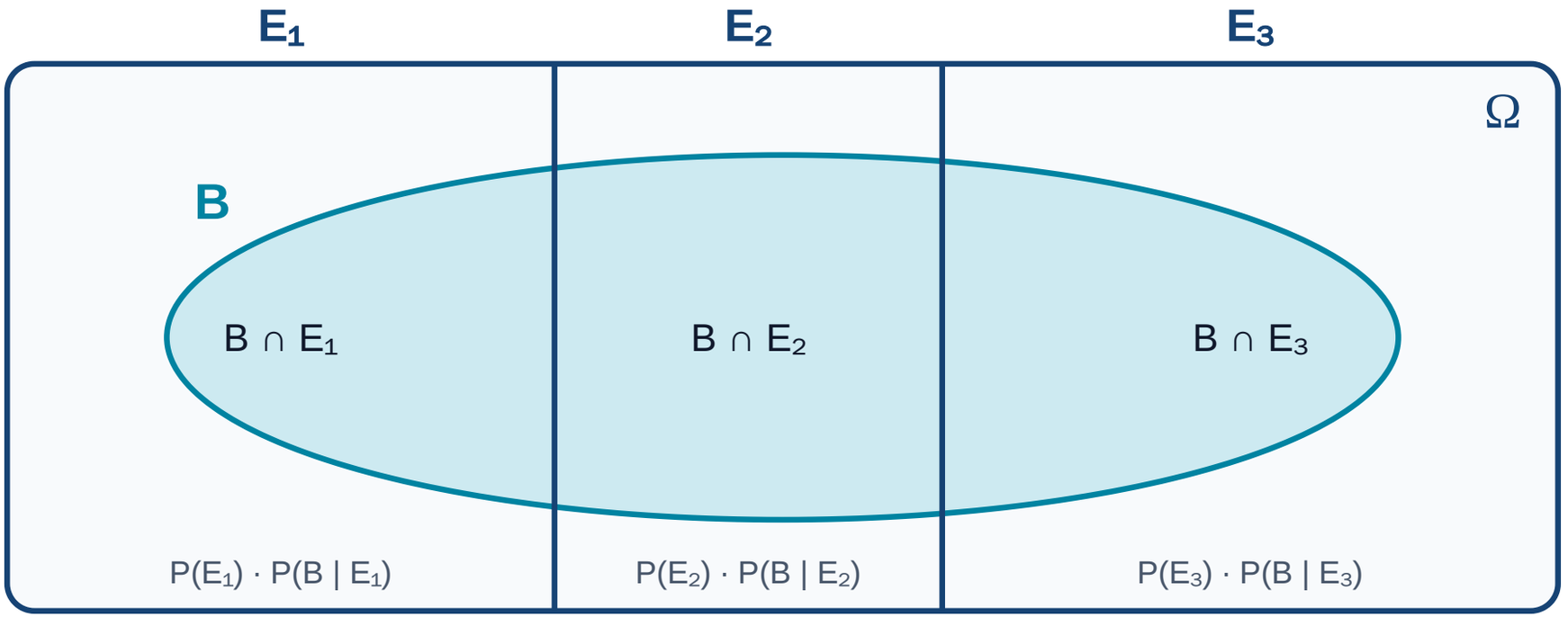
$$P(B) = P((E_1 \cup \ldots \cup E_k) \cap B) = \sum_{i=1}^{k} P(E_i \cap B) = \sum_{i=1}^{k} P(E_i) \cdot P(B \mid E_i)$$

Speaker notes

The proof uses three facts. First, $B = B \cap \Omega = \bigcup_i (E_i \cap B)$ because the E_i cover Ω. Second, the pieces $E_i \cap B$ are pairwise disjoint because the E_i are, so additivity (Axiom III) turns the union into a sum. Third, $P(E_i \cap B) = P(E_i)\,P(B \mid E_i)$ by the definition of conditional probability, which needs $P(E_i) > 0$.

The theorem computes a probability by cases: in the students-and-exams example, $P(\text{fail}) = 0.4 \cdot 0.25 + 0.6 \cdot 0.1 = 0.16$. It supplies the denominator $P(B)$ of Bayes' theorem whenever $P(B)$ is not known directly. The simplest partition is $\{E, \bar{E}\}$: $P(B) = P(E)P(B \mid E) + P(\bar{E})P(B \mid \bar{E})$.

Total Probability, Drawn



$$P(B) = P(B \cap E_1) + P(B \cap E_2) + P(B \cap E_3) = \Sigma_i\, P(E_i) \cdot P(B \mid E_i)$$

B is cut into disjoint pieces, one per E_i; each piece is "first E_i, then B within E_i" — the product rule.

Speaker notes

The strips E_1, E_2, E_3 are a partition: disjoint and together all of Ω. Each piece $B \cap E_i$ has probability $P(E_i)\,P(B \mid E_i)$ by the product rule, and the pieces add up to B by additivity. In the students-and-exams example the partition is the course $\{\mathrm{PML}, \mathrm{DB}\}$ and B = failed: $P(\mathrm{fail}) = 0.4 \cdot 0.25 + 0.6 \cdot 0.1 = 0.16$ — the sum rule of the count table, written with conditionals.

Rules That Follow

Product rule

$$P(A \cap B) = P(B) \cdot P(A \mid B)$$
$$= P(A) \cdot P(B \mid A)$$

Complement, same condition

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

But in general

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at B, $P_B(E) = P(E \mid B)$ for every event E satisfies the Kolmogorov axioms — conditioning yields a **new probability measure** P_B.

Speaker notes

The product rule is the definition of conditional probability multiplied out; it also holds when $P(B) = 0$ if both sides are read as 0. Its two forms, set equal and divided by $P(A) > 0$, turn $P(A \mid B)$ into $P(B \mid A)$: that is Bayes' theorem, which gets its own unit.

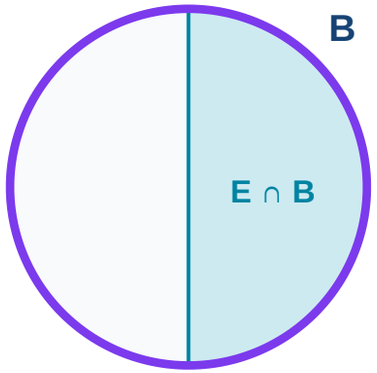
Complementing the event under a fixed condition is valid because P_B is a probability measure, and $P_B(\bar{A}) = 1 - P_B(A)$ holds for every measure. Complementing the condition is not valid: $P(A \mid B)$ and $P(A \mid \bar{B})$ belong to two different measures. In the dice example, $P(\text{sum } 8 \mid \text{first is } 3) = 1/6$, whereas $P(\text{sum } 8 \mid \text{first is not } 3) = 4/30 = 2/15$, not $5/6$.

That P_B satisfies the axioms is checked directly: $P(A \cap B)/P(B) \geq 0$; $P(\Omega \cap B)/P(B) = 1$; and disjoint E_1, E_2 give disjoint $E_1 \cap B, E_2 \cap B$. Every rule proved for P therefore holds for $P(\cdot \mid B)$.

The letter E is a placeholder for an arbitrary event; writing the map as $A \mapsto P(A \mid B)$ is equally correct, but A already names a specific event in the product rule on this slide.

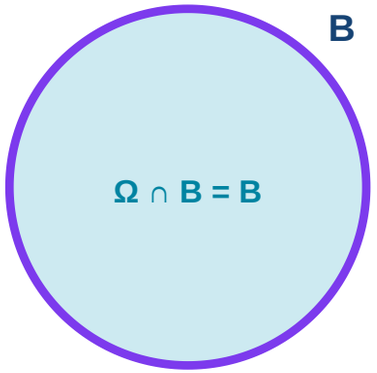
Conditioning Gives a New Measure

$P_B(E) = P(E \cap B) / P(B)$: every event E is measured by its piece inside B



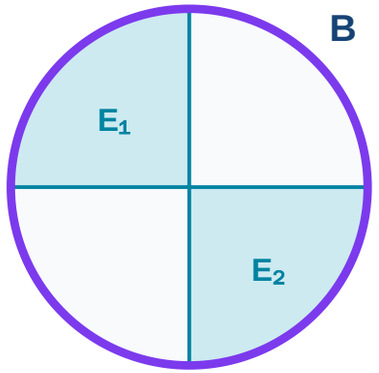
Axiom I: $P_B(E) \geq 0$

a piece of B is never negative



Axiom II: $P_B(\Omega) = 1$

all of B is the whole world



Axiom III: $P_B(E_1 \cup E_2) = P_B(E_1) + P_B(E_2)$

disjoint pieces of B add up

Inside B, the three axioms hold again — so **every rule proved for P holds for P_B**.

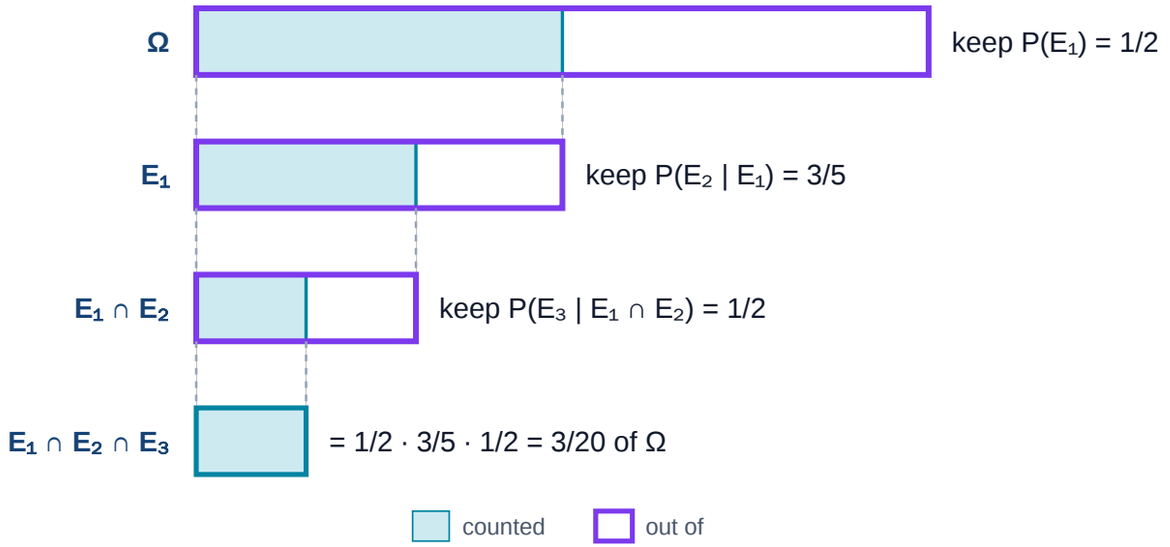
Speaker notes

Axiom I: $P_B(E) = P(E \cap B)/P(B)$ is a ratio of a non-negative number and a positive one. Axiom II: $P_B(\Omega) = P(\Omega \cap B)/P(B) = P(B)/P(B) = 1$; also $P_B(B) = 1$, so after conditioning B is certain. Axiom III: if $E_1 \cap E_2 = \emptyset$, the pieces $E_1 \cap B$ and $E_2 \cap B$ are disjoint too, and additivity of P carries over after dividing by $P(B)$. Events disjoint from B get $P_B(E) = 0$.

This licenses the complement rule under a fixed condition, $P(\bar{A} \mid B) = 1 - P(A \mid B)$, and every other consequence of the axioms. It does not license changing the condition: P_B and $P_{\bar{B}}$ are two different measures.

Chain Rule

$$P(E_1 \cap E_2 \cap \ldots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \ldots \cap E_{n-1})$$



Each factor keeps a share of what is left. Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.

Speaker notes

Proof for three events: $P(E_1 \cap E_2 \cap E_3) = P(E_1 \cap E_2)\, P(E_3 \mid E_1 \cap E_2) = P(E_1)\, P(E_2 \mid E_1)\, P(E_3 \mid E_1 \cap E_2)$; induction gives the general case. It requires $P(E_1 \cap \ldots \cap E_{n-1}) > 0$.

Any order of the events gives a valid factorisation. The chain rule holds without assumptions; simplification comes only from conditional independence, when a factor such as $P(E_3 \mid E_1 \cap E_2)$ equals $P(E_3 \mid E_2)$. For n binary variables the full joint needs $2^n - 1$ numbers; if each factor depends on at most k earlier variables, at most $n \cdot 2^k$ suffice. Bayesian networks are built on exactly this combination of an ordering and dropped conditions.

The figure uses $P(E_1) = \frac{1}{2}$, $P(E_2 \mid E_1) = \frac{3}{5}$ and $P(E_3 \mid E_1 \cap E_2) = \frac{1}{2}$: each bar is the shaded part of the bar above it, so the last one is $\frac{1}{2} \cdot \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{20}$ of Ω. Every conditional is measured against what is left, never against Ω.

Example: drawing two aces in a row from a 52-card deck without replacement has probability $\frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$.

Takeaway

> Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

Independence

Definition (Statistical independence). Two events A and B are **independent** iff

$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally $0 < P(B) < 1$, this is equivalent to

$$P(A \mid B) = P(A) \qquad \text{and} \qquad P(A \mid B) = P(A \mid \overline{B})$$

Independence means: **learning B changes nothing about A.**

Speaker notes

The conditional form carries the meaning: learning that B occurred leaves the probability of A unchanged. The product form is the standard definition because it is symmetric in A and B and remains defined when $P(B) = 0$; an event of probability 0 or 1 is independent of every event.

Equivalence for $0 < P(B) < 1$: dividing the product form by $P(B)$ gives $P(A \mid B) = P(A)$. Total probability gives $P(A) = P(B)P(A \mid B) + P(\bar{B})P(A \mid \bar{B})$, so $P(A)$ is a weighted average of the two conditionals, and it equals $P(A \mid B)$ exactly when $P(A \mid B) = P(A \mid \bar{B})$. If A and B are independent, so are A and $\bar{B}$, $\bar{A}$ and B, and $\bar{A}$ and $\bar{B}$.

Example: for two fair dice, "the first shows 6" and "the second shows 6" are independent, $\frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6}$.

The Classic Confusion

> **Statistical independence does not mean the events are disjoint.**

Disjoint events with positive probability are about as **dependent** as events can be: if A occurs, B certainly did not.

$$A \cap B = \emptyset \Rightarrow P(A \mid B) = 0 \neq P(A)$$

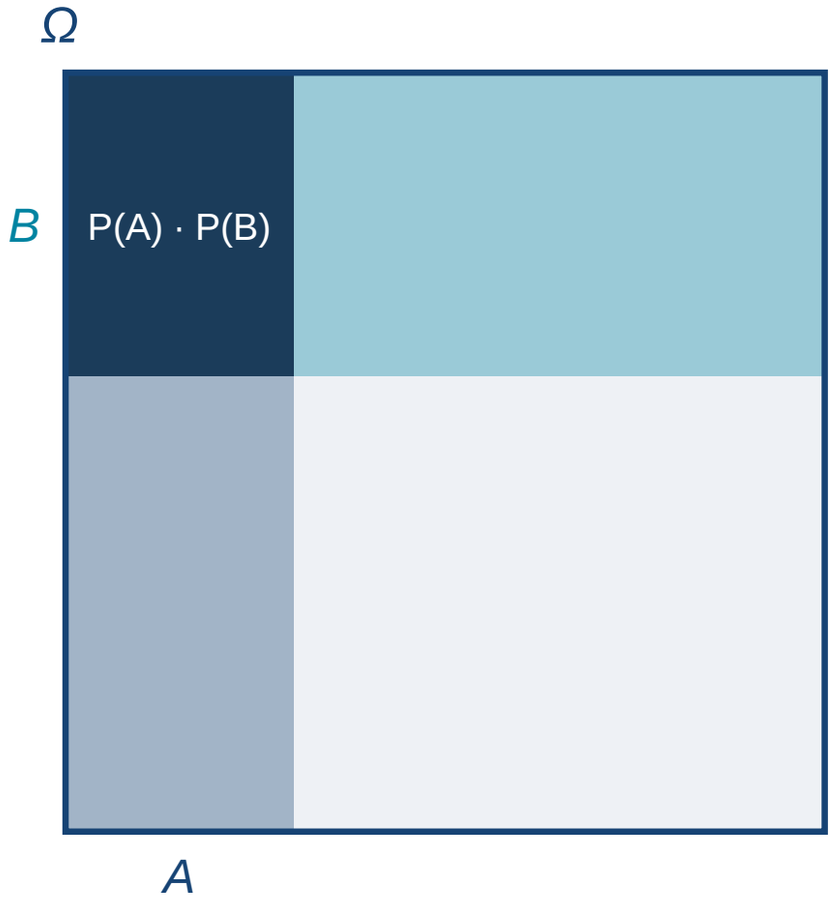
Speaker notes

Disjoint events with positive probability exclude each other: the occurrence of one rules out the other, which is maximal information about it. Formally, $P(A \cap B) = 0$ while $P(A)P(B) > 0$, so the product form fails. Learning B moves $P(A)$ to 0, the largest possible change downwards.

The Venn diagram misleads here because non-overlapping circles look "unrelated". Disjointness is a property of the sets; independence is a property of the probability measure and can change when the measure changes while the sets stay the same. The only case in which disjoint events are independent is when at least one of them has probability 0.

Independence Is About Areas, Not Overlap

P(A) · P(B) = 0.30 · 0.40 = 0.12 = P(A ∩ B) → independent



P(A) = 0.30

P(B) = 0.40

P(A ∩ B) = 0.12

P(A | B) = 0.12 / 0.40 = 0.30 = P(A)

Independent: **knowing B tells nothing about A.** A takes the same share of B as of all of Ω, so its probability stays the same.

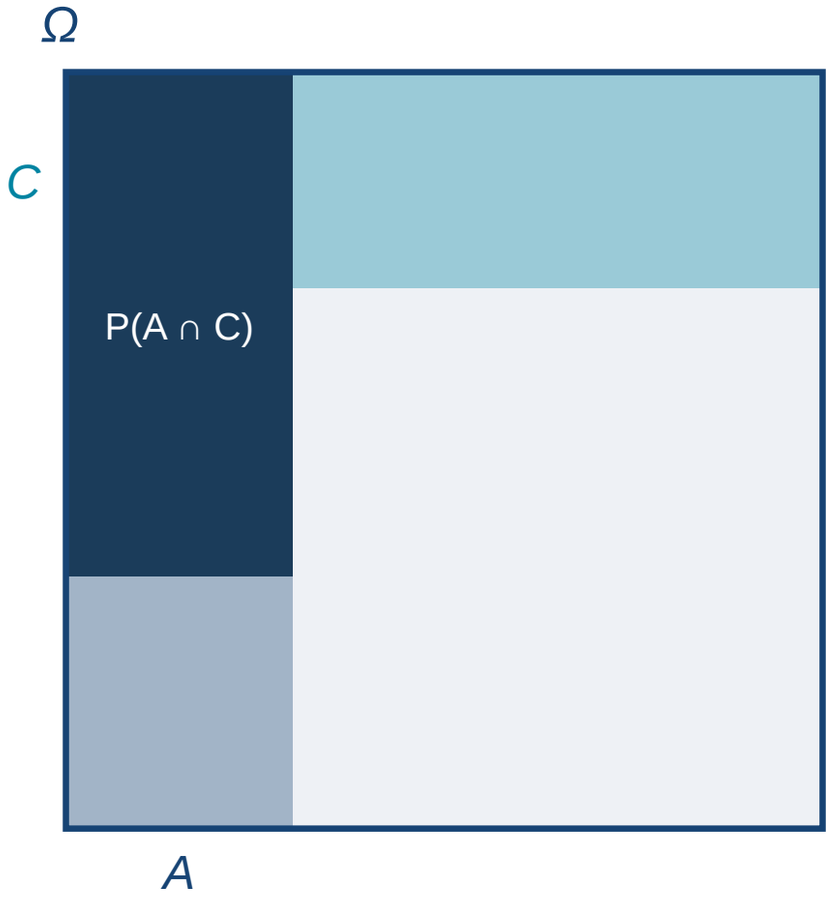
Speaker notes

The steps show the band A with area $P(A) = 0.3$, the band B with area $P(B) = 0.4$, their intersection with area $P(A \cap B) = 0.12$, and the comparison with $P(A)P(B) = 0.3 \cdot 0.4 = 0.12$: the events are independent. Equivalently $P(A \mid B) = 0.12/0.4 = 0.3 = P(A)$: A occupies the same fraction of B as of the whole of Ω .

Independence is an equality between numbers, not a property of the drawing: the shapes and positions of the regions are irrelevant, only their areas count. Overlapping regions can be independent, and so can regions that look strongly entangled.

The Same Picture, Dependent

P(A) · P(C) = 0.30 · 0.40 = 0.12 ≠ 0.20 = P(A ∩ C) → dependent



P(A) = 0.30

P(C) = 0.40

P(A ∩ C) = 0.20

P(A | C) = 0.20 / 0.40 = 0.50 ≠ P(A)

Here $P(A) \cdot P(C) = 0.12$ but $P(A \cap C) = 0.20$: the intersection is **larger** than independence would predict — A and C occur together more often than chance.

Speaker notes

Here $P(A \mid C) = 0.20/0.4 = 0.5 > P(A) = 0.3$: learning C raises the probability of A, so A and C are positively dependent. A smaller intersection, below 0.12, would give negative dependence.

The marginals $P(A) = 0.3$ and $P(C) = 0.4$ are the same as on the previous slide; only the joint probability $P(A \cap C)$ differs. Marginals therefore never determine whether events are independent. Given these marginals, $P(A \cap C)$ can take any value between $\max(0, P(A) + P(C) - 1) = 0$ and $\min(P(A), P(C)) = 0.3$ (the Fréchet bounds); independence is the single value 0.12 in that range. Answering questions about dependence requires the joint distribution.

More Than Two Events

Events $A_1, \ldots, A_n$ are (mutually) independent iff for **every** subset of indices $i_1, \ldots, i_l$:

$$P(A_{i_1} \cap \ldots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

Speaker notes

Standard counterexample: flip two fair coins; let A = "first is heads", B = "second is heads", C = "the coins agree". Each has probability $\frac{1}{2}$, and each pairwise intersection is $\{HH\}$ with probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$, so the three are pairwise independent. But $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$: any two of them determine the third.

For n events the definition imposes $2^n - n - 1$ equations, one for every subset of at least two events. The converse failure also exists: the triple product can factorise while a pair does not, so checking only the full intersection is not enough either. Models that assume features to be independent given a class (naive Bayes) assume mutual, not merely pairwise, conditional independence.

Takeaway

Independence is a statement about **numbers**, not pictures: $P(A \cap B)$ equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

Conditional Independence

Definition. Events A and B are **conditionally independent given** C (written $A \perp\!\!\!\perp B \mid C$) iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

equivalently

$$P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B carries no further information about A.**

Speaker notes

Math first (P5), but immediately give the reading. The second form is the one used in practice: it says B becomes redundant in the presence of C.

Dependent, but Independent Given C

Let $P(A) = \frac{8}{16}$, $P(B) = \frac{3}{16}$, $P(C) = \frac{4}{16}$ on a 16-cell sample space.

Unconditionally dependent

$$P(A \cap B) = \frac{1}{16}$$

$$P(A) \cdot P(B) = \frac{3}{32}$$

$\frac{1}{16} \neq \frac{3}{32}$, so $A \not\perp\!\!\!\perp B$

Conditionally independent given C

$$P(A \cap B \mid C) = \frac{1}{4}$$

$$P(A \mid C) \cdot P(B \mid C) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$$

so $A \perp\!\!\!\perp B \mid C$

Also directly: $P(A \mid B \cap C) = \frac{1}{2} = P(A \mid C)$.

Speaker notes

The whole idea in one comparison. Conditioning on C *created* independence that did not exist before. Ask for a real example: ice-cream sales and drownings are dependent, but conditionally independent given the temperature.

Why This Matters

Common cause
A common cause C can make two effects look related; given C, they are independent.

Common effect
Two independent causes become dependent once their common effect is observed — *explaining away*.

Model size
Conditional independence lets a joint distribution factorise into small pieces — naive Bayes and graphical models are built on it.

Conditional independence neither implies nor is implied by ordinary independence. Both directions can fail.

Speaker notes

The first two cards are the two directions of the takeaway: conditioning on a common cause *removes* a dependence (ice cream and drownings, given the temperature); conditioning on a common effect *creates* one (a wet lawn: rain and the sprinkler, once you know the grass is wet, compete to explain it). Name the models that rest on the definition without developing them. The warning box is worth a moment — students routinely assume one form gives the other.

Takeaway

> Conditioning can **create** independence as easily as it destroys it. "Given what I already know, does this extra fact still tell me anything?" is the question that makes large probabilistic models tractable.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

Speaker notes

The test has sensitivity $P(+ \mid \text{sick}) = 0.99$ and specificity $P(- \mid \text{healthy}) = 0.99$; the prevalence is 0.001. The exact answer is $P(\text{sick} \mid +) = \dfrac{0.99 \cdot 0.001}{0.99 \cdot 0.001 + 0.01 \cdot 0.999} = \dfrac{0.00099}{0.01098} \approx 0.090$.

The common intuitive answer is 99%, the test's accuracy. Physicians make the same error on comparable screening problems (Eddy, 1982, for mammography). The error is called base-rate neglect: the prevalence, 1 in 1,000, is ignored. Errors of this size occur because false positives from the large healthy group outnumber true positives from the small sick group about ten to one.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

> Only about **9%** — the answer most people miss by a factor of ten.

Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

P(sick | positive) = 1/11 ≈ 9%



Speaker notes

The steps show: a population of 1,000; the one person among them who is sick (prevalence 0.001); the healthy people the test flags, $1\% \cdot 999 \approx 10$; and only the positives kept, about 11 people of whom one is sick. Hence $P(\text{sick} \mid +) \approx 1/11 \approx 9\%$; exactly, $0.99/(0.99 + 9.99) \approx 0.090$.

Counting people in a fixed population ("natural frequencies") rather than multiplying percentages is known to reduce base-rate neglect substantially (Gigerenzer and Hoffrage, 1995). The computation is the same: restricting the population to the positives is conditioning on the event "positive".

What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space B; the single sick person among them is $A \cap B$. The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

The trap

"99% accurate" describes $P(\text{positive} \mid \text{sick})$.

You wanted $P(\text{sick} \mid \text{positive})$.

These are different numbers.

Speaker notes

Confusing $P(B \mid A)$ with $P(A \mid B)$ is known as the confusion of the inverse; in court it is called the prosecutor's fallacy, where the probability of the evidence given innocence is presented as the probability of innocence given the evidence. When $P(A \cap B) > 0$, the two conditionals agree only if $P(A) = P(B)$, since $P(A \mid B)\,P(B) = P(B \mid A)\,P(A)$.

Bayes' theorem converts one into the other, and the conversion requires the prior $P(A)$. A test's sensitivity and specificity are properties of the test; $P(\text{sick} \mid +)$, the positive predictive value, also depends on the population tested.

Bayes' Theorem — Two Events

Theorem (Bayes, 2 events). Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

posterior — what we believe after

likelihood — how well A explains B

evidence — how ordinary B is

prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

Speaker notes

The four steps highlight in turn the posterior $P(A \mid B)$, the likelihood $P(B \mid A)$, the prior $P(A)$ and the evidence $P(B)$. For the disease test, A = sick and B = positive: "99% accurate" is the likelihood $P(B \mid A) = 0.99$, and the term ignored by intuition is the prior $P(A) = 0.001$.

The likelihood is a probability of B, but it is read as a function of the hypothesis A: it need not sum to 1 over hypotheses ($P(+ \mid \text{sick}) + P(+ \mid \text{healthy}) = 1.00$ here only by coincidence of the chosen rates). The evidence $P(B)$ does not depend on A; it normalises the posterior. Equivalent odds form: $\frac{P(A|B)}{P(\bar{A}|B)} = \frac{P(B|A)}{P(B|\bar{A})} \cdot \frac{P(A)}{P(\bar{A})}$, i.e. posterior odds = likelihood ratio × prior odds. Reference: (Sheldon & others 2002).

Bayes' Theorem — Two Events

Theorem (Bayes, 2 events). Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

posterior — what we believe after

likelihood — how well A explains B

evidence — how ordinary B is

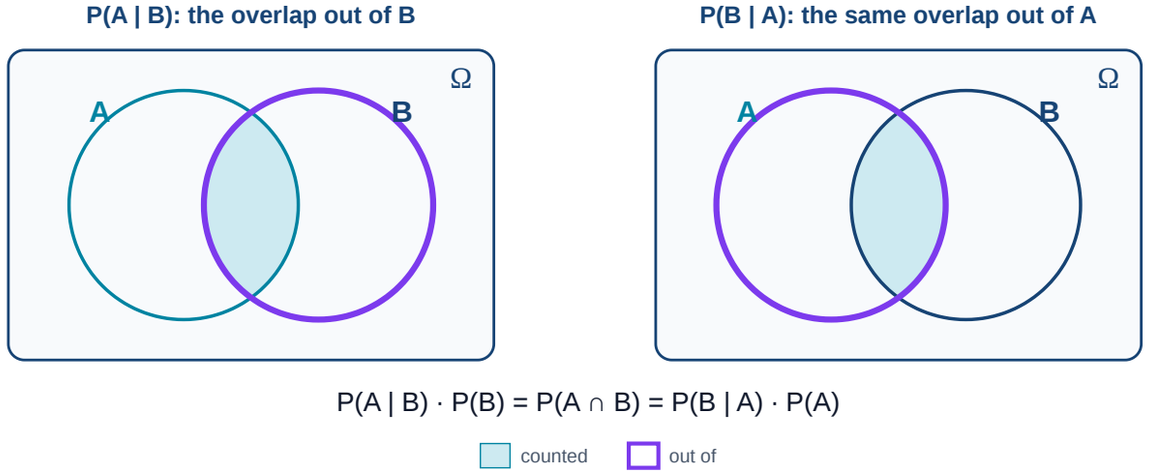
prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

"99% accurate" is the **likelihood** $P(+ \mid \text{sick})$; the intuition ignored the **prior** $P(\text{sick}) = 0.001$.

Proof — Two Events

Intuition



One overlap $A \cap B$, measured against two wholes: against B it is $P(A \mid B)$, against A it is $P(B \mid A)$.

Proof

step	because
$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$	definition
$= \frac{P(A \cap B)}{P(A)} \cdot \frac{P(A)}{P(B)}$	multiply and divide by $P(A)$
$= \frac{P(B \mid A)\, P(A)}{P(B)}$	definition, roles swapped

Needs $P(A), P(B) > 0$. ■

Speaker notes

The figure shows the same region $A \cap B$ measured against two different wholes: against B it is $P(A \mid B)$, against A it is $P(B \mid A)$. Both products with the size of the whole give back the overlap, $P(A \mid B)P(B) = P(A \cap B) = P(B \mid A)P(A)$; dividing by $P(B)$ is Bayes' theorem.

The first equality is the definition of $P(A \mid B)$ and needs $P(B) > 0$. The second multiplies and divides by $P(A)$ and needs $P(A) > 0$; the third recognises $P(A \cap B)/P(A) = P(B \mid A)$, the definition with the roles swapped. If $P(A) = 0$, $P(B \mid A)$ is undefined, but $P(A \mid B) = 0$ directly from the definition: a hypothesis with prior probability zero cannot gain probability from any evidence. This is why a prior should not assign zero to anything that may be true (Lindley's "Cromwell's rule").

Mini-Exercise

A high-risk group. Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

Speaker notes

2,000 is chosen so that both rates give whole people: 1 in 20 of 2,000 is 100, and 1% of the 1,900 healthy is 19. Only the positives count once the result is known — 118 people, the new sample space — and 99 of them are sick. With prevalence 1 in 1,000 the same column held about 1 true positive against 10 false alarms; now true positives outnumber false alarms five to one.

Neither result corrects the other. The posterior depends on the test (sensitivity, false-alarm rate) and on the population tested (prevalence); here the test is the same and only the prevalence differs, raising the positive predictive value from about 9% to about 84%. For the same reason a positive screening result is commonly confirmed by a second, independent test: the first positive raises the prior for the second.

Mini-Exercise

A high-risk group. Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

- 100 sick → **99** positive; 1,900 healthy → **19** positive.
- 99 of all **118** positives are sick: $P(\text{sick} \mid +) \approx 84\%$.

Mini-Exercise

A high-risk group. Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

- 100 sick → **99** positive; 1,900 healthy → **19** positive.
- 99 of all **118** positives are sick: $P(\text{sick} \mid +) \approx 84\%$.

	everyone	high-risk group
the test, $P(+ \mid \text{sick})$ / $P(+ \mid \text{healthy})$	99% / 1%	99% / 1%
who is tested, $P(\text{sick})$	1 in 1,000	1 in 20
$P(\text{sick} \mid +)$	$\approx 9\%$	$\approx 84\%$

Mini-Exercise

A high-risk group. Same test: positive for **99%** of the sick, for **1%** of the healthy. Now **1 in 20** is sick. Above or below 9%? Guess, then count 2,000 people.

- 100 sick → **99** positive; 1,900 healthy → **19** positive.
- 99 of all **118** positives are sick: $P(\text{sick} \mid +) \approx 84\%$.

	everyone	high-risk group
the test, $P(+ \mid \text{sick})$ / $P(+ \mid \text{healthy})$	99% / 1%	99% / 1%
who is tested, $P(\text{sick})$	1 in 1,000	1 in 20
$P(\text{sick} \mid +)$	$\approx 9\%$	$\approx 84\%$

> Both answers are right. Same test, same rule — **different people tested**.

The Same Answer, by Bayes' Theorem

Which known number goes where?

$$P(\text{sick} \mid +) = \frac{\overbrace{P(+ \mid \text{sick})}^{\text{likelihood: } 0.99} \cdot \overbrace{P(\text{sick})}^{\text{prior: } 0.05}}{\underbrace{P(+ \mid \text{sick})\,P(\text{sick})}_{\text{sick and positive: } 0.99 \cdot 0.05 = 0.0495} + \underbrace{P(+ \mid \text{healthy})\,P(\text{healthy})}_{\text{healthy and positive: } 0.01 \cdot 0.95 = 0.0095}}$$

Speaker notes

The denominator, the evidence $P(+)$, is total probability over the partition $\{\text{sick}, \text{healthy}\}$: a positive comes either from a sick person, $P(+ \mid \text{sick})\,P(\text{sick}) = 0.0495$, or from a healthy one, $P(+ \mid \text{healthy})\,P(\text{healthy}) = 0.01 \cdot 0.95 = 0.0095$. The posterior is the first share of the total. In odds form: prior odds $1 : 19$, likelihood ratio $0.99/0.01 = 99$, posterior odds $99 : 19$, i.e. $99/118 \approx 0.84$.

The Same Answer, by Bayes' Theorem

Which known number goes where?

$$P(\text{sick} \mid +) = \frac{\overbrace{P(+ \mid \text{sick})}^{\text{likelihood: } 0.99} \cdot \overbrace{P(\text{sick})}^{\text{prior: } 0.05}}{\underbrace{P(+ \mid \text{sick})\,P(\text{sick})}_{\text{sick and positive: } 0.99 \cdot 0.05 = 0.0495} + \underbrace{P(+ \mid \text{healthy})\,P(\text{healthy})}_{\text{healthy and positive: } 0.01 \cdot 0.95 = 0.0095}}$$

$$= \frac{0.0495}{0.0495 + 0.0095} = \frac{0.0495}{0.0590} \approx 0.84$$

The denominator is the **evidence** $P(+)$: every way to test positive.

The Same Answer, by Bayes' Theorem

Which known number goes where?

$$P(\text{sick} \mid +) = \frac{\overbrace{P(+ \mid \text{sick})}^{\text{likelihood: } 0.99} \cdot \overbrace{P(\text{sick})}^{\text{prior: } 0.05}}{\underbrace{P(+ \mid \text{sick})\,P(\text{sick})}_{\text{sick and positive: } 0.99 \cdot 0.05 = 0.0495} + \underbrace{P(+ \mid \text{healthy})\,P(\text{healthy})}_{\text{healthy and positive: } 0.01 \cdot 0.95 = 0.0095}}$$

$$= \frac{0.0495}{0.0495 + 0.0095} = \frac{0.0495}{0.0590} \approx 0.84$$

The denominator is the **evidence** $P(+)$: every way to test positive.

> Times 2,000, the fraction is **99 / 118**. Counting people and the formula are **one calculation**, written two ways.

Bayes' Theorem — Multiple Events

Theorem (Bayes, multiple events). Let $E_1, \ldots, E_k$ be mutually exclusive with $\Omega = E_1 \cup \ldots \cup E_k$ and $P(E_i) > 0$, and let B be an event with $P(B) > 0$. Then

$$P(E_i \mid B) = \frac{P(B \mid E_i) \cdot P(E_i)}{P(B)} = \frac{P(B \mid E_i) \cdot P(E_i)}{\sum_{j=1}^{k} P(B \mid E_j) \cdot P(E_j)}$$

posterior of E_i

evidence

every E_i explains B, weighted by its prior

likelihood — how well E_i explains B

prior of E_i

The denominator is the **total probability** of the evidence.

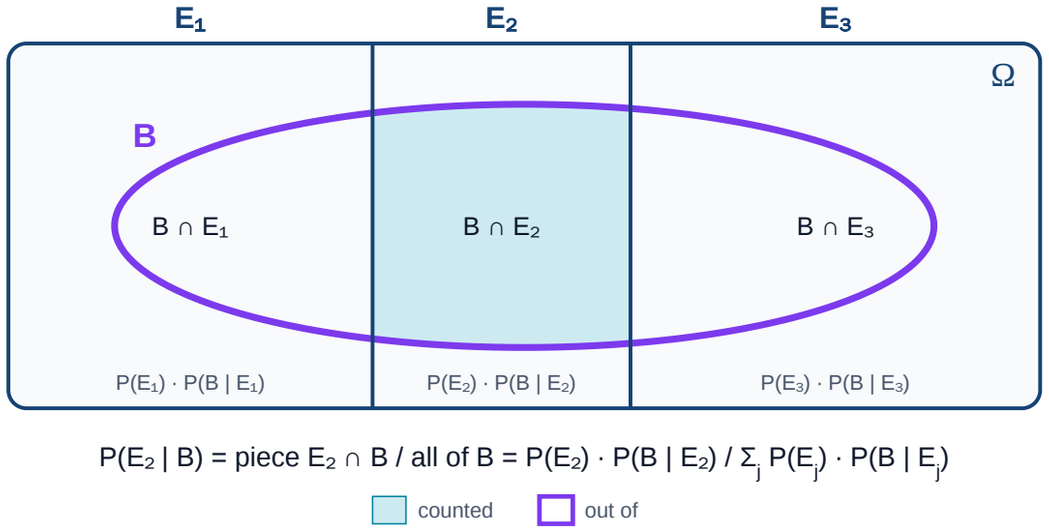
Speaker notes

The events $E_1, \ldots, E_k$ partition Ω: exactly one of them occurs. Read as competing explanations of B, they are called hypotheses. "Sick vs. healthy" is the case $k = 2$, and the students-and-exams example (PML vs. DB) is another. The posteriors $P(E_i \mid B)$ sum to 1 because the denominator is the sum of all numerators.

The procedure is: compute likelihood × prior for every E_i, then divide by the total. Since the denominator is the same for all i, ranking the events — choosing the most probable one — needs only the numerators. A Bayes classifier does exactly this, with the classes as the E_i and the observed features as B.

Proof — Multiple Events

Intuition



The events E_i cut B into pieces. The posterior of E_i is **its piece out of all pieces**.

Proof

step	because
$P(E_i \mid B) = \frac{P(B\|E_i)\,P(E_i)}{P(B)}$	Bayes, two events
$P(B) = \sum_j P(B \cap E_j)$	additivity
$= \sum_j P(B \mid E_j)\,P(E_j)$	product rule

Additivity holds because the E_j are disjoint and cover Ω. Insert $P(B)$ into the first line. ■

Speaker notes

The events E_j are the strips; the evidence B is the ellipse. Each piece $B \cap E_j$ has probability $P(E_j)\,P(B \mid E_j)$ (product rule), and the pieces add up to $P(B)$ (total probability). The posterior of E_2 is its piece out of the whole ellipse. An event E_j wins when its piece is large — because it was plausible beforehand (wide strip), or because it explains B well (much of its strip lies inside B), or both.

Takeaway

> Bayes' theorem is the arithmetic of changing your mind: **posterior ∝ likelihood × prior**. Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

Speaker notes

$\propto$ means "proportional to, as a function of the hypothesis": the posterior is the product likelihood × prior, rescaled so that it sums to 1 over all hypotheses. Two consequences follow. Ignoring the prior gives the base-rate error (99% instead of 9% for the disease test). The same evidence gives very different posteriors under different priors (9% at prevalence 1/1,000, 84% at 1/20).

Sequential updating is consistent: the posterior after one piece of evidence serves as the prior for the next, and if the pieces of evidence are conditionally independent given the hypothesis, the result equals a single update on all of them at once.

Module 5

Random Variables and Distributions

> ***Guiding Question:*** *We rarely care which outcome occurred, only what it was worth. How do we move from outcomes to quantities?*

Speaker notes

Session boundary: the second session starts here. Recall the three-card flow from the opening (assume → observe → check): Modules 1–4 built the model of chance for events; Modules 5–7 turn outcomes into quantities, describe a sample and check a difference.

Why Random Variables?

Three fair coins: what matters is rarely *which* sequence occurred, but the **number of heads** Y.

Y	outcomes ω with $Y(\omega) = k$	count	$P(Y = k)$
0	TTT	1	$\frac{1}{8}$
1	HTT · THT · TTH	3	$\frac{3}{8}$
2	HHT · HTH · THH	3	$\frac{3}{8}$
3	HHH	1	$\frac{1}{8}$

> Y compresses eight outcomes into four numbers — the probabilities come along.

Speaker notes

The eight outcomes are equally likely, $\frac{1}{8}$ each. Y maps them to $\{0, 1, 2, 3\}$, and $P(Y = k)$ is the number of sequences with k heads divided by 8, i.e. $\binom{3}{k}/8$: 1, 3, 3, 1 sequences. This is the binomial distribution with $n = 3$, $p = \frac{1}{2}$.

The compression loses information: which coins showed heads, and in which order. It gains a numerical quantity that can be averaged, compared and plotted; for example $E[Y] = (0 + 3 + 6 + 3)/8 = 1.5$. Whether the lost information matters depends on the question: for the number of heads it does not, for "did the first coin show heads" it does.

Definition

Definition (Random variable). A random variable X maps outcomes to a measurable range E: $X : \Omega \to E$. Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in ω, not in X.

Speaker notes

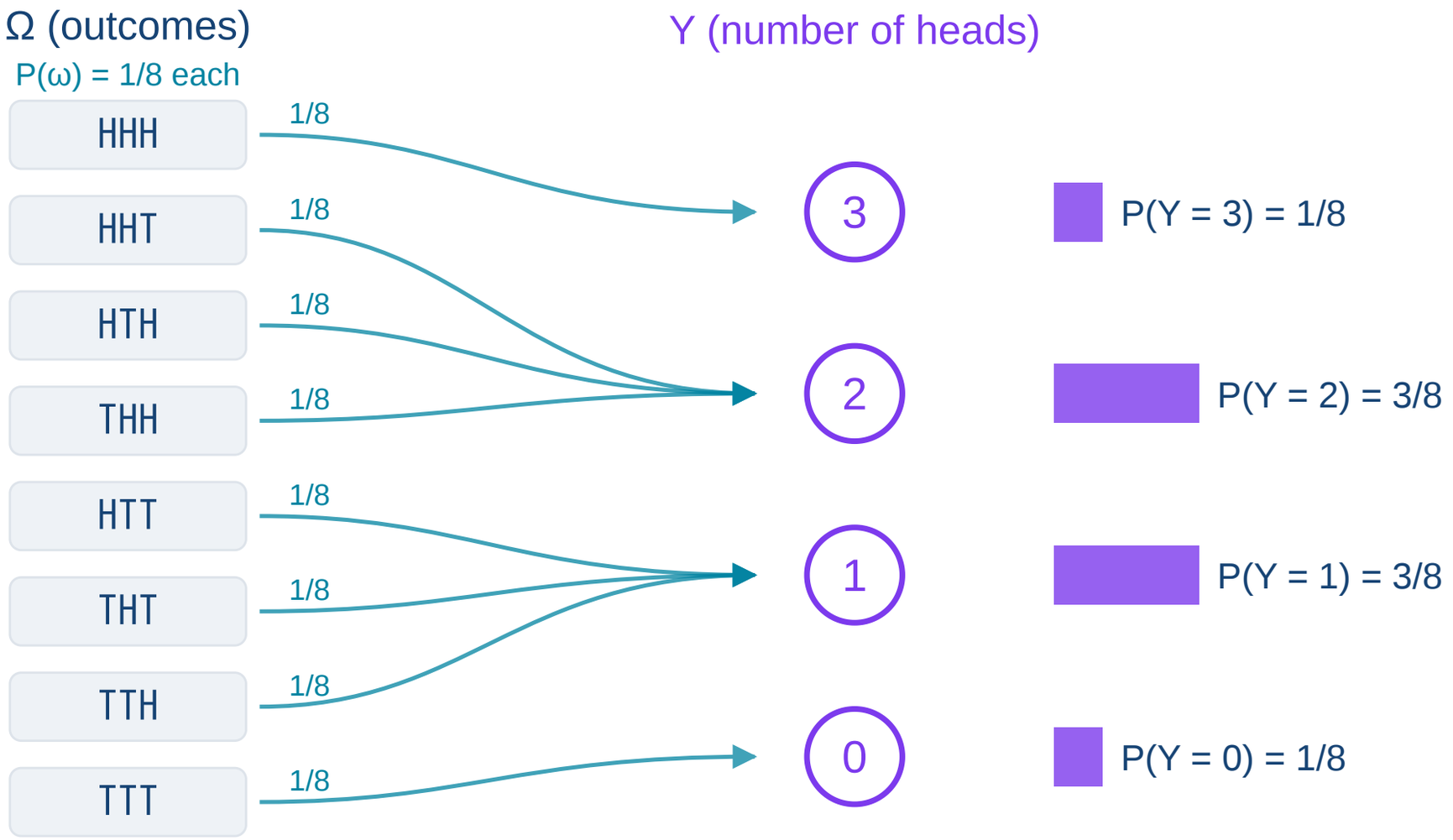
X is a deterministic function: the same outcome ω always gives the same value $X(\omega)$. The randomness is in which ω occurs. The notation $P(X \in S)$ abbreviates the probability of the event $X^{-1}(S) = \{\omega : X(\omega) \in S\}$, the preimage of S; for a single value, $P(X = x) = P(\{\omega : X(\omega) = x\})$.

"Measurable" means that every such preimage is an event, so that P is defined on it. On a finite Ω with the power set as event space every function is measurable. The convention is capital letters for random variables and lower case for their values: $P(X = x)$.

The resulting assignment $S \mapsto P(X \in S)$ is itself a probability measure on E, the distribution of X.

The Function, Drawn

Three arrows arrive at 1 and at 2: 3/8 each — the whole shape



Speaker notes

The steps show the eight outcomes of three coin tosses; one arrow from each outcome to its number of heads; the four values with the count of arrows arriving at each (1, 3, 3, 1); and the resulting probability mass function $P(Y = k)$.

Every outcome has exactly one outgoing arrow, which is what makes Y a function. A value may receive several arrows (the function need not be injective) or none (it need not be surjective onto E; a value with no preimage has probability 0). The values 1 and 2 are three times as probable as 0 and 3 solely because three outcomes map to each of them; no probability is assigned to Y directly, it is inherited from the measure on Ω.

Examples

experiment	sample space Ω	random variable $X(\omega)$	values
two coins	$\{HH, HT, TH, TT\}$	number of heads	$0, 1, 2$
two dice	pairs $\langle n_1, n_2 \rangle$	sum $n_1 + n_2$	$2, \ldots, 12$
draw a person	a population	height of the person	an interval of $\mathbb{R}$

The same outcome can carry many random variables: one per question asked of it.

Speaker notes

Two coins: X takes values 0, 1, 2 with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Two dice: the sum takes values 2 to 12; $P(X = 7) = 6/36$ is the most probable value, $P(X = 2) = P(X = 12) = 1/36$.

People: Ω is a population and the experiment is drawing one person at random. Height is then a random variable, and its distribution is the distribution of heights in the population. A column of a dataset is read the same way: a function from records to values, whose distribution is that of the attribute in the population the records were drawn from. Several columns on the same records are several random variables on the same Ω, which is what makes a joint distribution of them meaningful.

Discrete and Continuous

Discrete random variables

Countable range, usually $E \subseteq \mathbb{N}$.

- number of children of a person
- sum of two dice

Continuous random variables

Real-valued range, $E = \mathbb{R}$.

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — with a mass function or with a density.

Speaker notes

A discrete random variable takes countably many values (finitely many or indexed by $\mathbb{N}$); its distribution is fully described by the probability mass function $p(x) = P(X = x)$, and probabilities of sets are sums. A continuous random variable has a density f with $P(a \le X \le b) = \int_a^b f(x)\,dx$; every single value then has probability 0, and a density value $f(x)$ is not a probability and can exceed 1.

Real-valued range alone is not what makes a variable continuous: it is the existence of a density. Mixed cases occur, e.g. daily rainfall, which is exactly 0 with positive probability and continuous otherwise. Recorded data are always discrete because of finite measurement precision; a continuous model is an idealisation chosen for convenience, as with height.

A Random Variable With Infinitely Many Values

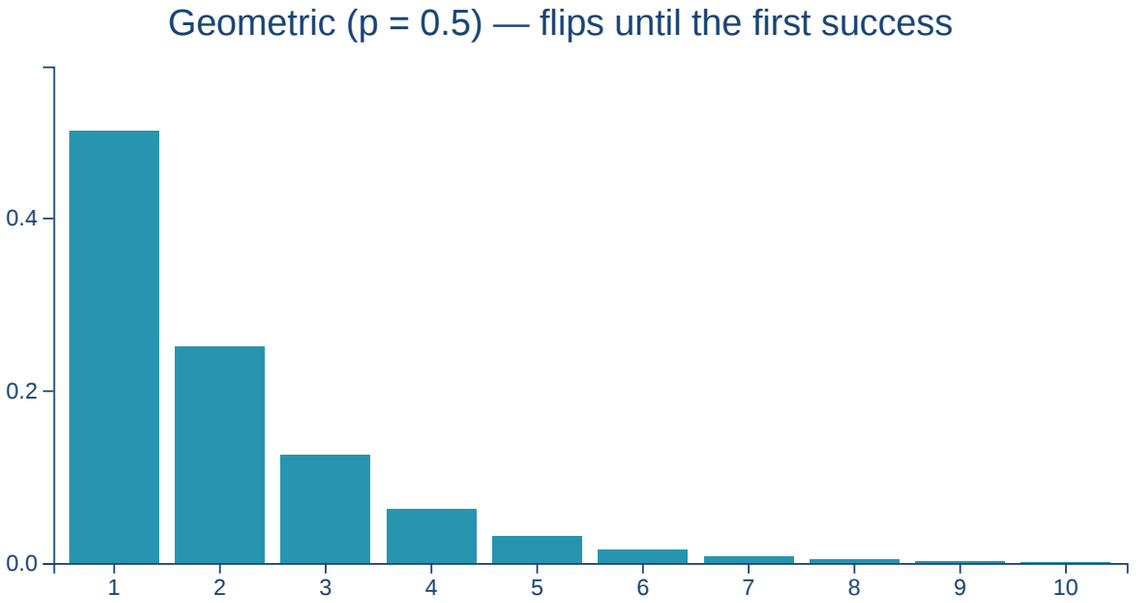
Flip a biased coin ($P(H) = p$) until the first head; X is the number of flips needed.

Geometric distribution, $X \sim \mathrm{Geom}(p)$:

$$P(X = 1) = p$$

$$P(X = 2) = (1 - p)p$$

$$P(X = n) = (1 - p)^{n-1}p$$



Countably infinite range, and the probabilities still sum to 1.

Speaker notes

The figure shows the geometric distribution for $p = \frac{1}{2}$ up to $n = 10$. $X = n$ means $n - 1$ tails followed by a head; with independent flips the probability of this sequence is the product $(1 - p)^{n-1}p$.

The probabilities sum to 1 by the geometric series, for $0 < p \leq 1$: $\sum_{n=1}^{\infty}(1 - p)^{n-1}p = p \cdot \frac{1}{1-(1-p)} = 1$. Countable additivity is what allows this infinite sum to be used. The expected number of flips is $E[X] = 1/p$, e.g. 2 for a fair coin. The distribution is memoryless: $P(X > m + n \mid X > m) = P(X > n)$, so past tails do not make a head more likely — the formal reason the gambler's fallacy is a fallacy for independent trials.

Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

Probability Mass Function

Definition (PMF). For a discrete random variable X, the **probability mass function** is

$$p(a) = P(X = a)$$

INTUITION

$p(a)$ is how much of the total probability sits exactly on the value a.

What follows from the axioms, with nothing new assumed:

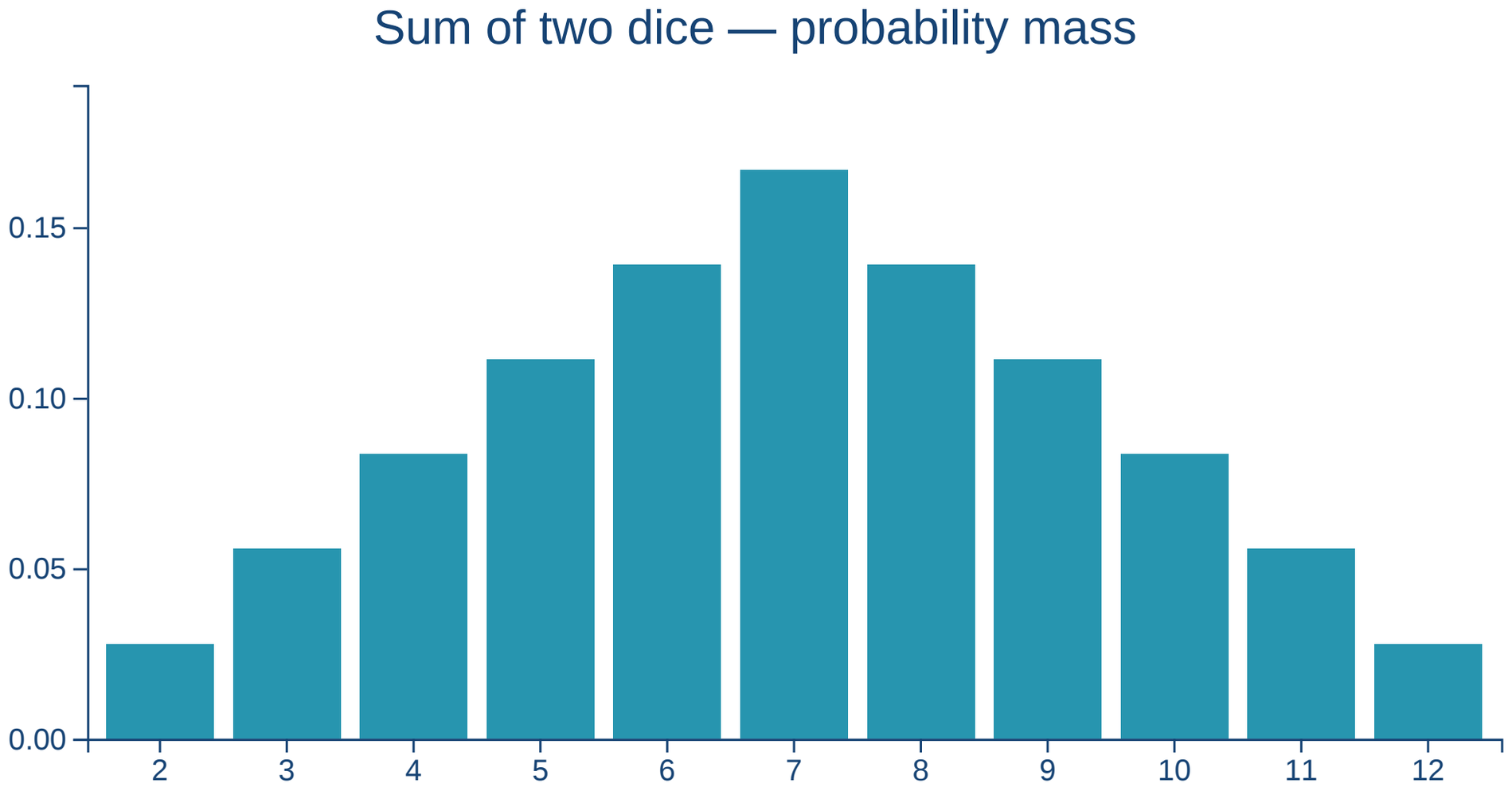
- $p(a) > 0$ for at most countably many values a
- $\sum_i p(a_i) = 1$ — the masses add up to the whole
- p defines a probability measure on the values: the **distribution** of X

Speaker notes

The PMF is the probability measure transferred from outcomes to values: $p(a) = P(\{\omega : X(\omega) = a\})$. Its properties are consequences of the axioms, not new assumptions. Non-negativity is inherited directly. The events $\{X = a_i\}$ are pairwise disjoint and together cover Ω, so countable additivity gives $\sum_i p(a_i) = 1$.

Strictly, p is a function on values; the probability measure it defines is $P_X(A) = \sum_{a_i \in A} p(a_i)$ for any set A of values, the **distribution** of X. A value outside the range of X has $p(a) = 0$. The cumulative distribution function $F(a) = P(X \leq a) = \sum_{a_i \leq a} p(a_i)$ carries the same information as the PMF and is a step function for a discrete variable.

Example: Sum of Two Dice



The triangular shape is counting, drawn: six of the 36 outcomes give 7, only one gives 2.

Speaker notes

The sample space has 36 equally likely **ordered** pairs (i, j). The number of pairs with sum s is $6 - |s - 7|$ for $s = 2, \ldots, 12$, so $p(s) = (6 - |s - 7|)/36$. The shape is symmetric about 7 because the map $(i, j) \mapsto (7 - i, 7 - j)$ is a bijection that sends sum s to $14 - s$; it peaks at 7 because middling sums can be formed in the most ways.

Treating the pairs as unordered (21 outcomes, e.g. $\{1, 2\}$ counted once) is a frequent error: the unordered outcomes are not equally likely, since $\{1, 2\}$ arises from two ordered pairs and $\{1, 1\}$ from only one.

The sum has $E = 7$ and variance $2 \cdot 35/12 = 35/6 \approx 5.83$. For the sum of n dice the standard deviation grows like $\sqrt{n}$ while the range grows like n, so the sum concentrates relative to its range — the first instance of the behaviour described by the law of large numbers and the central limit theorem.

Expectation

Definition (Expected value).

$$E[X] = \sum_i a_i \cdot p(a_i)$$

INTUITION

The **average outcome** if the experiment were repeated infinitely often — the balance point of the bar chart of p.

function of X	$E[g(X)] = \sum_i g(a_i)p(a_i)$
linear transformation	$E[aX + b] = aE[X] + b$
first moment	$\mu = E[X]$; the first *central* moment $E[X - \mu] = 0$

Speaker notes

The frequentist reading is justified by the law of large numbers: the average of n independent repetitions converges to $E[X]$. The expectation exists only if $\sum_i |a_i| p(a_i) < \infty$; for heavy-tailed variables the sum can diverge.

$E[g(X)] = \sum_i g(a_i)p(a_i)$ computes the expectation of $g(X)$ without first deriving the distribution of $g(X)$. Linearity, $E[aX + b] = aE[X] + b$, extends to sums: $E[X + Y] = E[X] + E[Y]$ holds for any two random variables, independent or not.

In general $E[g(X)] \neq g(E[X])$. With X uniform on $\{-1, 1\}$ and $g(x) = x^2$: $E[X^2] = 1$, but $(E[X])^2 = 0$. For convex g, Jensen's inequality gives $E[g(X)] \geq g(E[X])$. The first central moment vanishes because $E[X - \mu] = E[X] - \mu = 0$.

Variance

Definition (Variance).

$$\mathrm{Var}(X) = E\big[(X-\mu)^2\big] = E[X^2] - \big(E[X]\big)^2$$

Spread

The second central moment: the expected squared distance from the mean. Larger variance, larger typical deviation.

Scale and unit

$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$

$\sigma = \sqrt{\mathrm{Var}(X)}$, the **standard deviation**, has the unit of X.

Speaker notes

The computational identity follows from linearity: $E[(X-\mu)^2] = E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2$. Since the variance is non-negative, $E[X^2] \geq (E[X])^2$, with equality exactly when X is constant with probability 1. In floating-point arithmetic the identity can lose precision through cancellation when the mean is large relative to the spread; numerical code therefore subtracts the mean first.

$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$ because $(aX+b) - (a\mu+b) = a(X-\mu)$: a shift moves the distribution without changing its spread, and a scaling by a scales σ by $|a|$. The standard deviation is reported because it has the unit of X; the variance has the squared unit (cm² for heights in cm).

Fair die: $E[X] = 3.5$, $E[X^2] = 91/6$, so $\mathrm{Var}(X) = 91/6 - 12.25 = 35/12 \approx 2.92$ and $\sigma \approx 1.71$.

Takeaway

A discrete distribution is a PMF; expectation and variance compress it into **where it sits** and **how much it spreads**.

Bernoulli and Binomial

	models	PMF	$E[X]$	$\mathrm{Var}(X)$
Bernoulli(p)	one trial: $X = 1$ success, $X = 0$ failure	$p(x) = p^x(1-p)^{1-x}$	p	$p(1-p)$
Binomial $B(n, p)$	successes in n independent trials	$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts the sequences with k successes; the PMF multiplies it by the probability of one such sequence — combinatorics and independence in a single line.

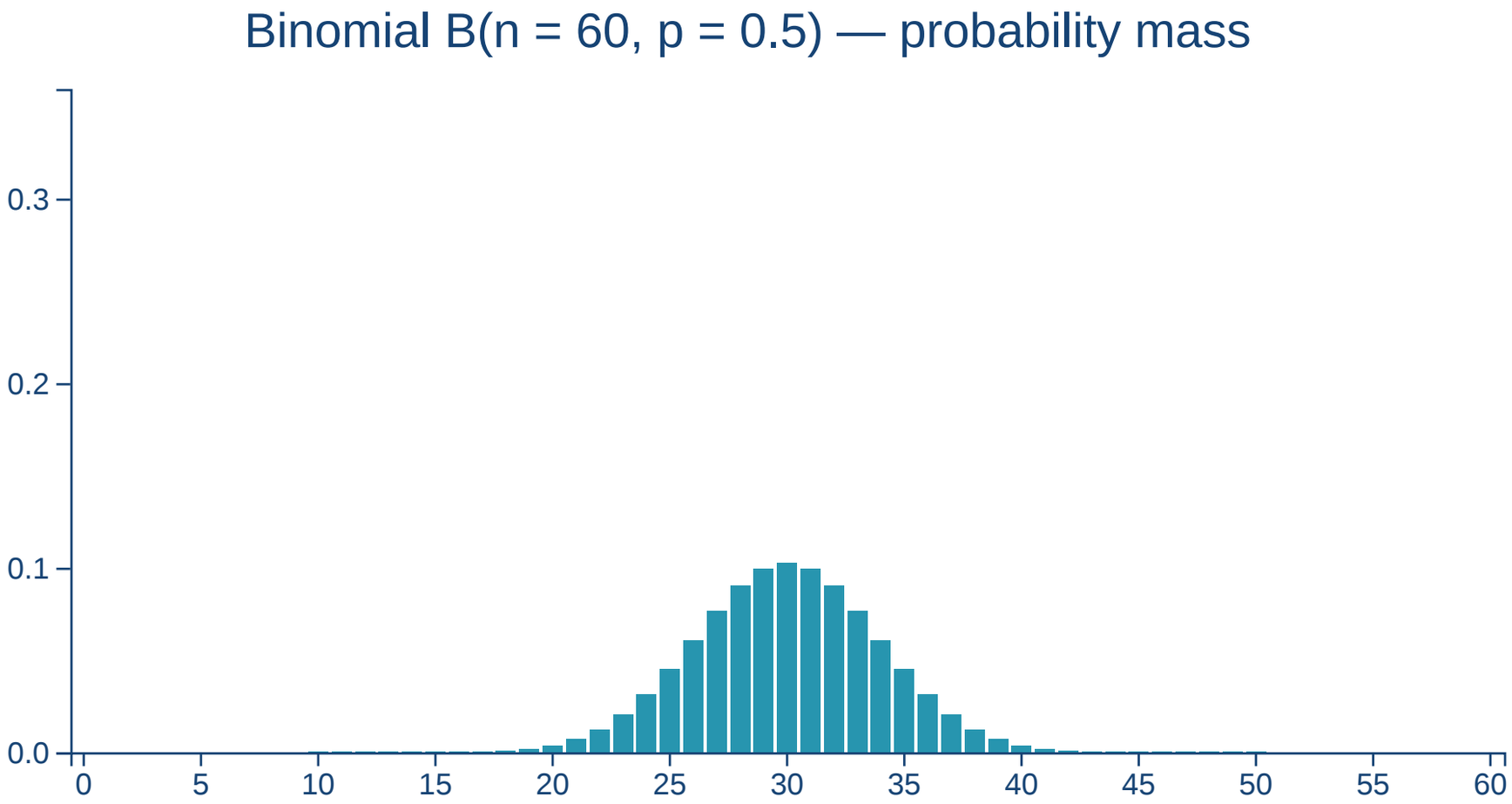
Speaker notes

Bernoulli moments: $E[X] = 1 \cdot p + 0 \cdot (1-p) = p$; since $X^2 = X$, $E[X^2] = p$ and $\operatorname{Var}(X) = p - p^2 = p(1-p)$, which is largest ($\frac{1}{4}$) at $p = \frac{1}{2}$ and 0 at $p \in \{0, 1\}$.

The binomial count is a sum of n independent Bernoulli(p) variables. A particular sequence with k successes has probability $p^k(1-p)^{n-k}$ by independence, and $\binom{n}{k}$ such sequences exist. The PMF sums to 1 by the binomial theorem, $\sum_k \binom{n}{k} p^k (1-p)^{n-k} = (p + 1 - p)^n$. The mean np follows from linearity alone; the variance $np(1-p)$ uses independence, because only for uncorrelated summands do the variances add.

The model requires independent trials with the same success probability p. If p varies between trials or the trials influence one another, the count is not binomial.

How the Shape Depends on the Parameters



Predict before each click: as n grows, does the distribution get wider or narrower — and what happens to its shape?

Speaker notes

With $p = 0.5$ the mean is $n/2$ and the standard deviation $\sqrt{np(1-p)} = \sqrt{n}/2$. For the five values of n shown: $n = 5$: $\sigma \approx 1.12$; $n = 10$: 1.58; $n = 20$: 2.24; $n = 40$: 3.16; $n = 60$: 3.87. Each step shows the next value of n.

Two effects occur together. In absolute terms the distribution gets wider, growing like $\sqrt{n}$. Relative to n it gets narrower: the proportion of successes k/n has standard deviation $\sqrt{p(1-p)/n}$, falling from 0.22 at $n = 5$ to 0.065 at $n = 60$. The shape also becomes bell-like, because a binomial is a sum of independent Bernoulli variables and the central limit theorem applies. For $p = 0.5$ the PMF is symmetric for every n; for other p it is skewed at small n and becomes symmetric only as n grows.

Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

Speaker notes

Model: $X \sim B(10, 0.25)$, so $E[X] = 2.5$ and $\mathrm{Var}(X) = 10 \cdot 0.25 \cdot 0.75 = 1.875$, $\sigma \approx 1.37$. Five correct answers lie about 1.8σ above the mean. The exact tail is $P(X \geq 5) = \sum_{k=5}^{10} \binom{10}{k} 0.25^k 0.75^{10-k} \approx 0.078$, a sum of six PMF values (or $1 - P(X \leq 4)$).

The binomial model assumes that the ten guesses are independent and each succeeds with the same probability 0.25. It fails for a real exam taker who partly knows the material (the success probability then differs between questions), who can rule out some options (raising p above 0.25), or when questions build on each other (dependence).

Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

$B(10, 0.25)$: on average $10 \cdot 0.25 = 2.5$ correct. Five or more is twice the expectation — unlikely (about 8%).

Takeaway

A yes/no attribute is a **Bernoulli** variable with one parameter p; counting the yeses in n independent trials gives a **binomial**. Both are counting in probabilistic clothing — and p is the number data will have to tell us.

From Mass to Density

Definition (PDF). A continuous random variable X has a **probability density function** f with

$$P(b \le X \le c) = \int_b^c f(a)\,da, \qquad \int_{-\infty}^{\infty} f(a)\,da = 1$$

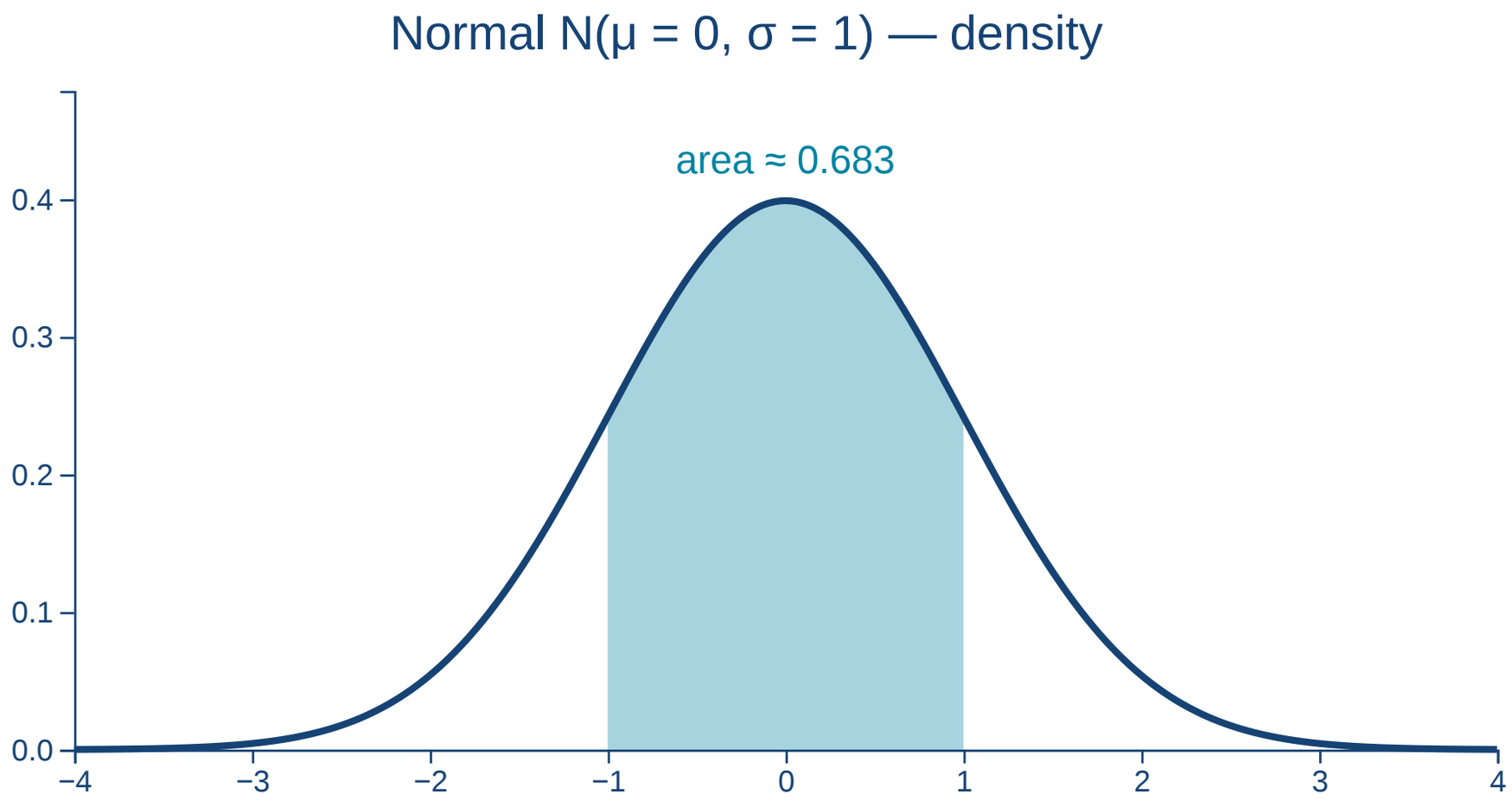
$f(a)$ is **not** a probability — it is probability *per unit interval*. Consequently $f(a) > 1$ is perfectly possible, and $P(X = a) = 0$ for every single value a.

Speaker notes

For small h, $P(a \le X \le a+h) \approx f(a)h$, so f has the unit "probability per unit of X". A density can therefore exceed 1: the uniform distribution on $[0, 0.5]$ has $f = 2$ on that interval. The probability of a single value is an integral over an interval of length zero, so $P(X = a) = 0$; a person being exactly 180.000… cm tall has probability zero. As a consequence, $\le$ and $<$ give the same interval probabilities. Probability zero does not mean impossible: some value is always observed.

The cumulative distribution function $F(a) = P(X \le a) = \int_{-\infty}^{a} f(t)dt$ links the two views; where f is continuous, $f = F'$. Not every random variable is either discrete or continuous — daily rainfall, for instance, has a point mass at 0 and a density above it.

Probability Is Area Under the Curve



The shaded region is $P(-1 \le X \le 1)$ — the same "probability as area" intuition from the axioms unit, now with a curve instead of a Venn diagram.

Speaker notes

In terms of the CDF, $P(b \le X \le c) = F(c) - F(b)$. For the standard normal the shaded area is $P(-1 \le X \le 1) \approx 0.683$; within two standard deviations it is ≈ 0.954, within three ≈ 0.997. The normal CDF has no closed form and is computed numerically (via the error function or tables).

The total area under any density is 1. This is what fixes the constant in the normal density: $\int_{-\infty}^{\infty} e^{-(x-\mu)^2/(2\sigma^2)} dx = \sigma\sqrt{2\pi}$ (the Gaussian integral), so the factor $1/(\sigma\sqrt{2\pi})$ is exactly the normalisation.

Expectation for Continuous Variables

	discrete — mass p	continuous — density f
total	$\sum_i p(a_i) = 1$	$\int_{-\infty}^{\infty} f(a)\,da = 1$
expectation	$E[X] = \sum_i a_i p(a_i)$	$E[X] = \int_{-\infty}^{\infty} a f(a)\,da$
function of X	$E[g(X)] = \sum_i g(a_i) p(a_i)$	$E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a)\,da$

Only the sum turns into an integral. The rules are unchanged: $E[aX + b] = aE[X] + b$ and $\mathrm{Var}(X) = E[(X - \mu)^2]$.

Speaker notes

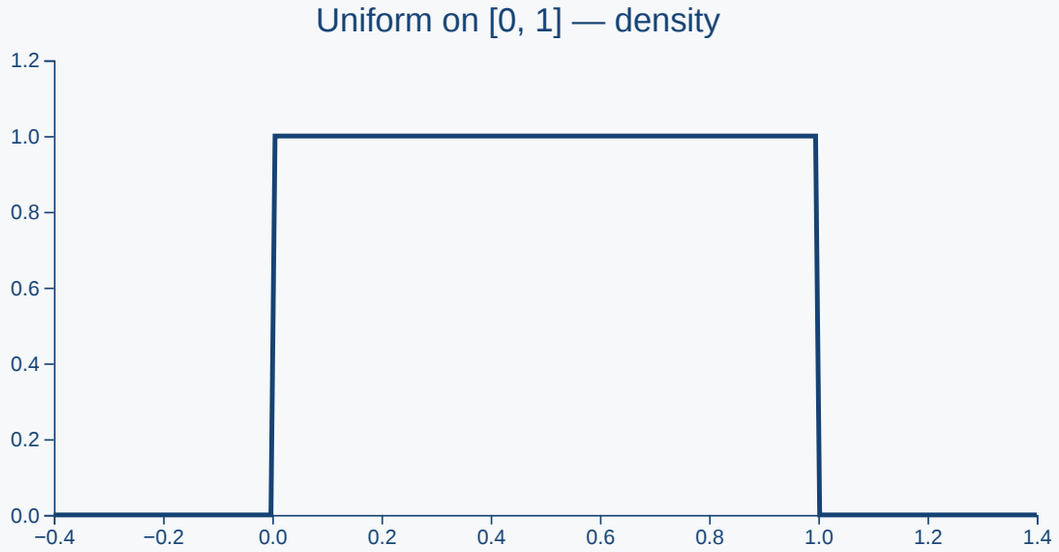
Only the summation becomes an integral; linearity of expectation, the variance identity $\operatorname{Var}(X) = E[X^2] - (E[X])^2$, and $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$ carry over with the same proofs. In practice the integral runs over the support of f, the region where $f > 0$.

Existence is again a condition: the expectation is defined only if $\int |a| f(a)\, da < \infty$. The Cauchy distribution, $f(x) = 1/(\pi(1 + x^2))$, is a standard example of a density without a mean (and without a variance), even though it is symmetric about 0.

Two Basic Continuous Distributions

Uniform on $[a, b]$

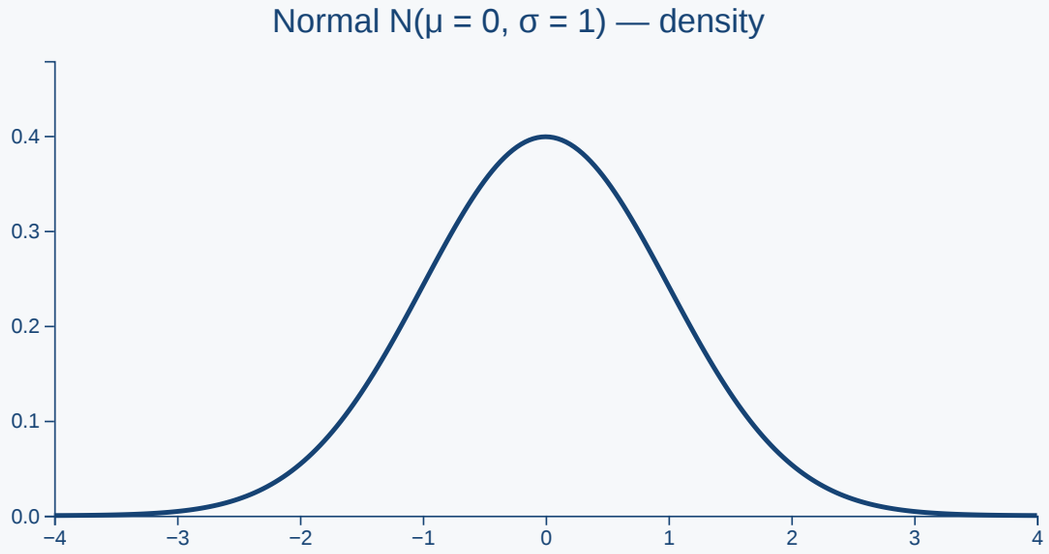
$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$



Equal-length intervals are equally likely — the Laplace assumption, continuous.

Normal $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



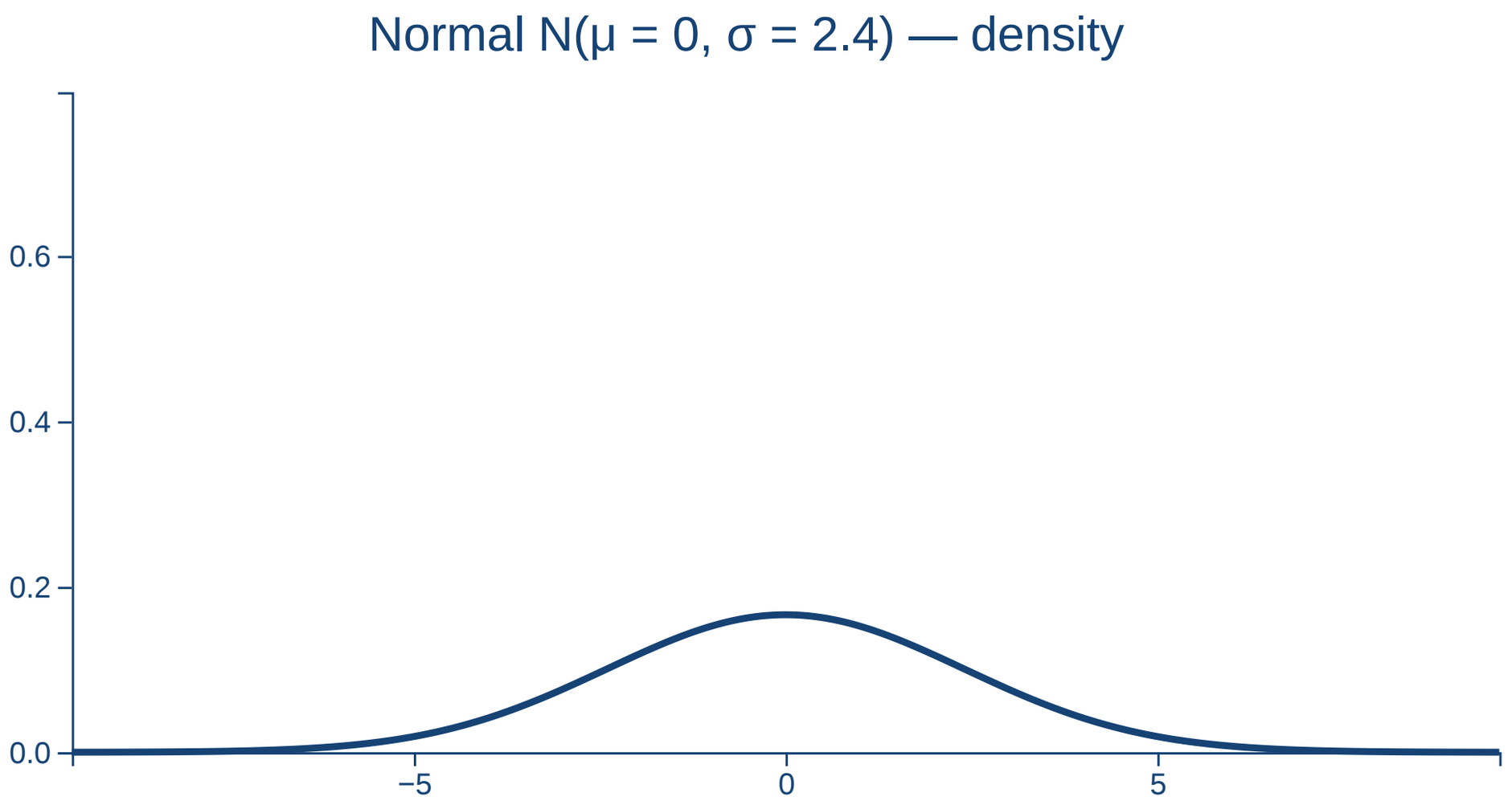
$E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$; $N(0, 1)$ is the **standard** normal.

Speaker notes

The figures show $[a, b] = [0, 1]$ and $N(0, 1)$. The uniform density is $1/(b - a)$ on $[a, b]$ and 0 outside, with $E[X] = (a + b)/2$ and $\mathrm{Var}(X) = (b - a)^2/12$. It is the continuous counterpart of the Laplace assumption: equal-length subintervals receive equal probability.

The normal density is symmetric about μ, has its maximum $1/(\sigma\sqrt{2\pi})$ at μ and its inflection points at $\mu \pm \sigma$. Linear transformations preserve normality: if $X \sim N(\mu, \sigma^2)$ then $aX + b \sim N(a\mu + b, a^2\sigma^2)$. In particular the standardisation $Z = (X - \mu)/\sigma$ is $N(0, 1)$, which is why a single table of the standard normal suffices. The notation $N(\mu, \sigma^2)$ uses the variance, not the standard deviation, as second parameter.

What σ Does to the Shape



The peak falls as the curve widens — the area is always 1.

Speaker notes

The peak height of $N(\mu, \sigma^2)$ is $1/(\sigma\sqrt{2\pi})$, inversely proportional to σ. For the four values shown: $\sigma = 0.6$ gives ≈ 0.665, $\sigma = 1$ gives ≈ 0.399, $\sigma = 1.6$ gives ≈ 0.249, $\sigma = 2.4$ gives ≈ 0.166. Widening by a factor k lowers the peak by the same factor, so the area stays 1.

Independently of σ, about 68% of the mass lies within $\mu \pm \sigma$: changing σ only rescales the horizontal axis. For $\sigma < 1/\sqrt{2\pi} \approx 0.40$ the peak exceeds 1, another instance of a density larger than 1.

Why the Normal Is Everywhere

> The **central limit theorem**: under mild conditions, the average of n independent observations is approximately normal for large n — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown — and why a sum of many independent coin flips, the binomial, looks bell-shaped.

Speaker notes

Precise statement: for independent, identically distributed $X_1, \ldots, X_n$ with mean μ and finite variance σ^2, the standardised mean $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ converges in distribution to $N(0, 1)$. The "mild conditions" are independence and finite variance; versions exist that relax identical distribution. Without finite variance the theorem fails: the mean of independent Cauchy variables is again Cauchy, however large n.

The mean $\bar{X}_n$ has variance σ^2/n, so its standard deviation, the **standard error**, is $\sigma/\sqrt{n}$. This is the basis of error bars and confidence intervals computed from a sample.

A binomial $\mathrm{Bin}(n, p)$ is a sum of n independent Bernoulli variables and is therefore approximately $N(np, np(1-p))$ for large n. The approximation is poor when p is close to 0 or 1; rules of thumb require both np and $n(1-p)$ to be reasonably large (commonly at least 5 or 10).

Other Distributions You Will Meet

distribution	values	typically models
Exponential	$[0, \infty)$	time until an event occurs (e.g. the next earthquake)
Gamma	$[0, \infty)$	time until n events have occurred
Beta	$[0, 1]$	an unknown probability
Dirichlet	probability vectors	distributions *over* probabilities

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

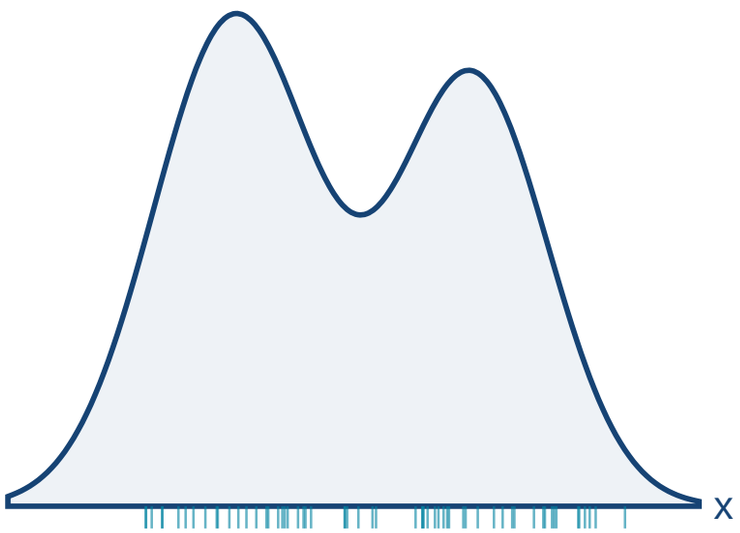
Speaker notes

The exponential distribution with rate λ has density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and mean $1/\lambda$. It is memoryless, $P(X > s + t \mid X > s) = P(X > t)$, which encodes a constant event rate. The gamma distribution with integer shape n and rate λ is the distribution of a sum of n independent exponential waiting times, with mean n/λ.

The Beta(α, β) distribution lives on $[0, 1]$ with mean $\alpha/(\alpha + \beta)$; $\mathrm{Beta}(1, 1)$ is the uniform distribution. It is the standard prior for an unknown probability, because a Beta prior combined with Bernoulli or binomial data yields a Beta posterior. The Dirichlet distribution generalises the Beta to vectors of probabilities that sum to 1 and plays the same role for categorical data. None of these four is needed for the rest of this unit.

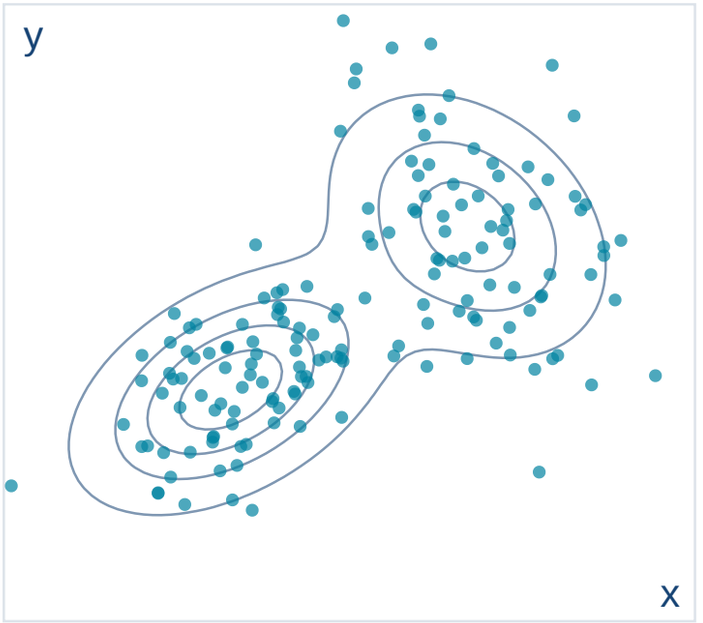
More Than One Random Variable

Univariate



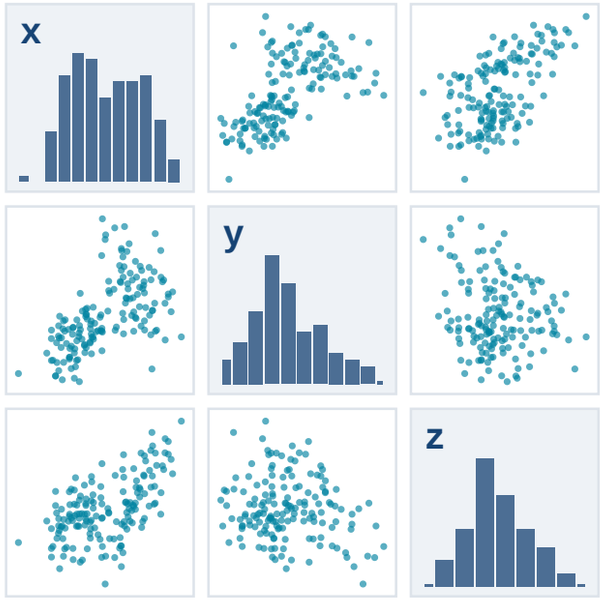
p(x) — one column

Bivariate



p(x, y) — two columns

Multivariate



p(x, y, z, …) — three or more

A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

Speaker notes

The figure shows one variable as a density curve, two as a point cloud with the contour lines of their joint density, and three as a scatter matrix: every pair of columns as a cloud, every single column as a histogram on the diagonal. A scatter matrix shows only the two-dimensional projections; the joint of three or more variables can no longer be drawn directly.

A table with columns age, income and churn is a sample from a trivariate joint distribution: each row is one realisation of the vector $(X_{\text{age}}, X_{\text{income}}, X_{\text{churn}})$, and the joint distribution assigns probability to combinations of values.

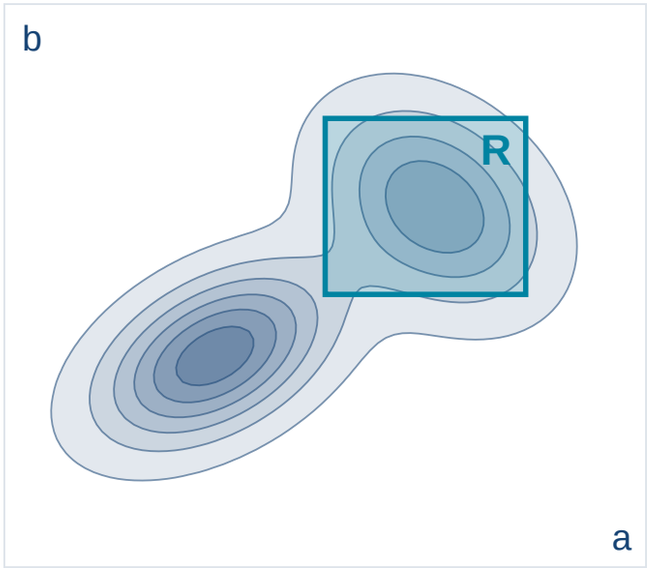
The joint determines every marginal and every conditional distribution. The converse fails: the marginals alone do not determine the joint, except under independence, where the joint is the product of the marginals. The size of a joint grows exponentially: d binary variables need $2^d - 1$ free probabilities ($d = 30$ already gives more than 10^9). Any practical model therefore imposes structure, such as independence or conditional independence assumptions, to reduce this number.

Joint Mass and Density

joint PMF (discrete)	joint PDF (continuous)
$p_{X,Y}(a,b) = P(X=a, Y=b)$	$P((X,Y)\in R) = \iint_R f_{X,Y}(a,b)\,da\,db$

	G=0	G=1	G=2	G=3
B=0	.15	.1	.0875	.0375
B=1	.1	.175	.1125	0
B=2	.0875	.1125	0	0
B=3	.0375	0	0	0

one probability per cell · all sum to 1



darker = denser · volume over R ≈ 0.33

From here on we write $p(a,b,c,\ldots)$ for the joint of $X_1, X_2, X_3, \ldots$ — a mass or a density, as the variables require.

Speaker notes

Left: the joint PMF of the number of boys and girls in a family (the example that follows), one probability per cell. Right: a joint density drawn as contour lines; the probability of the region R is the volume under the density surface above R, here computed numerically.

Both joint forms are normalised: $\sum_a \sum_b p_{X,Y}(a,b) = 1$ and $\iint f_{X,Y}(a,b)\,da\,db = 1$. A joint may also mix the two kinds, e.g. a class label and a measurement; then one sums over the discrete and integrates over the continuous variable.

The convention $p(a,b,c,\ldots)$ is safe because marginalisation, conditioning and factorisation follow the same rules in both cases; only summing versus integrating differs. Two caveats apply. For continuous variables p is a density, so its values are not probabilities and can exceed 1; reading $p(a,b,c)$ as "the probability of state a,b,c" is then shorthand for a density value. And p is overloaded: $p(a)$ and $p(b)$ denote different functions, distinguished only by which variable the argument refers to.

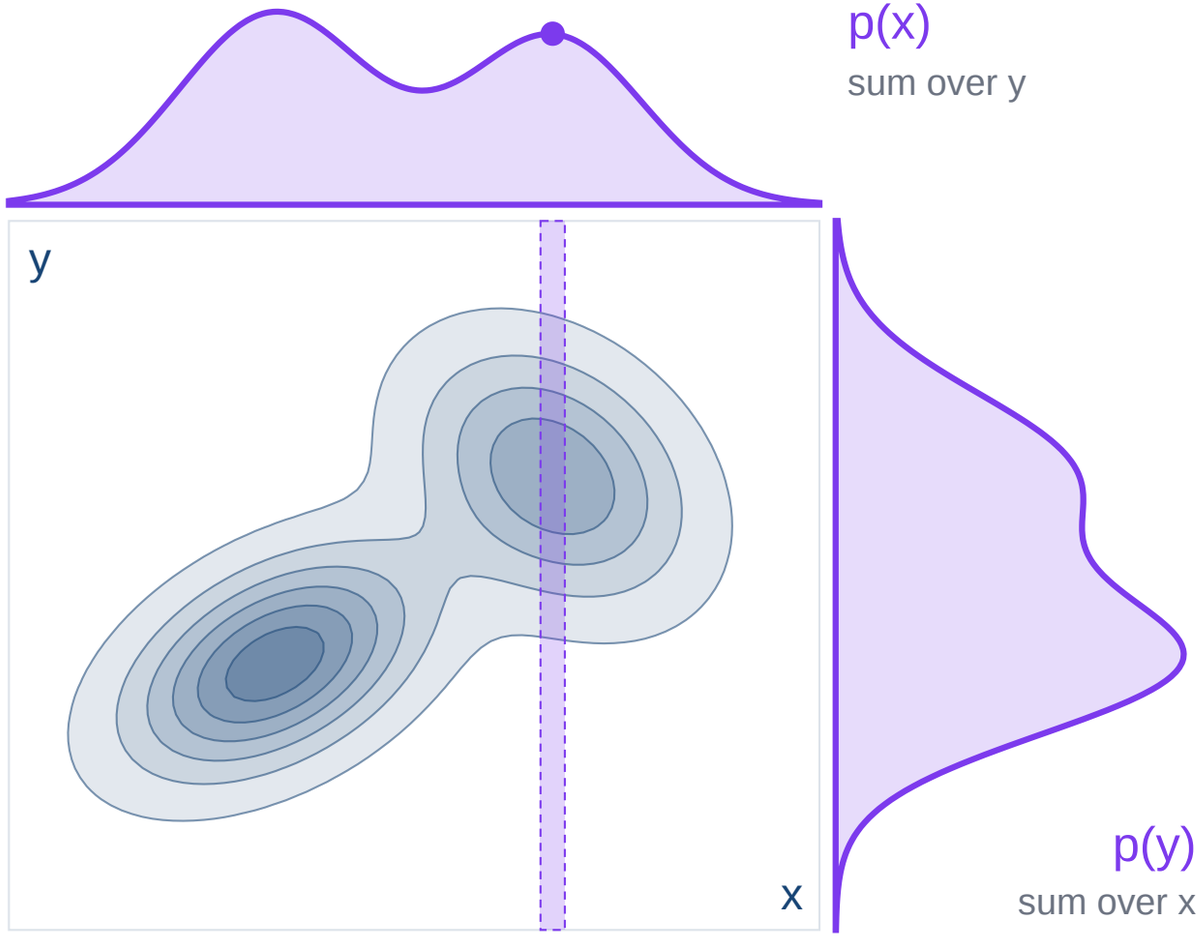
Marginalisation

Marginalisation eliminates a random variable by summing over all its states. For $p(x_1, \ldots, x_n)$, a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \ldots, x_n)$$

INTUITION

Forget about a variable, but keep its influence correctly accounted for: collapse the joint onto the axis of the variable you keep.



Speaker notes

The steps project the joint density from before onto its axes: (1) summing (integrating) every vertical strip gives one point of $p(x)$, (2) summing every horizontal strip gives $p(y)$. The marginals are computed from the joint, not assumed; the two bumps of the joint reappear in the marginals, as two peaks in $p(x)$ and as a peak with a shoulder in $p(y)$.

Marginalisation is the law of total probability applied to random variables: the events $\{X_2 = x_2\}$ for all states x_2 partition the sample space, so $p(x_1) = \sum_{x_2} p(x_1, x_2) = \sum_{x_2} p(x_1 \mid x_2)p(x_2)$. The second form shows how the eliminated variable keeps its influence: each conditional is weighted by how probable that state is. For continuous variables the sum becomes an integral.

In general, the marginal of any subset of variables is obtained by summing over all the remaining ones. Done naively, marginalising $n - 1$ variables with k states each takes k^{n-1} terms; factorised models allow the sums to be rearranged and computed far more cheaply.

Example: Boys and Girls

	G=0	G=1	G=2	G=3	p(B)
B=0	.15	.1	.0875	.0375	.375
B=1	.1	.175	.1125	0	.3875
B=2	.0875	.1125	0	0	.2
B=3	.0375	0	0	0	.0375
p(G)	.375	.3875	.2	.0375	total 1



Families: 15% have no children, 20% one, 35% two, 30% three. Each child is a boy or a girl with probability $\frac{1}{2}$, independently.

B boys, G girls.

> The margins of the table *are* the marginals — hence the name.

Speaker notes

The steps sum each row into $p(B)$ on the right and each column into $p(G)$ below; both sum to 1.

Each cell follows from family size times a binomial probability. One child (20%): one boy or one girl, 10% each. Two children (35%): BB and GG with probability $\frac{1}{4}$ each ($.0875$), one of each with $\frac{1}{2}$ ($.175$). Three children (30%): 3 boys or 3 girls with $\frac{1}{8}$ each ($.0375$), 2+1 in either direction with $\frac{3}{8}$ each ($.1125$). The cell $(B = 1, G = 2) = .1125$ reads: 11.25% of families have one boy and two girls. Rows and columns sum to the marginals, which are identical for B and G by symmetry.

B and G are dependent, because family size couples them: $p(B = 3, G = 3) = 0$, whereas $p(B = 3)p(G = 3) = .0375^2 \approx .0014$. The covariance is negative: $E[B] = E[G] = 0.9$, $E[BG] = .175 + 2 \cdot .1125 + 2 \cdot .1125 = .625$, so $\text{cov}(B, G) = .625 - .81 = -.185$.

Covariance

$$\operatorname{cov}(A, B) = E\big[(A - E[A])(B - E[B])\big]$$

$\operatorname{cov} > 0$
A large when B large — positively related

$\operatorname{cov} < 0$
A large when B small — negatively related

$\operatorname{cov}(A, A) = \operatorname{Var}(A)$
variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *linear* association.

Speaker notes

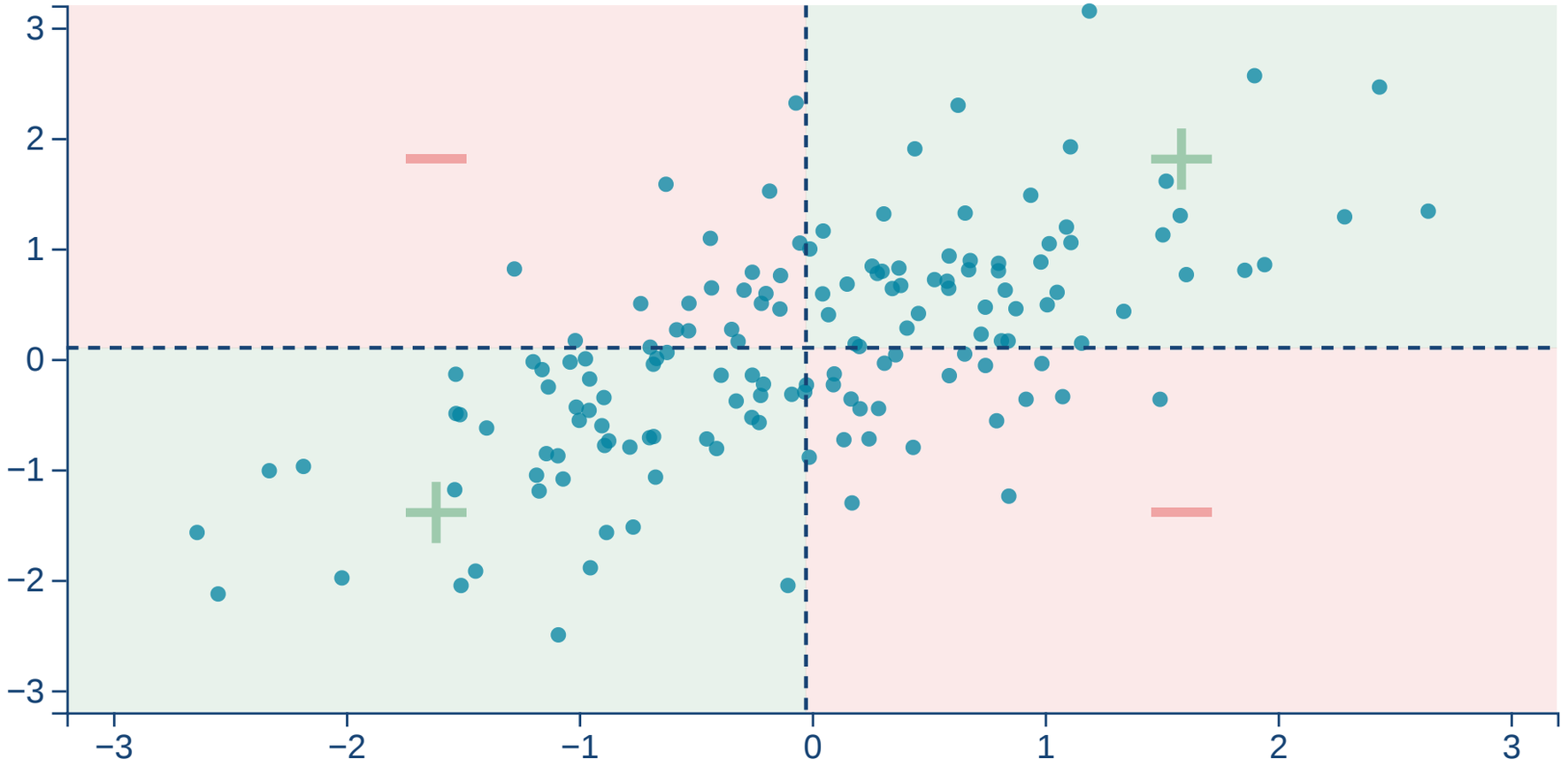
Expanding the definition gives the computational form $\operatorname{cov}(A, B) = E[AB] - E[A]E[B]$. For independent variables $E[AB] = E[A]E[B]$, hence zero covariance. The converse fails: X uniform on $[-1, 1]$ and $Y = X^2$ have $E[XY] = E[X^3] = 0$ and $E[X] = 0$, so $\operatorname{cov}(X, Y) = 0$, although Y is a function of X. Methods that rely on covariance or correlation alone can therefore miss real dependence. An important exception: for jointly normal variables, zero covariance does imply independence.

Covariance measures linear association. Its size depends on the units; the correlation $\rho = \operatorname{cov}(A, B)/(\sigma_A \sigma_B)$ removes them and lies in $[-1, 1]$, with $|\rho| = 1$ exactly when B is a linear function of A. Covariance also governs the variance of a sum: $\operatorname{Var}(A + B) = \operatorname{Var}(A) + \operatorname{Var}(B) + 2\operatorname{cov}(A, B)$.

Covariance Sees Only One Kind of Structure

Slide ρ, then switch the relation. Predict the covariance first.

cov(X, Y) = 0.62



Predict first: with $Y = X^2 - 1$, every point is fixed by its x. The covariance?

Speaker notes

The steps show (1) a cloud of points with correlation $\rho = 0.7$, (2) the two means as crosshairs, from which deviations are measured, and (3) the four quadrants: the product $(x - \bar{x})(y - \bar{y})$ is positive in the upper-right and lower-left quadrants and negative in the other two. The covariance is the average of these products, so it is large and positive when most points fall in the positive quadrants. The final step allows changing ρ and switching to the relation $Y = X^2 - 1$.

For $Y = X^2 - 1$ with X symmetric about 0, each point at x has a mirror point at $-x$ with the same y and an opposite-signed product, so the products cancel: $\operatorname{cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$. The sample value is close to, not exactly, zero. The dependence is complete — Y is fixed by X — yet the covariance is about 0. This is why "uncorrelated" and "independent" are different properties.

Takeaway

The joint distribution holds everything; **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

Module 6

Describing Data

> ***Guiding Question:*** *The distribution is unknown and all we have is a sample. What can we honestly say about it?*

We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

	what it is	its numbers	do we see them?
Population	the entire group — all customers, all transactions, all documents	**parameters** μ, σ^2	no — the truth we want
Sample	the observations we actually have	**statistics** $\overline{x}$, s^2	yes — and we estimate the parameters from them

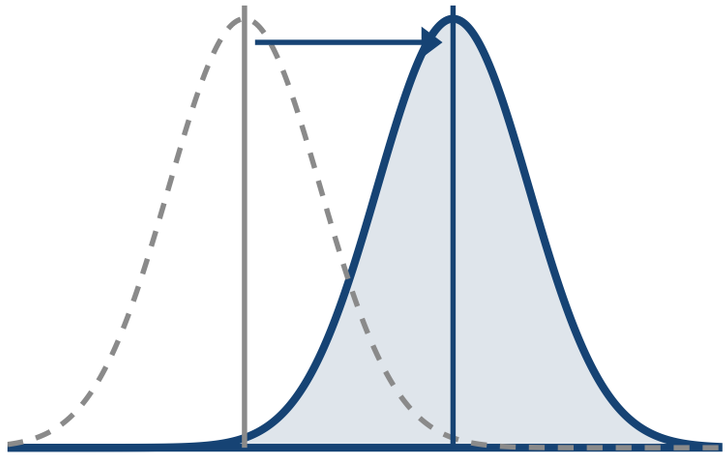
Descriptive statistics summarise a sample, usually while assuming something about the underlying distribution.

Speaker notes

Descriptive statistics summarise the observed data; inferential statistics draw conclusions about the population from them, with quantified uncertainty. The two vocabularies match: a **parameter** (μ, σ^2) is a fixed but unknown property of the population; a **statistic** ($\overline{x}$, s^2) is a function of the sample. Because the sample is random, a statistic is itself a random variable, with a distribution of its own across possible samples.

The population can be a finite set (all customers of a shop) or a conceptual one (all future transactions, the process that generates documents). In the latter case the population parameters are properties of the generating distribution, e.g. $\mu = E[X]$.

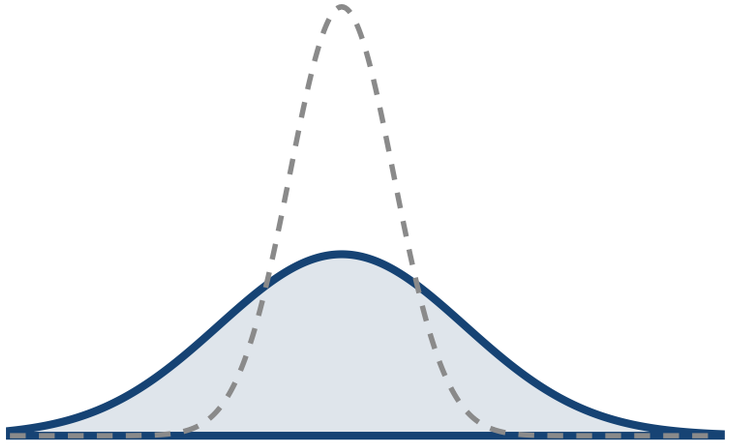
What We Summarise



Centre

1st moment

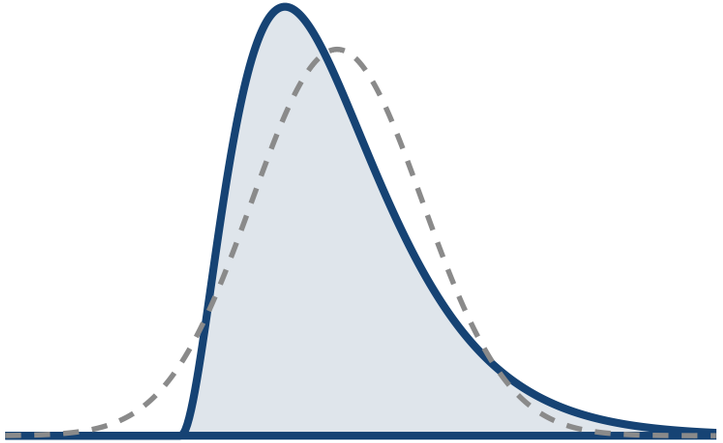
mean, median



Dispersion

2nd central moment

variance, std. dev., IQR



Shape

3rd central moment

skewness, quantiles

Univariate statistics describe one random variable; **multivariate** statistics describe $X_1, \ldots, X_d$ together.

Speaker notes

Each panel changes one aspect of a distribution against the dashed grey reference: the centre shifts, the spread widens, the shape becomes lopsided with a long right tail. The k-th moment of X is $E[X^k]$; the k-th central moment is $E[(X-\mu)^k]$. The mean is the first moment, the variance the second central moment. Skewness is the standardised third central moment, $E[(X-\mu)^3]/\sigma^3$: positive for a long right tail (incomes), negative for a long left tail, zero for symmetric distributions. The standardised fourth moment (kurtosis) describes the weight of the tails.

Moment-based summaries are not robust: a single extreme value can move them arbitrarily far. Quantile-based summaries are: the median for location, the interquartile range $Q_3 - Q_1$ (spread of the middle 50%) for dispersion. The multivariate counterparts of mean and variance are the mean vector and the covariance matrix.

Means

mean	formula	use it for	example
sample	$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$	values that add up	average salary
weighted	$\overline{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$	values that count unequally	grade over courses of different credits
geometric	$\left(\prod_{i=1}^{n} x_i\right)^{1/n}$	factors that multiply	+50% then −50%: factor 0.87 per period
harmonic	$\frac{n}{\sum_i 1/x_i}$	rates over equal amounts	60 km/h out, 40 back: 48 km/h
trimmed	average after dropping the k smallest and k largest	robustness on the cheap	judges' scores

Speaker notes

The sample mean estimates $E[X]$; it is also the value c that minimises the sum of squared deviations $\sum_i (x_i - c)^2$. Which mean is appropriate depends on how the quantities combine.

Speeds over equal distances average harmonically: 60 km/h out and 40 km/h back give $2/(1/60 + 1/40) = 48$ km/h, not 50, because more time is spent at the lower speed. Growth factors average geometrically: +50% followed by −50% gives the factor $1.5 \cdot 0.5 = 0.75$, so the average factor is $\sqrt{0.75} \approx 0.866$ (−13.4% per period), while the arithmetic mean of the rates, 0%, is wrong. For positive values, harmonic $\le$ geometric $\le$ arithmetic mean, with equality only when all values are equal; the geometric and harmonic means require positive values. A trimmed mean with k values dropped at each end tolerates up to k outliers per side; judged sports use it when the highest and lowest score are discarded. A grade average weighted by course credits is a weighted mean with the credits as weights.

Median

The **median** is the middle value of the sorted sample: for odd n the $\frac{n+1}{2}$-th value, for even n the average of the two middle values.

INTUITION

Half of the values lie below it, half above. Only their **order** counts, not how far out they lie.

✓ **Robust** against outliers — one extreme value moves the median by at most one position

✗ **Costlier** to compute — it needs sorting or a selection algorithm, not one pass with a running sum

For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.

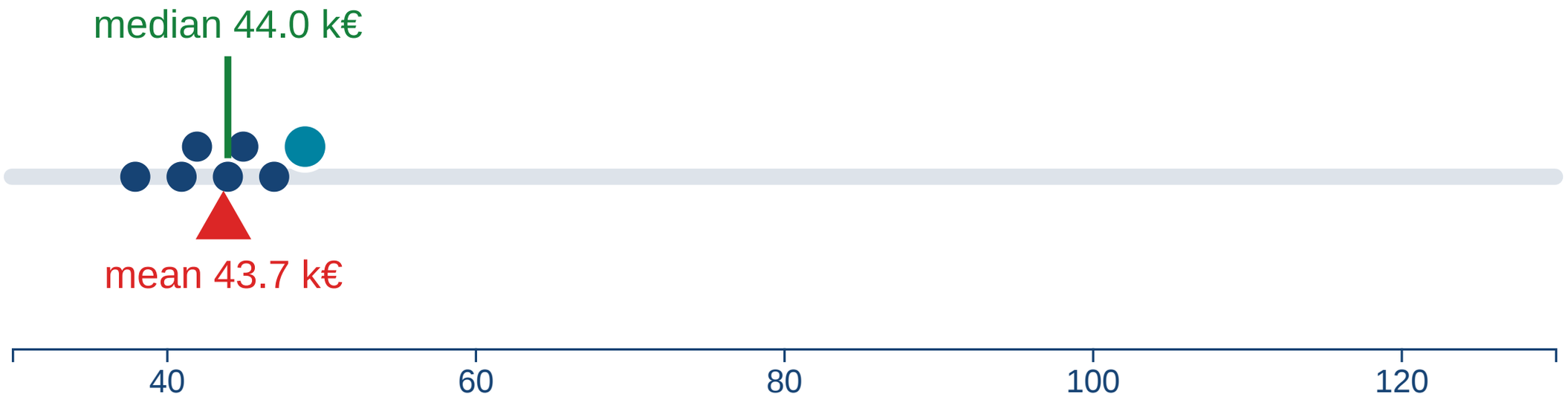
Speaker notes

The median minimises the sum of absolute deviations $\sum_i |x_i - c|$, as the mean minimises the sum of squares. Its robustness can be quantified by the breakdown point, the fraction of values that can be made arbitrarily large before the statistic follows: 50% for the median, 0% for the mean.

Full sorting ($O(n \log n)$) is sufficient but not necessary: selection algorithms such as quickselect find the median in expected $O(n)$ time. What the median lacks is the one-pass, constant-memory computation of the mean; streaming systems therefore use approximate quantiles. For binned data the median is interpolated as $L + \frac{n/2 - F}{f} \cdot w$, where L is the lower border of the interval containing the median, F the cumulative count below it, f the count in it, and w its width; this assumes the values are spread evenly within the interval.

One Value, Two Very Different Reactions

Drag the light point — which marker follows it?



Commit first: drag the top salary to the far right. Which marker follows?

Speaker notes

The steps show (1) the mean as the point where the beam balances, (2) the median as a position in the sorted order, and (3) the top salary made draggable. Moving one of n values by Δ moves the mean by Δ/n, so the mean follows the dragged value at a seventh of its speed. The median does not move at all while the dragged value stays above the middle value: it remains 44 k€.

With the top salary far to the right, the mean lies above six of the seven salaries, a "typical" value almost nobody earns. Which summary is quoted is therefore not neutral: an employer advertising pay has an incentive to quote the mean, an employee representative the median.

Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

Mini-Exercise — Solution

Speaker notes

Mean: $(38+41+42+44+45+47+260)/7 = 517/7 \approx 73.9$. Median: the 4th of the 7 sorted values, 44. Without the top salary the six remaining values have mean $257/6 \approx 42.8$ and median 43, so the two summaries nearly agree once the outlier is gone.

Neither number is wrong; each answers a different question. The mean is payroll divided by headcount and is the right figure for total cost ($7 \times 73.9 \approx 517$ k€). The median is the right figure for the salary of a typical employee. Quoting the mean in a recruitment ad suggests a salary only one employee earns.

Mini-Exercise — Solution

Mean ≈ 73.9, median $= 44$.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee; the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

Dispersion

Sample variance and **standard deviation** of the observed data:

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2, \qquad s = \sqrt{s^2}$$

INTUITION

The average squared distance from the centre: how stretched or squeezed the data is. s brings it back to the unit of the data.

Computable in **one pass**, $O(n)$, from the running totals n, $\sum x_i$ and $\sum x_i^2$ — which matters when the data does not fit in memory.

Speaker notes

Expanding the square gives $s^2 = \frac{1}{n}\sum_i x_i^2 - \overline{x}^2$, so the counts n, $\sum x_i$ and $\sum x_i^2$ suffice. These three totals can be accumulated in one pass and merged across partitions by simple addition, which is why streaming and distributed systems can compute variances without holding the data.

The formula is numerically unstable in floating point when the mean is large relative to the spread: two large, almost equal numbers are subtracted and the significant digits cancel (values around 10^9 with standard deviation 1 lose essentially all precision in double arithmetic). Welford's algorithm is the standard numerically stable one-pass alternative; it updates the running mean and the running sum of squared deviations with each new value.

Why the Denominator Changes

The **population variance** σ^2 is the variance over the whole population. With the true mean μ unknown, it is estimated by

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

INTUITION

$\overline{x}$ is computed from the same data, so it sits closer to the data than μ does: the deviations come out **systematically too small**. Dividing by $n-1$ instead of n compensates — one degree of freedom was spent on the mean.

Naming is not uniform: many textbooks, and pandas, call the $n-1$ version s^2; NumPy's `var` divides by n unless told otherwise.

Speaker notes

Derivation of the correction for i.i.d. data with mean μ and variance σ^2. Algebraically, $\sum_i (x_i - \overline{x})^2 = \sum_i (x_i - \mu)^2 - n(\overline{x} - \mu)^2$. Taking expectations, $E[\sum_i (x_i - \mu)^2] = n\sigma^2$ and $E[(\overline{x} - \mu)^2] = \mathrm{Var}(\overline{x}) = \sigma^2/n$, so $E[\sum_i (x_i - \overline{x})^2] = (n-1)\sigma^2$. Dividing by $n-1$ makes the estimator unbiased. The deviations are too small on average because $\overline{x}$ is, by construction, the value closest to the data in squared distance.

With $n = 1$ the sample variance is 0 (the point equals its own mean) and the corrected formula is $0/0$, undefined: one observation carries no information about spread. If μ is known, $\frac{1}{n}\sum_i (x_i - \mu)^2$ is already unbiased. Unbiasedness does not carry over to the square root: $\hat{\sigma}$ underestimates σ slightly on average. For large n the two denominators differ negligibly. In pandas, `var` uses `ddof=1` by default; in NumPy, `ddof=0`.

Takeaway

s^2 describes **the sample**; $\hat{\sigma}^2$ estimates **the population**. The $n-1$ is not a convention — it is the price of having estimated the mean from the same data.

Quantiles

> The k-th q-**quantile** is the value x such that k/q of the values are smaller and $(q-k)/q$ are larger ($0 < k < q$).

Percentiles

$q = 100$

"the 12th percentile"

Quartiles

$q = 4$

Q_1 = 25th, Q_2 = 50th (= median),
Q_3 = 75th percentile

IQR

$$\text{IQR} = Q_3 - Q_1$$

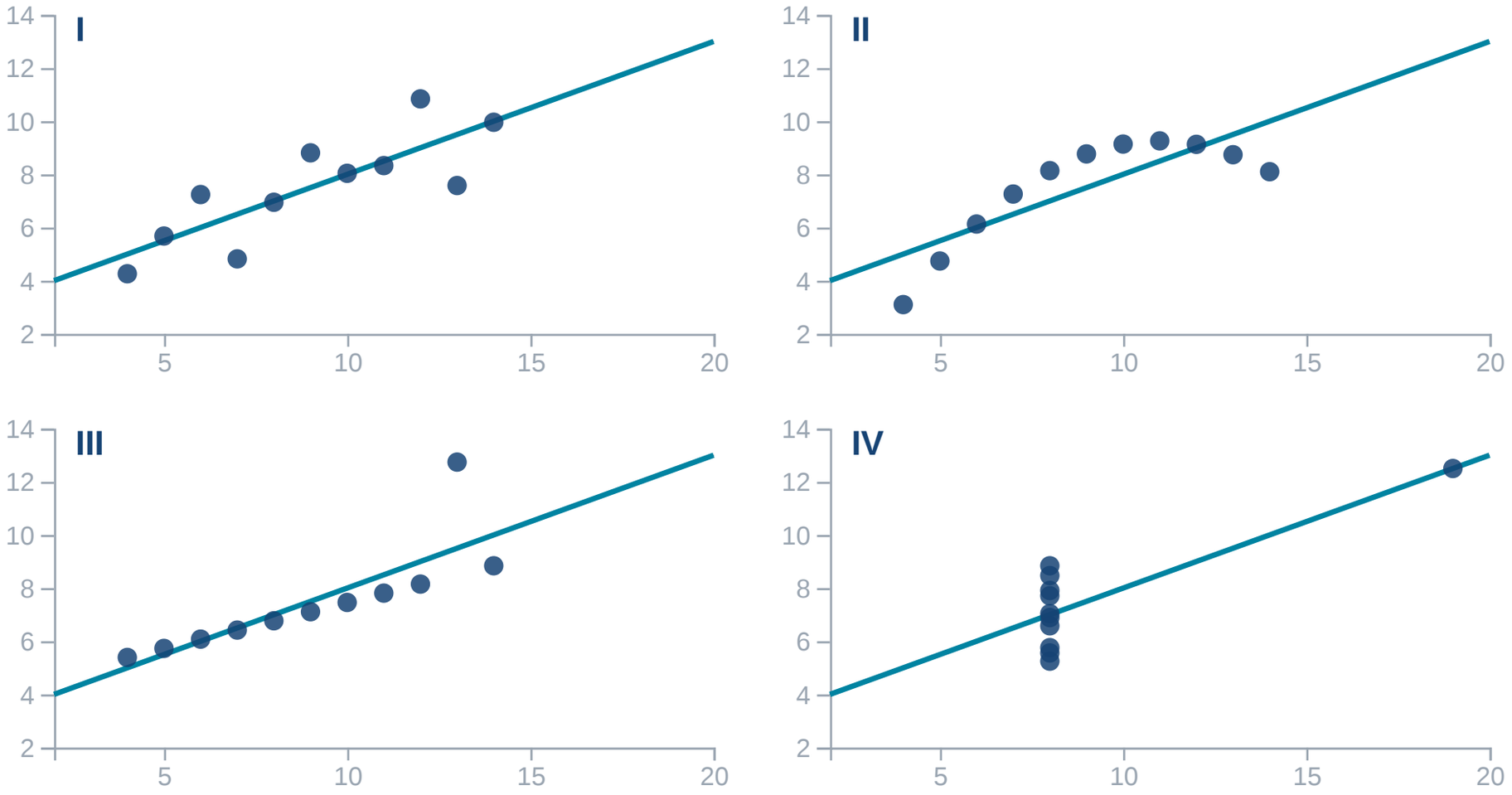
the middle half of the data

Quantiles summarise **shape and skewness** without assuming a distribution.

Speaker notes

Quantiles are the distribution-free alternative to moments: no assumption of symmetry, no sensitivity to tails. The boxplot is their standard graphical form — minimum, Q1, median, Q3, maximum, with outliers marked separately.

Anscombe's Quartet



Same mean, same variance, same correlation, same regression line — different data

all four: $\bar{x} = 9$, $s^2_x = 11$, $\bar{y} = 7.50$, $s^2_y \approx 4.12$, $r \approx 0.82$, $\hat{y} = 3.00 + 0.500 \cdot x$

Speaker notes

Stepped reveal (P6): one dataset per click, then the shared regression line and the shared statistics. All four have the same mean, variance, correlation and least-squares line. Ask after the fourth click, before revealing: "these four have identical summary statistics — what does that tell you about summary statistics?" Anscombe (1973), the origin of "always plot your data".

What the Quartet Teaches

Summaries compress — and compression loses.
Identical moments can hide a curve, an outlier, or a variable that barely varies at all.

Plot first, summarise second.
Visual inspection is not a soft alternative to statistics; it is how you find out which statistics are meaningful.

Quantiles describe a distribution without assuming its shape — and no set of summary numbers can replace looking at the data.

Speaker notes

This is the bridge to the data-exploration chapter, where visual analytics gets its own treatment. Stabilise the lesson as a working rule, not a slogan: every summary answers one question and is silent about the rest.

You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\overline{x} = 73.9$ k€.

What is the probability that the population mean μ is exactly 73.9 k€?

Speaker notes

The question has no useful answer. If uncertainty about μ is described by any continuous distribution, the probability of one exact value is zero; in the frequentist view μ is a fixed number and carries no probability at all (it is either 73.9 or not). In both readings, $\overline{x} = \mu$ exactly is not to be expected: a different sample of seven people would give a different mean.

The well-posed question concerns the size of the error $\overline{x} - \mu$. Its answer needs the distribution of $\overline{x}$ across samples, which is the subject of this unit. With seven values, one of them an outlier, the error can be large.

You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\overline{x} = 73.9$ k€.

What is the probability that the population mean μ is exactly 73.9 k€?

Zero. $\overline{x}$ is not μ — it is one **estimate** of it, computed from the seven people who happened to be in the sample.

The honest question is not *what is* μ but **how far from** μ **can this number be?**

Estimator and Estimate

Definition. An **estimator** is a function of the sample, $\hat{\theta} = g(X_1, \ldots, X_n)$, used to approximate an unknown quantity θ of the population. The number it takes on for one particular sample is an **estimate**.

INTUITION

The estimator is the **recipe**, $\overline{X} = \frac{1}{n}\sum_i X_i$; the estimate is what one batch of data cooks from it, $\overline{x} = 73.9$.

The sample is random, so the estimator is a **function of random variables**: $\hat{\theta}$ is itself a random variable, with a distribution of its own.

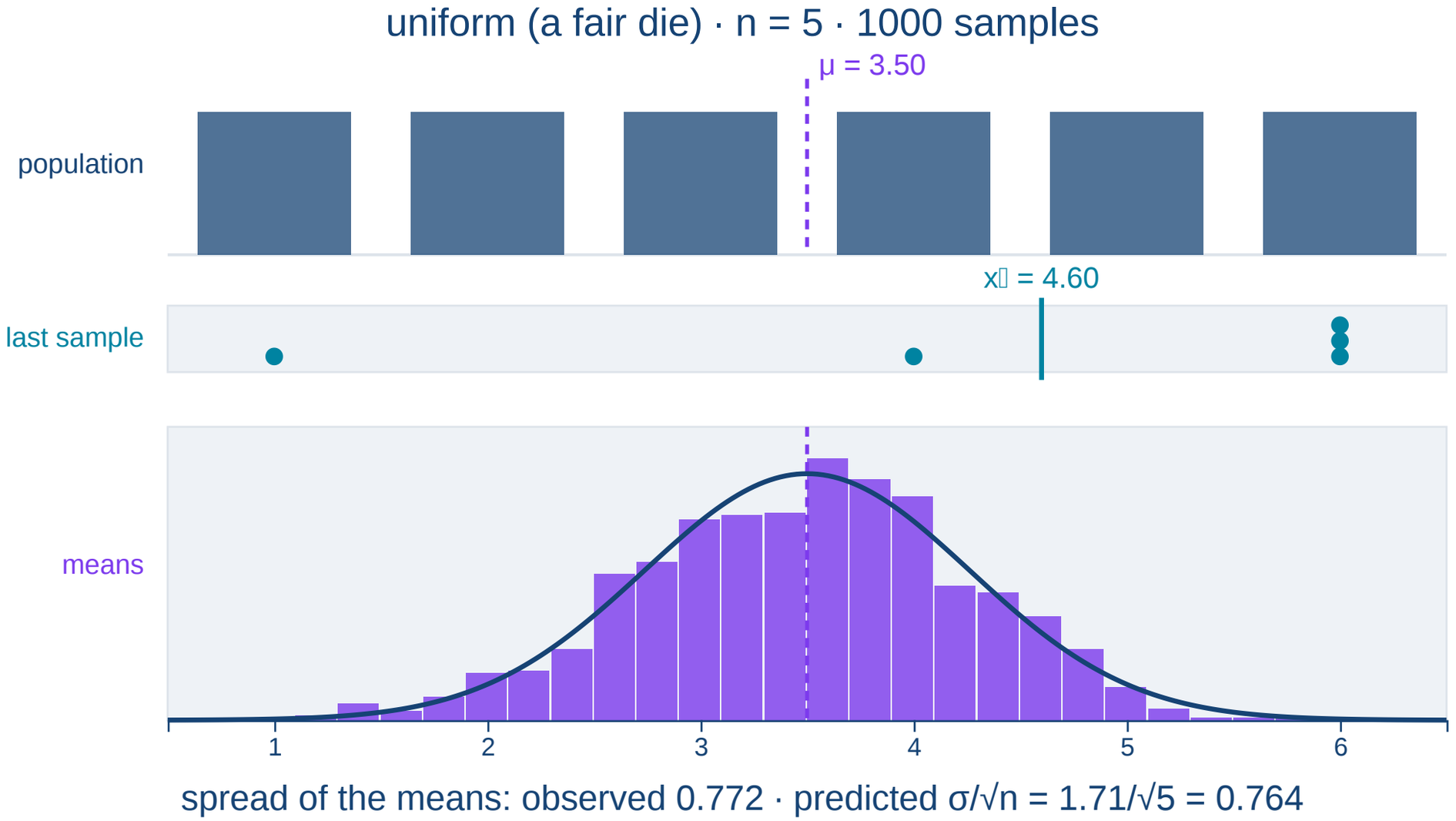
$\overline{x}$, s^2 and $\hat{\sigma}^2$ from descriptive statistics are all estimators — used so far without saying what kind of object they are.

Speaker notes

The distinction is the one between a rule and its output. The estimator $\overline{X} = \frac{1}{n}\sum_i X_i$ is written with capital letters because it is a function of the random variables $X_1, \ldots, X_n$; the estimate $\overline{x}$ is the number obtained by plugging in the observed values. A statistic inherits randomness from the sample, so properties such as bias and spread belong to the estimator, not to a single estimate.

Several estimators may target the same θ: the sample mean, the median and a trimmed mean all estimate the centre of a symmetric population. They differ in their sampling distributions, and comparing those distributions is how one estimator is judged better than another.

An Estimator Has a Distribution



Speaker notes

The steps show (1) one sample of $n = 5$ draws from the population in the upper panel, (2) its mean recorded as a single point in the lower panel, (3) the means of fifty samples, and (4) the means of a thousand samples with a normal curve and the standard error overlaid. Both panels share the horizontal axis, so their widths can be compared directly. Afterwards the population shape and n can be changed.

The populations live on the values 1 to 6. The uniform one is a fair die, with $\mu = 3.5$ and $\sigma = \sqrt{35/12} \approx 1.71$. For $n = 5$ the means have standard deviation $1.71/\sqrt{5} \approx 0.76$, less than half the population's. A frequent wrong prediction is that the means are spread as widely as the observations.

Three Things the Picture Shows

- → The means pile up **around** μ — the estimator is not systematically off
- → The pile is **narrower than the population** — averaging cancels noise
- → The pile is **bell-shaped**, even though the population was not

> **Definition (Sampling distribution).** The distribution of an estimator $\hat{\theta}$ over all possible samples of size n. $\hat{\theta}$ is **unbiased** if $E[\hat{\theta}] = \theta$.

Speaker notes

The three observations are unbiasedness ($E[\overline{X}] = \mu$, by linearity), the standard error ($\mathrm{Var}(\overline{X}) = \sigma^2/n$, by independence) and the central limit theorem (approximately normal shape). The first two hold exactly for every n; the third is an approximation that improves with n.

For the sample median the picture differs. In a symmetric population the median is centred on μ; in a skewed population it targets the population median, which differs from μ, and is in general not exactly unbiased even for that. Its spread also shrinks like $1/\sqrt{n}$, and for large n it is approximately normal, but this follows from a separate asymptotic result, not from the CLT for sums.

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

Speaker notes

Unbiasedness and the standard error $\sigma/\sqrt{n}$ do not depend on the population's shape at all; they need only independence and a finite variance. The bell shape is the approximate part. The bimodal population is the clearest case: two separate humps, yet the means of five draws already form a single hump.

How fast the bell shape appears depends on the population. For a symmetric population, a handful of draws suffices; for a strongly skewed one, the sampling distribution of the mean stays visibly skewed at small n, its skewness falling like $1/\sqrt{n}$. At $n = 1$ the sampling distribution is the population itself, and at $n = 2$ the population shape is still clearly visible. The CLT is a limit statement; "large enough n" depends on the population.

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

All three. The shape of the sampling distribution stops depending on the shape of the population as soon as n is more than a handful — which is why one formula can cover cases we never inspect.

Standard Error

Definition (Standard error). The standard deviation of an estimator's sampling distribution. For the sample mean of independent draws:

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$

INTUITION

How far off a typical **estimate** is — where σ says how far apart typical **observations** are. Both in the unit of the data.

σ and SE are different quantities. More data does nothing to σ — the world stays as variable as it was — and shrinks SE.

Speaker notes

$\sigma/\sqrt{n}$ follows from independence: $\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum_i \mathrm{Var}(X_i) = \sigma^2/n$. For dependent observations (e.g. a time series with positive correlation) the true standard error is larger, and the formula is overconfident. When sampling without replacement from a finite population of size N, the standard error is multiplied by $\sqrt{(N-n)/(N-1)}$, which is negligible for $n \ll N$ and gives 0 when the whole population is sampled.

In practice σ is unknown and replaced by the sample estimate $\hat{\sigma}$, giving $\hat{\sigma}/\sqrt{n}$. A weather forecast's "±3 °C" generally mixes both kinds of uncertainty: the inherent variability of the weather and the forecaster's limited knowledge.

Precision Costs Data

Multiply the data by k, and the standard error is multiplied by $1/\sqrt{k}$:

0.71	→	**½**	→	**⅕**	→	**1⁄10**
2 × the data		4 × the data		25 × the data		100 × the data

The $\sqrt{n}$ is why the last digit of an estimate is so expensive: halving the error costs four times the data, and a tenfold improvement costs a hundredfold.

Speaker notes

The table entries are $1/\sqrt{k}$ for k times the data: $1/\sqrt{2} \approx 0.71$, $1/\sqrt{4} = \frac{1}{2}$, $1/\sqrt{25} = \frac{1}{5}$, $1/\sqrt{100} = \frac{1}{10}$. Equivalently, the sample size needed for a target standard error e is $n = (\sigma/e)^2$.

Example: an A/B test comparing two conversion rates of about 5% that should detect a difference of 0.1 percentage points (at 5% significance and 80% power) needs roughly $2 \cdot 0.05 \cdot 0.95 \cdot (2.8/0.001)^2 \approx 750{,}000$ users per variant; the factor $2.8 \approx 1.96 + 0.84$ combines the two normal quantiles. Since precision bought with data is expensive, reducing σ itself — through better measurement or a model that explains part of the variation — is often the cheaper route.

Two Theorems We Have Already Seen Happen

For i.i.d. observations with mean μ and finite variance σ^2:

Law of large numbers	$\overline{X}_n \to \mu \text{ as } n \to \infty$	*the pile keeps narrowing onto the truth*
Central limit theorem	$\overline{X}_n \approx N(\mu, \sigma^2/n) \text{ for large } n$	*and we know the shape of what is left*

The first says the error goes away; the second says how much of it is left after n observations. Only the second lets us put a number on our ignorance.

Speaker notes

Law of large numbers: for i.i.d. observations with finite mean, $\overline{X}_n$ converges to μ. It states that the error vanishes but not how quickly. The central limit theorem adds the rate and the shape: the error $\overline{X}_n - \mu$ is approximately $N(0, \sigma^2/n)$. Only the second allows probability statements such as "within $\pm 2\,\mathrm{SE}$ in about 95% of samples".

A frequent error is to invoke the LLN where the CLT is needed, e.g. to justify an error bar. Another is to read the LLN as a balancing law ("after many heads, tails is due"): the mean converges because the influence of early deviations is diluted by later observations, not because they are compensated.

From a Number to an Interval

If $\overline{X} \approx N(\mu, \sigma^2/n)$, then for 95% of samples $\overline{X}$ lands within $1.96\,\mathrm{SE}$ of μ — and therefore μ lands within $1.96\,\mathrm{SE}$ of $\overline{X}$.

$$\overline{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

the estimate — our best single guess

how confident we insist on being

how noisy the estimate is

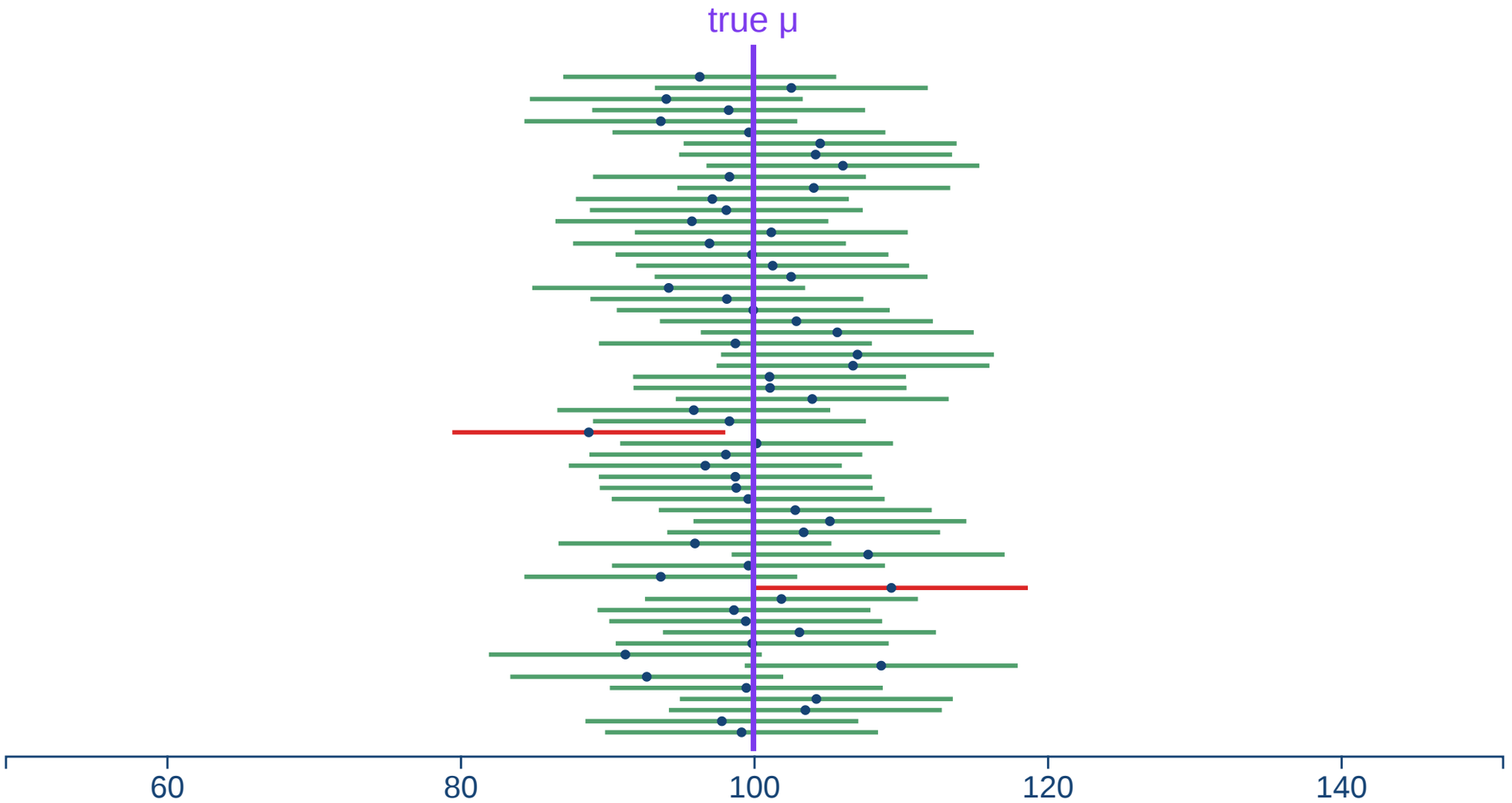
Speaker notes

The interval follows from the normal approximation: $P(|\overline{X} - \mu| \leq 1.96\,\mathrm{SE}) \approx 0.95$, and the event $|\overline{X} - \mu| \leq 1.96\,\mathrm{SE}$ is the same as $\mu \in [\overline{X} - 1.96\,\mathrm{SE}, \overline{X} + 1.96\,\mathrm{SE}]$. The value 1.96 is the 97.5% quantile of $N(0, 1)$; 90% intervals use 1.645, 99% intervals 2.576.

The interval widens with a higher confidence level, with larger σ, and with smaller n. The formula treats σ as known. When σ is replaced by $\hat{\sigma}$, the quantile 1.96 is replaced by a quantile of the t distribution with $n - 1$ degrees of freedom, which is larger for small samples (2.262 for $n = 10$) and approaches 1.96 as n grows.

What Does "95% Confident" Mean?

Change the level or n — which number follows, and which does not?



58 of 60 intervals cover μ (97%) — nominal 95%

Speaker notes

The steps show (1) one sample's interval against the fixed true mean, (2) twenty intervals, (3) all sixty with the coverage count, misses drawn in red, and (4) controls for the confidence level and n. With $\mu = 100$, $\sigma = 15$ and $n = 10$, the standard error is $15/\sqrt{10} \approx 4.74$ and each 95% interval has half-width ≈ 9.3; all intervals have the same width because σ is known, and only their centres move.

Of sixty 95% intervals, on average 3 miss μ; the count is itself binomial, $B(60, 0.05)$, so anything from 0 to about 6 is unremarkable. Changing the level changes the width and the miss rate; changing n changes the width but not the miss rate. A missing interval cannot be recognised from the data: it looks exactly like one that covers μ.

The Sentence Almost Everyone Gets Wrong

WHAT IT DOES NOT MEAN

"There is a 95% probability that μ lies in **this** interval."

μ is a fixed number. It is either in this interval or it is not — there is no probability left to have.

WHAT IT MEANS

"95% of the intervals produced **by this procedure** contain μ."

The probability is a property of the recipe, evaluated before the data is seen.

Speaker notes

Before the data are seen, the interval's end points are random variables, and $P(\mu \in [\overline{X} \pm 1.96\,\mathrm{SE}]) = 0.95$ is a genuine probability statement. After the data are plugged in, both end points and μ are fixed numbers, and the statement is either true or false. The 95% is therefore the coverage rate of the procedure under repeated sampling.

A Bayesian credible interval does support the first sentence: it is an interval to which the posterior distribution of μ, given the observed data, assigns 95% probability. This requires a prior distribution for μ. For the mean of normal data with a flat prior, the 95% credible interval coincides numerically with the 95% confidence interval, although the two statements mean different things.

Where This Leaves Us

Two kinds of uncertainty meet in one formula. We can now measure one of them.

- → σ — how variable the world is. **Aleatoric.** Unchanged by data.
- → $\mathrm{SE} = \sigma/\sqrt{n}$ — how badly we know our own estimate. **Epistemic.** Shrinks with data.

"More data helps" was a slogan.
It is now a rate: $1/\sqrt{n}$.

Speaker notes

Aleatoric uncertainty is the variability inherent in the data-generating process; it is described by σ and does not shrink with more data. Epistemic uncertainty is uncertainty due to limited knowledge; for the sample mean it is the standard error, which shrinks like $1/\sqrt{n}$. A prediction of a single new observation carries both: its variance is approximately $\sigma^2 + \sigma^2/n$, so for large n the epistemic part vanishes and the aleatoric part remains.

The claim that a model is uncertain "because it has not seen enough data" therefore refers to epistemic uncertainty, and the $1/\sqrt{n}$ rate gives its order of magnitude.

Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

Speaker notes

$\mathrm{SE} = 60/\sqrt{100} = 6$ s. The 95% interval is $180 \pm 1.96 \cdot 6 = 180 \pm 11.8$, i.e. about $[168, 192]$ s. Halving the SE requires doubling $\sqrt{n}$, so $n = 400$. In general, a target standard error e requires $n = (\sigma/e)^2$; for $e = 3$ s this gives $n = 400$.

If σ had to be estimated from the same 100 customers, the interval would use $\hat{\sigma}$ and a t quantile with 99 degrees of freedom (≈ 1.98), practically the same as 1.96 at this sample size. The calculation also assumes independent customers; customers served in the same queue at the same time can be correlated, which would make the true standard error larger.

Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

$\mathrm{SE} = 60/\sqrt{100} = 6$ s, so the 95% interval is about 180 ± 12 s. Halving the SE needs twice the $\sqrt{n}$ — **four times the data**, $n = 400$. $\sigma = 60$ s does not change: customers stay as variable as they were; only our knowledge of their average improves.

Takeaway

A statistic is a **random variable**. Its distribution over repeated samples is the sampling distribution; its spread is the **standard error**, $\sigma/\sqrt{n}$. A confidence interval reports that spread — and its percentage describes the **procedure**, not the one interval in front of you.

Speaker notes

The three points are: a statistic computed from a sample is a random variable; its standard error shrinks like $1/\sqrt{n}$; and the confidence level of an interval is a property of the procedure. They apply to every reported measurement with an error bar, including an accuracy or error rate estimated on a test set: an accuracy measured on 1,000 test cases at around 90% has a standard error of $\sqrt{0.9 \cdot 0.1/1000} \approx 0.95$ percentage points. A Bayesian treatment answers the same question with a posterior distribution over the unknown quantity instead of a sampling distribution.

Module 7

Hypothesis Testing

> ***Guiding Question:*** *Your new model scores higher than the baseline. How do you know the improvement is real?*

Motivation: Comparing Two ML Experiments

Experiment A — baseline model accuracy measured over n_A runs

Experiment B — new architecture accuracy measured over n_B runs

B scored higher. Is the improvement systematic, or just randomness?

H_0: the two means are equal — $\mu_A = \mu_B$
H_1: they differ — $\mu_A \neq \mu_B$

Speaker notes

Guiding question of the module, in the students' own working context. Every one of them has compared two model runs; almost none has asked whether the difference survives resampling. Keep the example running through the unit.

The Idea

> Compare what we **observed** with what H_0 **predicts**. If the disagreement is large enough to be implausible under H_0, reject H_0.

We never prove H_1. We only measure how badly H_0 fails to explain the data.

Speaker notes

The asymmetry is the hardest idea in the unit. "Fail to reject" is not "accept". Ask what a court verdict of "not guilty" claims, and what it does not.

Test Statistic and p-Value

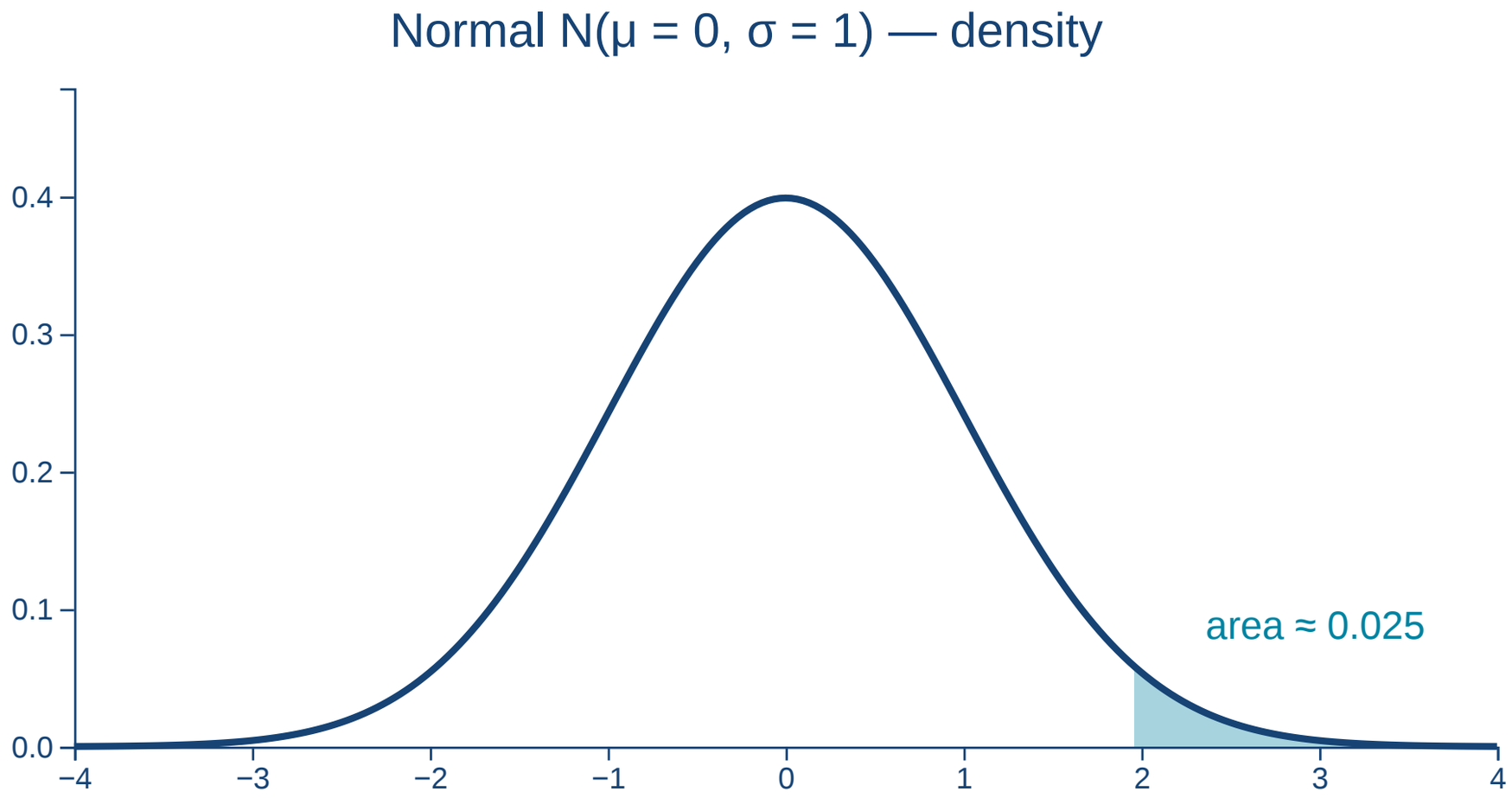
A **test statistic** $T = T(X_1, \ldots, X_n)$ compresses the data so that its distribution **under** H_0 is known.
The **p-value** is the probability, under H_0, of observing a statistic at least as extreme as the one we got.
Small p-value → the data is surprising under H_0.

> The p-value is **not** the probability that H_0 is true. It is $P(|T| \geq |t_{\text{obs}}| \mid H_0)$ — a tail probability of the statistic — not $P(H_0 \mid \text{data})$.

Speaker notes

The warning box is the Bayes unit returning: swapping the two conditionals is the same error as the disease test, now with a career's worth of published consequences. Make that link explicit — students remember it.

The Rejection Region



Under H_0 the statistic follows a known distribution. The shaded tail carries probability $\alpha/2$: if T lands there, we reject.

Speaker notes

One click shades the tail. Emphasise: α is chosen *before* seeing the data, and it is the rate of false alarms we are willing to tolerate — a decision, not a discovery.

Two Ways to Be Wrong

Type I error — false alarm
Reject H_0 when it is true.
Probability α (we choose it).

Type II error — miss
Fail to reject H_0 when H_1 holds.
Probability β; **power** $= 1 - \beta$.

Lowering α raises β for a fixed design. To improve both: more data, less noise, or a **paired** design that cancels shared noise.

Speaker notes

The trade-off is the practical content of the slide. In an ML context: a strict α means real improvements get dismissed; a lax α means noise gets published. Power analysis answers "how many runs do I need", which is the question people actually have. The paired design is the cheap one in ML: evaluate both models on the same seeds and test splits, and the split-to-split noise cancels in the differences.

One Sample: z or t?

Known variance — z-test

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$$

Reject (two-sided) if $|Z| > z_{1-\alpha/2}$.

Unknown variance — t-test

$$T = \frac{\overline{X} - \mu_0}{s/\sqrt{n}}$$

s estimated from the data → **heavier tails** (t with $n-1$ d.f.).

$\sigma/\sqrt{n}$ is the **standard error**: the standard deviation of the sample mean $\overline{X}$. $s = \sqrt{s^2}$ is the sample standard deviation. Both statistics measure the distance from μ_0 in standard errors.

Speaker notes

The only difference between the two is the denominator. Plugging in an estimate s for σ adds uncertainty, which the t distribution's heavier tails pay for; with $n \geq 30$ the two are practically the same.

Two Samples: Which t?

Situation	Test	Statistic
independent runs, equal variances credible	pooled t	$\frac{\overline{X}_B - \overline{X}_A}{s_p\sqrt{1/n_A + 1/n_B}}$
independent runs, variances may differ	**Welch's t**	$\frac{\overline{X}_B - \overline{X}_A}{\sqrt{s_A^2/n_A + s_B^2/n_B}}$
both models on the **same** seeds or splits	**paired t**	t-test on the differences $d_i = x_{B,i} - x_{A,i}$

> Welch is the default for independent runs. When both models can be run on the same splits, pair them: the shared noise cancels in d_i.

Speaker notes

A table because the three tests differ along the same dimensions. Welch by default because the equal-variance assumption is rarely justified across architectures. The paired test is the one ML comparisons usually need and rarely use: a cross-validation comparison evaluates both models on the same folds, so the fold results are not independent samples.

Testing Variances

Single variance (normal model), χ^2 statistic with $n-1$ degrees of freedom.

Is run-to-run variability within specification?

Two variances, $F = s_A^2 / s_B^2$.

If the variances differ, **do not pool —** go back to Welch's t for the means.

Speaker notes

Variance tests matter in ML more than people think: a model that is equally accurate on average but wildly less stable across seeds is a different product.

Back to the A/B Question — Worked

Ten runs each. A: mean accuracy 80 %, $s_A = 2$. B: mean 82 %, $s_B = 1$.

$$t = \frac{82 - 80}{\sqrt{4/10 + 1/10}} = \frac{2}{0.71} \approx 2.83$$

Speaker notes

The numbers are chosen so the standard error is $\sqrt{0.5} \approx 0.71$. Welch's d.f.: $0.25/((0.16 + 0.01)/9) \approx 13.2$. The paired numbers are hypothetical — the point is that the per-seed differences vary far less than either model's accuracies do, because a hard seed is hard for both.

Back to the A/B Question — Worked

Ten runs each. A: mean accuracy 80 %, $s_A = 2$. B: mean 82 %, $s_B = 1$.

$$t = \frac{82 - 80}{\sqrt{4/10 + 1/10}} = \frac{2}{0.71} \approx 2.83$$

Welch's degrees of freedom ≈ 13; the critical value $t_{0.975} \approx 2.16$. Since $2.83 > 2.16$, **reject** H_0 at $\alpha = 0.05$ ($p \approx 0.014$).

Back to the A/B Question — Worked

Ten runs each. A: mean accuracy 80 %, $s_A = 2$. B: mean 82 %, $s_B = 1$.

$$t = \frac{82 - 80}{\sqrt{4/10 + 1/10}} = \frac{2}{0.71} \approx 2.83$$

Welch's degrees of freedom ≈ 13; the critical value $t_{0.975} \approx 2.16$. Since $2.83 > 2.16$, **reject** H_0 at $\alpha = 0.05$ ($p \approx 0.014$).

> Paired on the same ten seeds, with differences of mean 2 and $s_d = 0.5$: $t = 2/(0.5/\sqrt{10}) \approx 12.6$. Pairing buys far more power than more runs.

The Recommendation

Default

Welch's two-sample t on the mean difference — or paired t if both ran on the same splits

Care about stability

F-test on s_A^2 / s_B^2

Normality dubious

Permutation (randomisation) test on the mean difference

Hypothesis testing compares data to a model through a statistic whose null distribution is known. Known vs. unknown variance changes that distribution (z vs. t); the design changes the test (Welch for independent runs, paired for shared splits).

Speaker notes

Close by returning to the opening question with a concrete recommendation — the module started with a decision they have to make, and ends with the decision rule. Mention that the permutation test needs no distributional assumption at all, which is why it is a good fallback when in doubt.

The Churn Shop: Is the Difference Real?

Lecture 01's dashboard, with counts. Over one year:

	customers	churned	rate
app users	400	40	10%
no app	400	60	15%

H_0: both groups churn at the same rate. Pooled rate $\hat{p} = 100/800 = 0.125$.

Speaker notes

The running example of the course (a subscription business, introduced in the Lecture 01 reasoning-cycle exercise), now with numbers. The statistic is the z-test of this lecture applied to two proportions: a proportion is the mean of a Bernoulli variable, so its standard error is $\sqrt{(p(1-p)/n)}$, and under H_0 both groups share one p, estimated from all 800 customers. Exact values: SE ≈ 0.0234, z ≈ 2.14, two-sided p ≈ 0.033.

Say the limit clearly: the test says the gap is unlikely under "no difference"; it says nothing about which way the arrow points. Loyal customers may simply be the ones who install the app.

The Churn Shop: Is the Difference Real?

Lecture 01's dashboard, with counts. Over one year:

	customers	churned	rate
app users	400	40	10%
no app	400	60	15%

H_0: both groups churn at the same rate. Pooled rate $\hat{p} = 100/800 = 0.125$.

$$z = \frac{0.15 - 0.10}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{400} + \frac{1}{400}\right)}} = \frac{0.05}{0.023} \approx 2.1 \qquad p \approx 0.03$$

The Churn Shop: Is the Difference Real?

Lecture 01's dashboard, with counts. Over one year:

	customers	churned	rate
app users	400	40	10%
no app	400	60	15%

H_0: both groups churn at the same rate. Pooled rate $\hat{p} = 100/800 = 0.125$.

$$z = \frac{0.15 - 0.10}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{400} + \frac{1}{400}\right)}} = \frac{0.05}{0.023} \approx 2.1 \qquad p \approx 0.03$$

> Chance alone would rarely produce a gap this large: at $\alpha = 0.05$ the difference is **real**. Whether the app *causes* it is a different question — Lecture 03 draws both answers.

Image Credits & References

Sheldon, Ross and others (2002). A first course in probability. *Pearson Education India*.